Optimocracy: Evidence-Based Governance Through Outcome-Bound Optimization

Mike P. Sinn

Abstract

Politicians allocate trillions annually. What gets you reelected is often not what benefits voters. This misalignment costs an estimated 20% (95% CI: 9%-39.3%) of potential welfare.

Optimocracy: (1) Analyze historical data on policies across jurisdictions, (2) Calculate which predict above-average median income and healthy life years, (3) Publish recommendations for every vote, (4) Track politician alignment, (5) Fund accordingly via SuperPAC. Politicians vote however they want. Following evidence gets funded; ignoring it doesn't.

The structural advantage: current governance has thousands of capture points (committees, agencies, regulations). Optimocracy consolidates these to a single, harder target: the verification layer. Corrupting five independent data sources requires coordination that corrupting individual committee decisions does not.

Table of contents

1 The Mechanism

Optimocracy is simple:

1. **Analyze historical data**: Examine every policy and budget allocation across thousands of jurisdictions (municipal, state, federal, international) going back decades. For each, measure what happened to median real after-tax income and healthy life years.

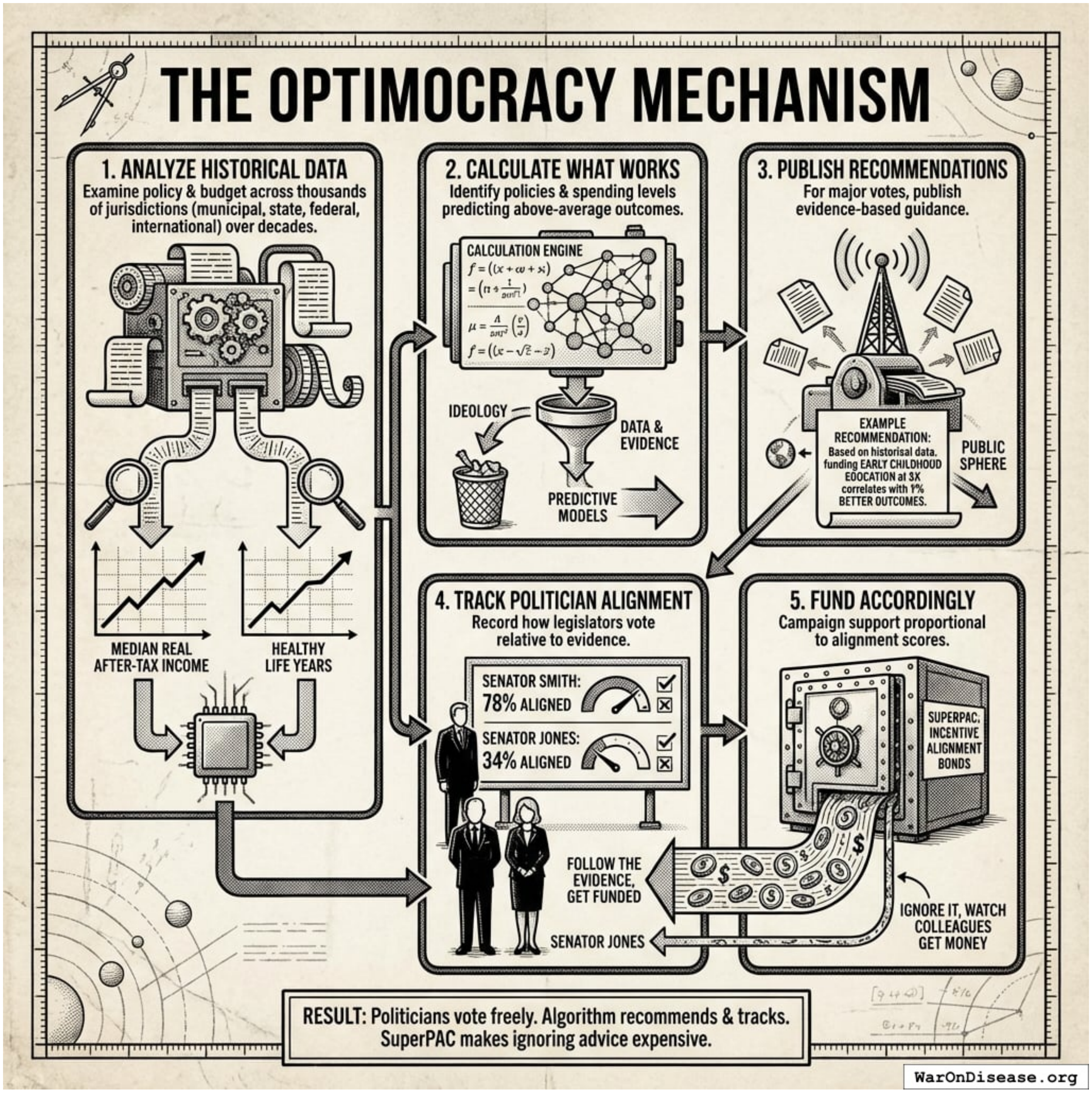

Figure 1: A flowchart illustrating the Optimocracy cycle: from historical data analysis and outcome calculation to evidence-based recommendations, alignment tracking, and incentive-based funding.

2. **Calculate what works**: Which policies and spending levels actually predicted above-average outcomes? Not ideology. Data.
3. **Publish recommendations**: For every major vote, publish what the evidence suggests. "Based on historical data, funding early childhood education at $X correlates with Y% better outcomes."
4. **Track politician alignment**: When legislators vote, record how often they align with evidence-based recommendations. Senator Smith: 78% aligned. Senator Jones: 34% aligned.
5. **Fund accordingly**: A SuperPAC allocates campaign support to maximize expected gover-

nance improvement. The algorithm weighs alignment differences between candidates, position power, race competitiveness, and marginal funding effectiveness. Close races with large alignment gaps get priority. Follow evidence, get funded. Ignore it, and resources flow to races where they matter more.

That's it. Politicians still vote however they want. The algorithm just recommends and tracks. The SuperPAC makes ignoring good advice expensive.

1.1 Why This Works

Capture resistance: Currently, lobbyists have thousands of capture points: committee earmarks, agency decisions, regulatory rulings. Each is relatively cheap to influence. Optimocracy consolidates these to a single target: the verification layer measuring health and wealth. Corrupting five independent data sources (Census Bureau, Federal Reserve, BLS, academic institutions, citizen surveys) requires coordination across institutions with different governance structures, funding sources, and methodologies. Corruption doesn't disappear; it just becomes prohibitively expensive.

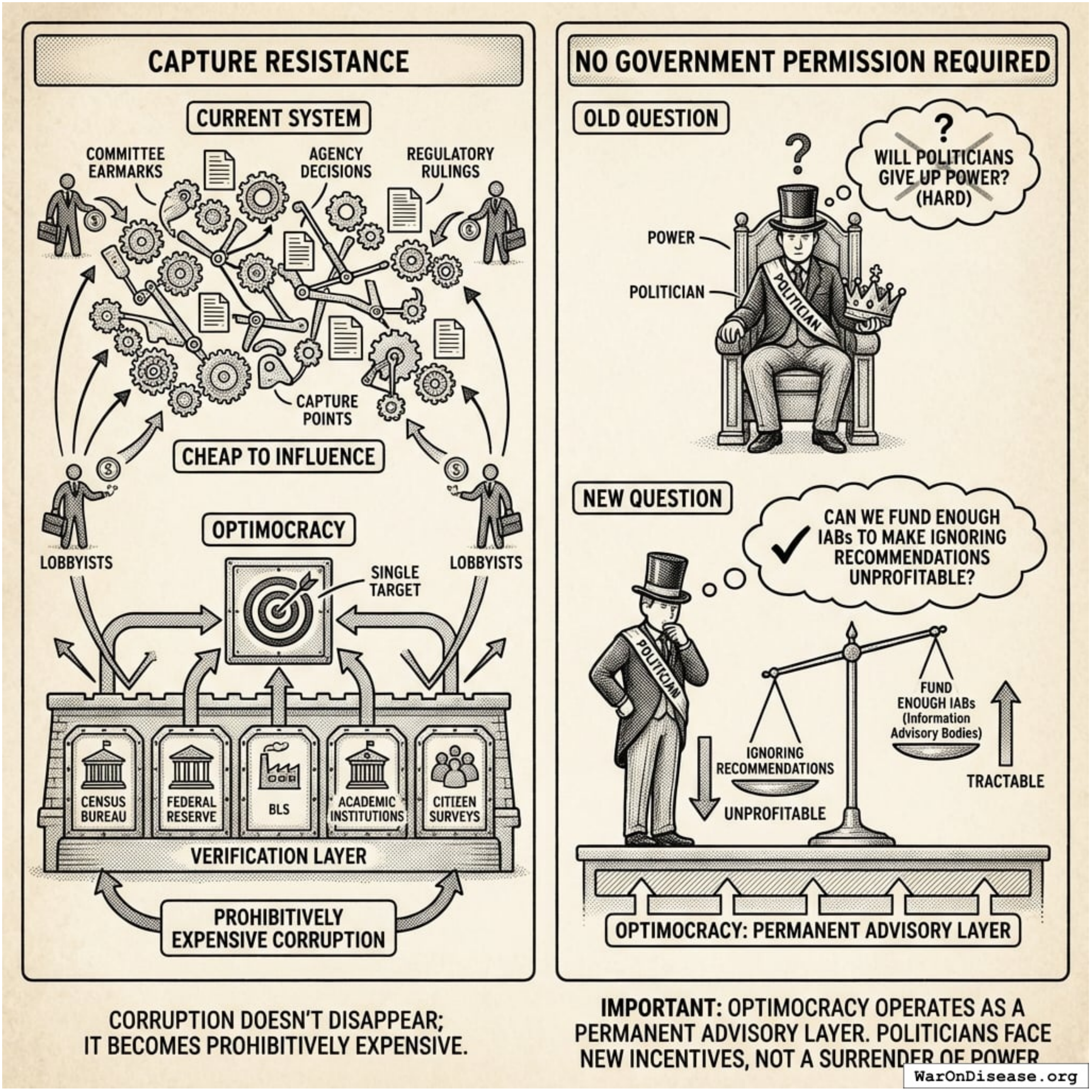

Figure 2: A comparison between the current fragmented system of thousands of lobbyist capture points and Optimocracy's consolidated verification layer protected by multiple independent data sources.

No government permission required: Optimocracy operates as a permanent advisory layer. Politicians don't surrender power; they just face new incentives. The question changes from "Will politicians give up power?" (hard) to "Can we fund enough IABs to make ignoring recommendations unprofitable?" (tractable).

2 The Problem: Political Dysfunction Tax

Political actors optimize for what gets them reelected: campaign contributions, constituent services, ideological positioning. These differ systematically from what improves measured outcomes. The result: systematic resource misallocation costing an estimated 20% (95% CI: 9%-39.3%) of potential welfare. For full derivation, see The Political Dysfunction Tax.

3 Why Information Alone Fails

Rankings of government programs by cost-effectiveness already exist. The Copenhagen Consensus publishes rigorous benefit-cost analyses: childhood vaccinations (101:1 BCR), e-government procurement (125:1), maternal health interventions (87:1). GiveWell, Open Philanthropy, and academic institutions produce similar analyses.

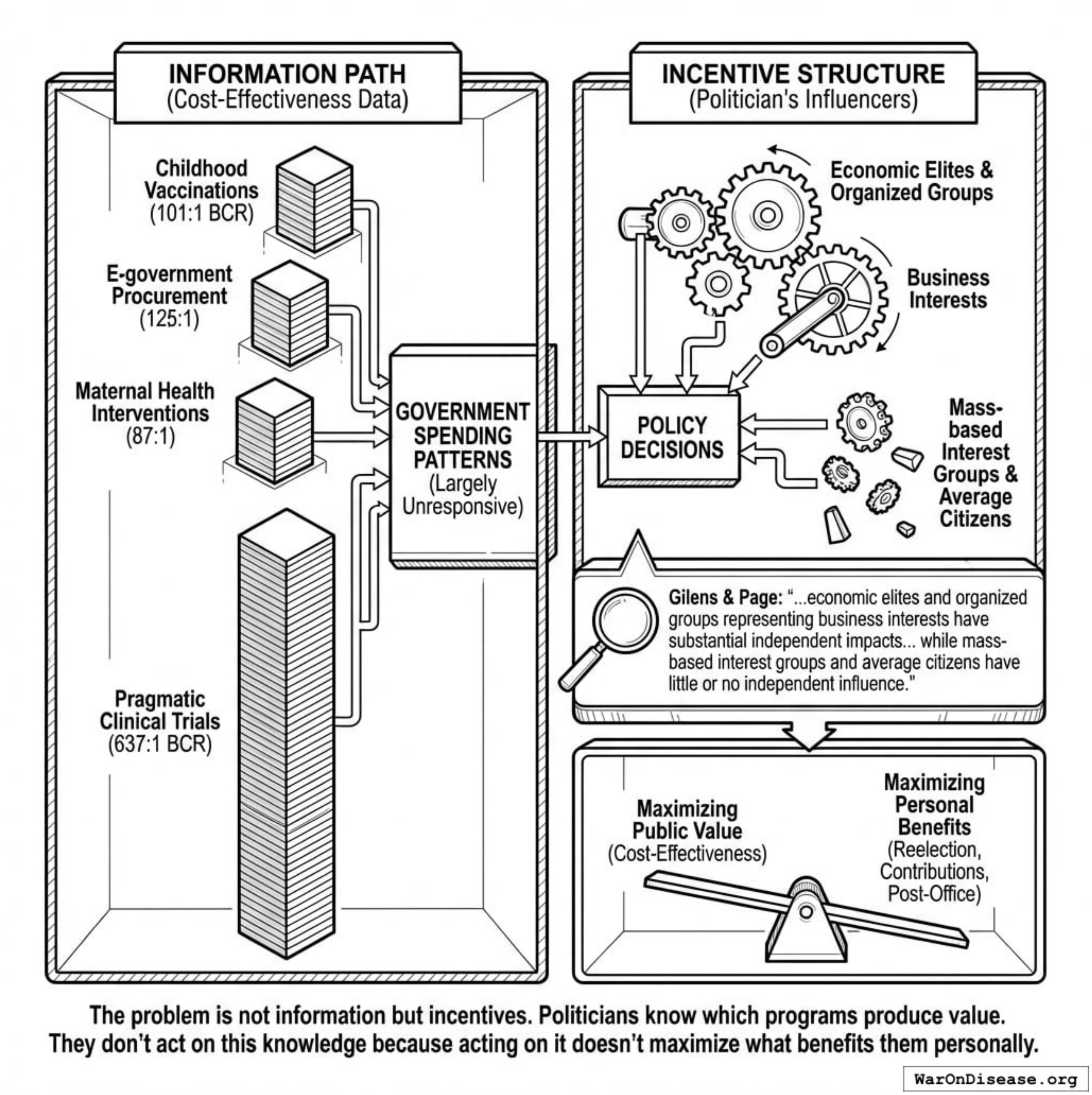

Figure 3: A conceptual diagram showing the disconnect between high-return evidence and government action, where political incentives such as reelection and campaign contributions act as a barrier to implementing cost-effective programs.

Yet government spending patterns remain largely unresponsive.[138] analyzed 1,779 policy decisions: "economic elites and organized groups representing business interests have substantial independent impacts on U.S. government policy, while mass-based interest groups and average citizens have little or no independent influence."

The problem is not information but incentives. Politicians know which programs produce value. They don't act because acting doesn't maximize reelection probability, campaign contributions, or post-office career prospects.

4 The Scale of Welfare Loss: Empirical Foundations

Before proposing solutions, we must establish the magnitude of the problem. This section synthesizes peer-reviewed research documenting welfare losses from suboptimal policy. The metaphor "trillion dollar bills on the sidewalk" comes from[87], who showed that differences between rich and poor countries are primarily due to institutions and policies, not factors of production.

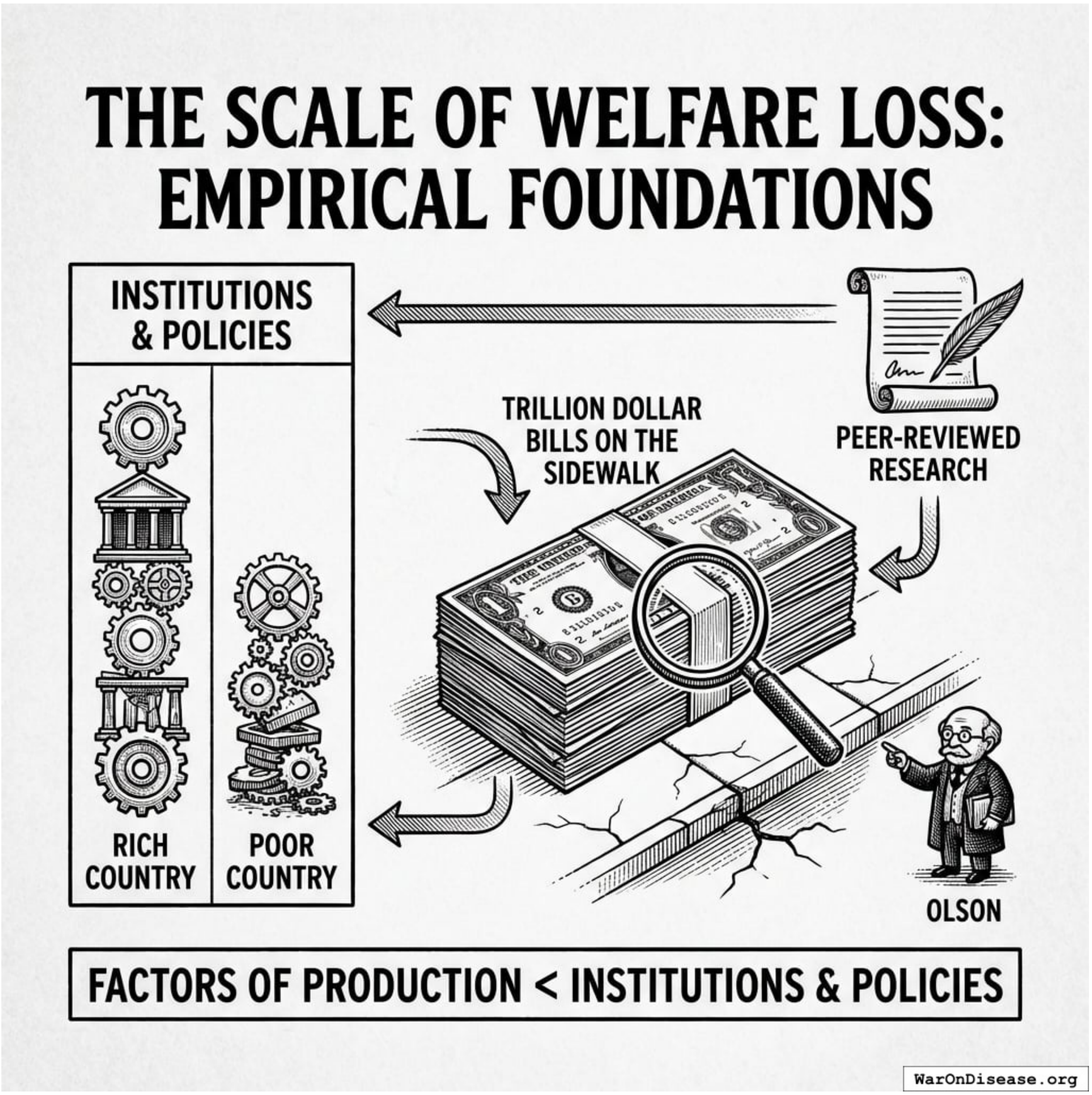

Figure 4: A conceptual diagram illustrating how institutional and policy quality, rather than raw factors of production, serve as the primary drivers for the disparity in economic welfare between nations.

4.1 A Formal Model of the Political Dysfunction Tax

Let W^* represent maximum achievable welfare under optimal policy, and W represent actual welfare under current policy. We define the **Political Dysfunction Tax** as:

$$\tau_{dysfunction} = \frac{W^* - W}{W^*} = 1 - \frac{W}{W^*}$$

This can be decomposed into sources:

$$\tau_{dysfunction} = \tau_{crony} + \tau_{time} + \tau_{information} + \tau_{coordination}$$

Substituting empirical estimates:

$$\begin{aligned} &\tau_{dysfunction} \\ &= \tau_{crony} + \tau_{time} + \tau_{info} + \tau_{coord} \\ &= 10\% + 3\% + 2\% + 5\% \\ &= 20\% \end{aligned}$$

Where:

- τ_{crony}: **Crony Tax**, welfare loss from resources flowing to concentrated interests rather than outcome-maximizing alternatives (10% (95% CI: 5%-20%))
- τ_{time}: **Short-Termism Cost**, welfare loss from short electoral horizons causing underinvestment in long-term goods (3% (95% CI: 1%-6%)). Politicians facing re-election in 4 years systematically underinvest in goods that pay off over 10-30 years: basic research, infrastructure maintenance, pandemic preparedness, climate mitigation.
- $\tau_{information}$: **Ignorance Cost**, welfare loss from decision-makers lacking relevant knowledge (2% (95% CI: 1%-4%))
- $\tau_{coordination}$: **Ideological Gridlock Cost**, welfare loss from inability to commit to welfare-improving policies (5% (95% CI: 2%-10%))

Total Political Dysfunction Tax: 20% (95% CI: 9%-39.3%)

Optimocracy primarily addresses τ_{crony} and τ_{time} by making evidence-based recommendations public and rewarding politicians (via campaign support and career opportunities) in proportion to their alignment with those recommendations.

4.2 Documented Welfare Losses by Policy Domain

The following table synthesizes estimates from peer-reviewed research:

Policy Domain	Source	Methodology	Estimated Welfare Loss	Confidence
US regulatory accumulation (1980-2012)	[139]	Counterfactual growth trajectory	25% of GDP ($4T annually)	Low†
US regulation (1949-2005)	[140]	Panel regression, regulatory index	GDP would be 3.5x higher	Low†

Policy Domain	Source	Methodology	Estimated Welfare Loss	Confidence
Global corruption	[141]	Multiple estimation approaches	5% of GDP (~$5T/year)	Medium
FDA drug delays (1960-2001)	[142]	Consumer/producer surplus	140M life-years lost	Medium
Trade barriers	[143]	Gravity models	5-10% of GDP	High
Occupational licensing	[14]	Labor market distortion	2-3% of GDP	High

†Think tank source with weak causal identification; achievable gains likely smaller.

Note on interpretation: These estimates are not additive; many inefficiencies interact and overlap. However, even conservative aggregation suggests the Political Dysfunction Tax exceeds 20% (95% CI: 9%-39.3%) of potential welfare.

This pattern (massive welfare losses persisting due to political economy constraints) recurs across policy domains. The question is not whether trillion-dollar bills exist, but why they remain on the sidewalk.

Calibrating the estimates: These theoretical maxima require heroic assumptions. Regulatory estimates from think tanks use weak causal identification; 5-10% is more defensible than 25%. After adjusting:

- Regulatory reform: 5-10% of GDP (not 25%)
- Corruption reduction: 3-5% of GDP
- Other allocative improvements: 5-10% of GDP

Conservative aggregate: 20% (95% CI: 9%-39.3%) of global GDP in achievable welfare improvements from better policy. This is still enormous ($20-40 trillion annually) and sufficient to motivate Optimocracy, without relying on heroic assumptions.

4.3 Alternative Estimation: Bottom-Up Policy Cost Accounting

The theoretical estimate of 20% (95% CI: 9%-39.3%) can be triangulated with bottom-up accounting of documented US policy costs. Our Federal Resource Allocation Efficiency Audit enumerates specific, measurable policy failures across defense, healthcare, justice, regulatory, and subsidy subsystems, totaling $5.03T (95% CI: $3.44T-$7.69T) (17.5% (95% CI: 11.9%-26.7%) of GDP, 95% CI: 11.9%-26.7%).

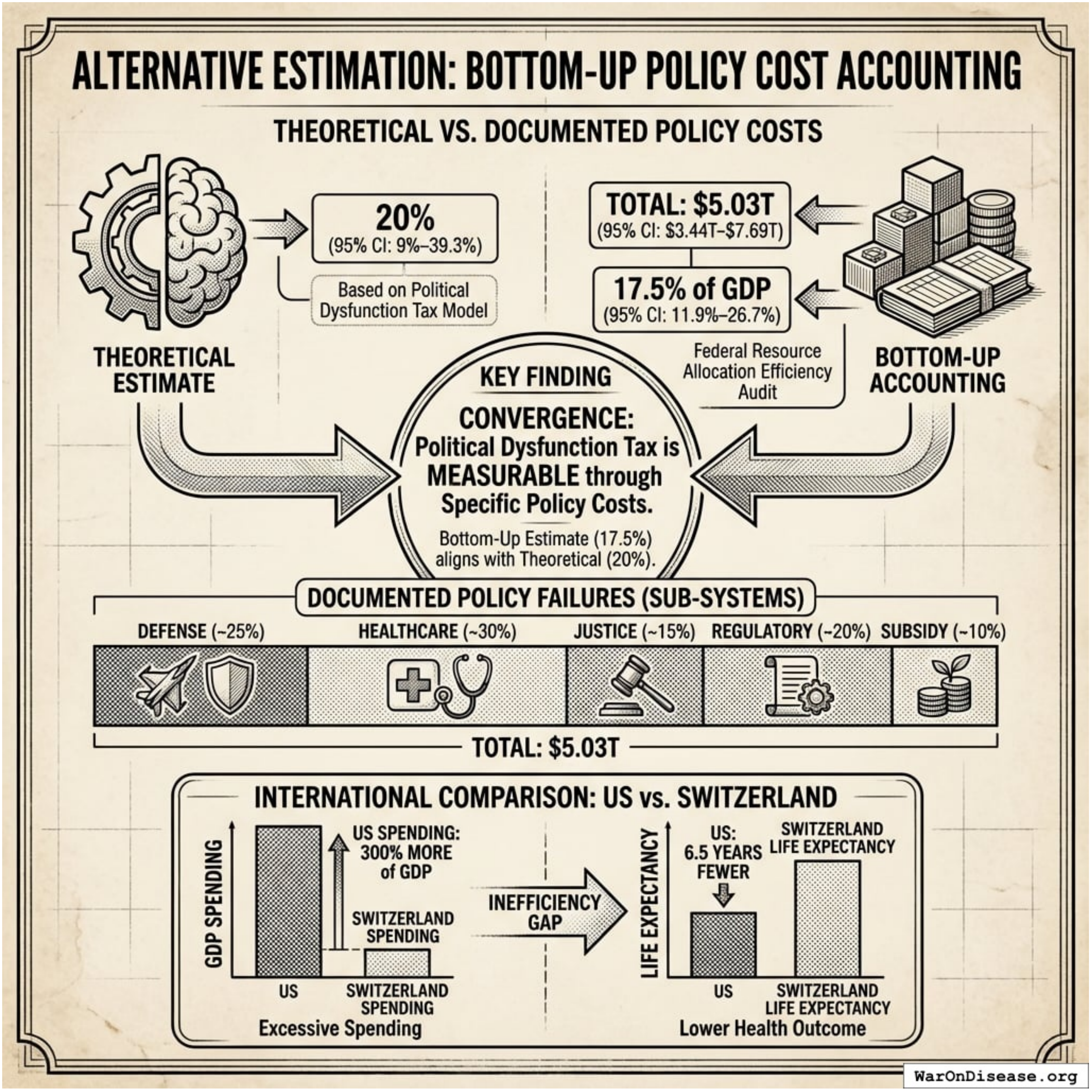

Figure 5: A comparison showing the convergence between theoretical and bottom-up estimates of the Political Dysfunction Tax, illustrating the overlap in their confidence intervals and the disparity between US and Switzerland spending vs. health outcomes.

Key finding: The bottom-up estimate converges with the theoretical estimate of 20% (95% CI: 9%-39.3%), suggesting the Political Dysfunction Tax is not merely theoretical but measurable through specific policy costs. International comparisons reinforce this: the US spends 300% percentage points MORE of GDP than Switzerland yet achieves 6.5 years FEWER years of life expectancy.

4.4 Why Information Doesn't Solve the Problem

If the welfare losses are documented, why don't governments act? As established in the Introduction, the problem is incentives, not information.[138] found that economic elites and organized interests, not average citizens, drive policy outcomes. This explains the persistence of obvious inefficiencies:

Intervention	Benefit-Cost Ratio	Political Economy
Pragmatic clinical trials	637:1 (95% CI: 569:1-790:1)	No concentrated beneficiary; competes with traditional trial industry
Childhood vaccination	101:1	No concentrated beneficiary to lobby
Pandemic preparedness	100:1+	Benefits are diffuse and probabilistic
Medical research	45:1	Competes with defense spending
Agricultural subsidies	<1:1	Concentrated beneficiaries, effective lobby

Information about optimal policy is freely available. The Copenhagen Consensus, GiveWell, and academic researchers publish rigorous benefit-cost analyses. Governments ignore this information because acting on it is not politically rewarded.

4.5 Implications for Mechanism Design

The empirical evidence suggests:

1. **The welfare loss is massive**: Conservative estimates suggest 20% (95% CI: 9%-39.3%) of potential welfare is foregone due to the Political Dysfunction Tax.
2. **Information alone is insufficient**: Better data does not change political incentives.
3. **The inefficiency is systematic**: It recurs across domains and countries, suggesting structural rather than contingent causes.
4. **Capture is the primary mechanism**: Policies systematically favor concentrated interests over diffuse citizen welfare.

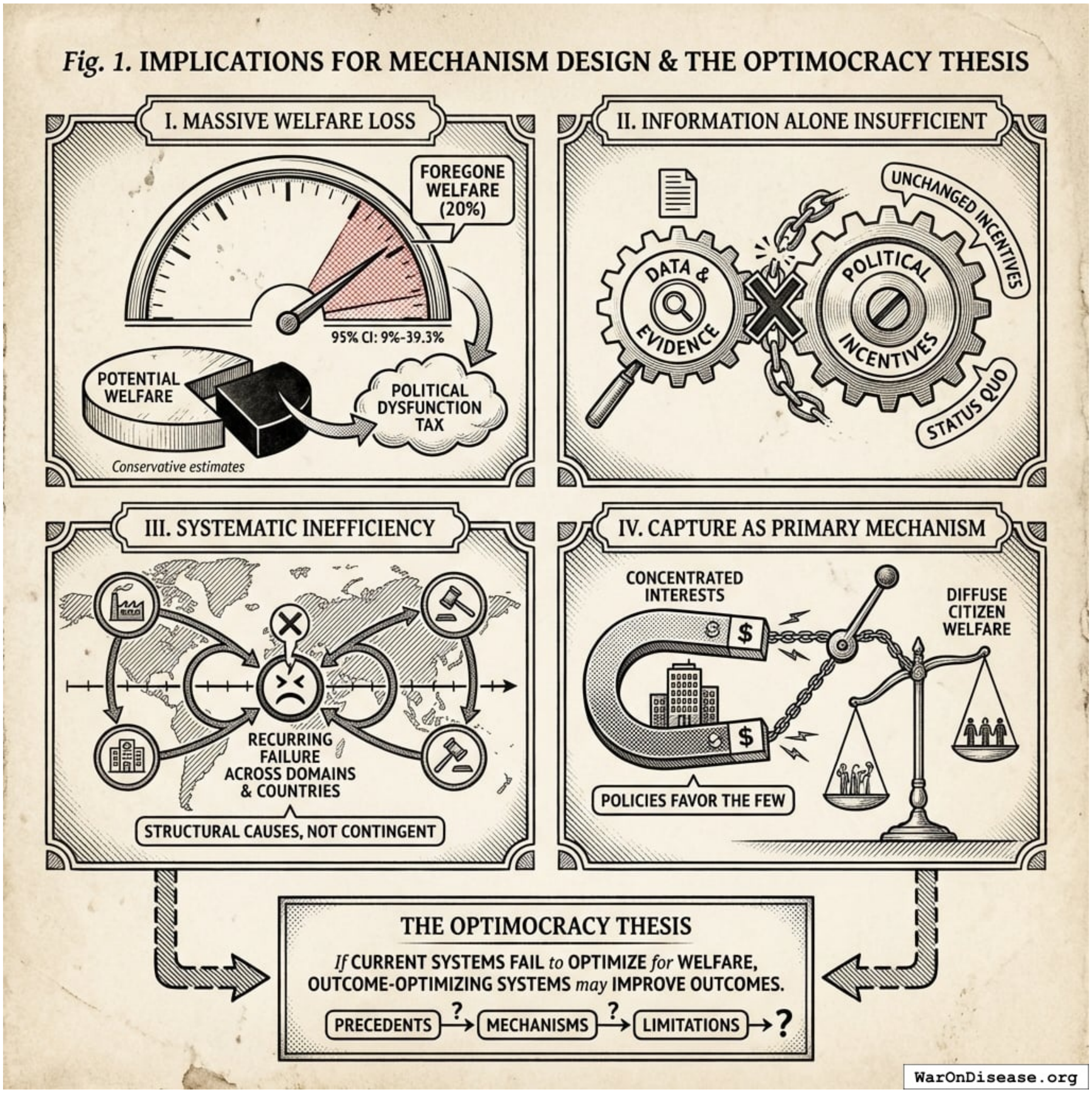

Figure 6: A conceptual diagram showing how political dysfunction and interest capture lead to systemic welfare loss, highlighting the shift toward outcome-optimizing mechanism design.

This motivates the Optimocracy thesis: if current systems systematically fail to optimize for welfare, outcome-optimizing systems may improve outcomes. The following sections examine precedents, mechanisms, and limitations.

4.6 When Does Optimocracy Beat the Status Quo?

Optimocracy's structural advantage comes from consolidating capture opportunities. The only attack surface is the verification layer itself: coordinating multiple independent institutions to misreport the same metrics, without detection.

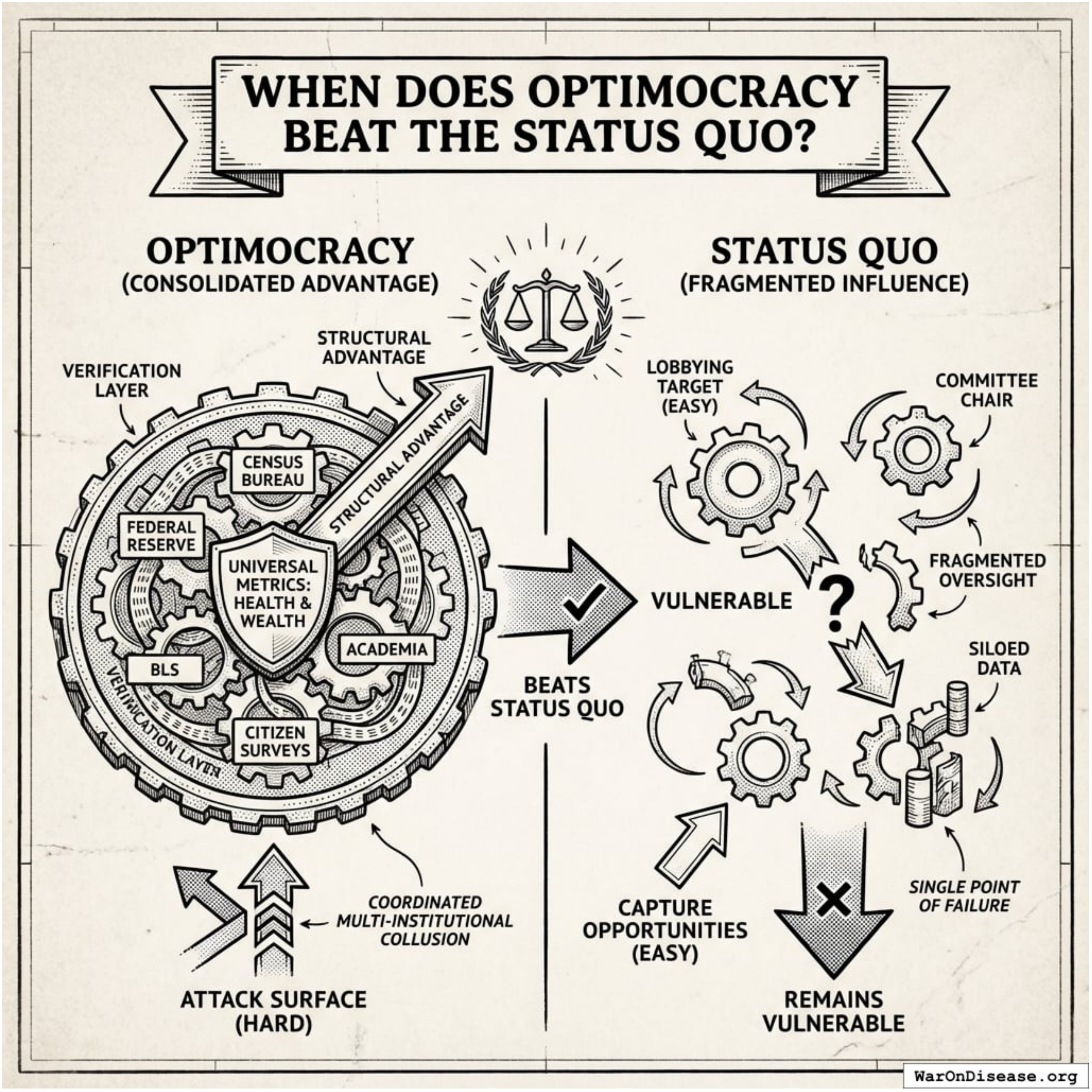

Figure 7: A conceptual comparison between the status quo's single-point-of-failure (lobbying a committee chair) and Optimocracy's multi-institutional verification layer requiring coordination across independent bodies.

Health and wealth are not arbitrary metrics. They are what humans universally value. The challenge for would-be corruptors: coordinate the Census Bureau, Federal Reserve, BLS, academic researchers, and citizen surveys to all report the same biased figure, without any whistleblowers. These institutions have different governance structures, funding sources, and methodologies. Collusion among strangers who lose credibility if caught is fundamentally harder than lobbying a single committee chair.

4.6.1 Formal Model

Let N denote the number of capture-prone allocation decisions under the status quo. Each decision i has capture probability p_i and capture cost c_i (welfare loss when captured). Under Optimocracy, capture opportunity collapses to oracle manipulation: K independent oracles, each with capture probability p_O, requiring majority collusion.

Proposition 1 (Optimocracy Dominance Condition):

Optimocracy beats capture-prone governance when:

$$\underbrace{\binom{K}{\lceil K/2 \rceil} p_O^{\lceil K/2 \rceil} \cdot c_O}_{\text{Expected capture cost under Optimocracy}} \quad < \quad \underbrace{\sum_{i=1}^{N} p_i \cdot c_i}_{\text{Expected capture cost under status quo}}$$

In plain English: the left side is the cost of corrupting Optimocracy (coordinating oracle collusion across multiple independent sources). The right side is the cost of corrupting the status quo (bribing all those committee decisions). Optimocracy wins when the left side is smaller.

When this works:

1. **N is large**: More decision points = more capture opportunities = higher status quo corruption cost
2. **Many independent oracles**: More oracles require exponentially harder collusion (5 sources means capturing 3; probability scales as p_O^3)
3. **Oracles are genuinely independent**: Different governance structures, funding sources, methodologies

What the model shows: The magnitude of Optimocracy's advantage depends on empirical parameters (capture probabilities, number of decision points, oracle independence). The model doesn't prove Optimocracy always wins. It identifies the *conditions* under which consolidating decision points reduces capture.

Limitations: This model assumes capture probabilities are independent (may underestimate coordinated attacks) and oracle collusion requires simple majority (stronger thresholds reduce risk further). Real-world calibration is essential before drawing quantitative conclusions.

5 Theoretical Foundations

Previous attempts at technocratic governance (Soviet central planning, credit rating agencies) failed due to knowledge problems, incentive misalignment, and scope creep. Optimocracy learns from these failures by using median aggregation across independent data sources, limiting scope to budget/policy recommendations (not comprehensive planning), and preserving market mechanisms for production decisions. For detailed historical analysis and engagement with Arrow's impossibility theorem, mechanism design theory, and democratic legitimacy concerns, see Appendix B.

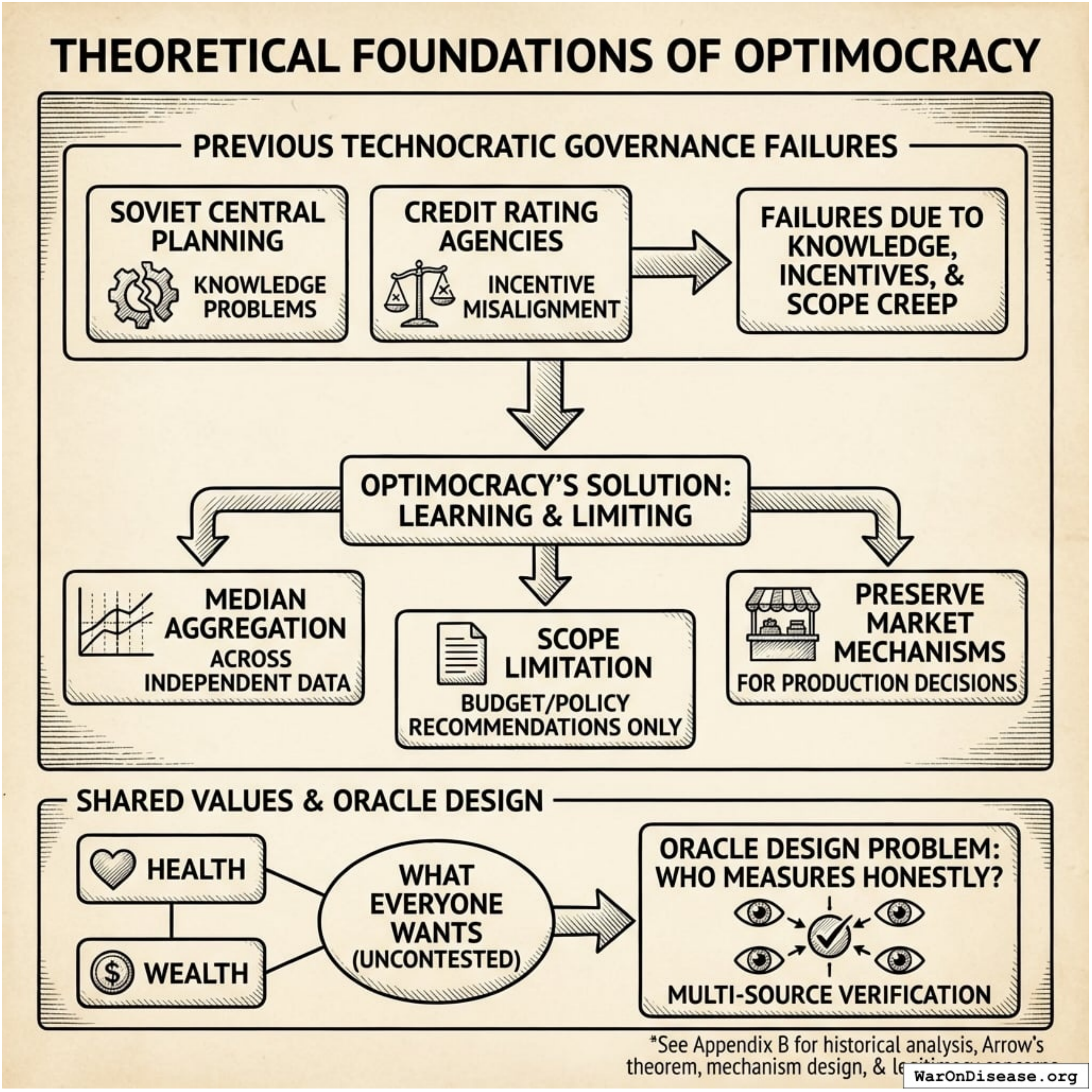

Figure 8: A comparison diagram illustrating how Optimocracy addresses historical technocratic failures through multi-source verification and median aggregation to generate policy recommendations.

Health and wealth aren't contested values requiring democratic selection. They're what everyone already wants. The hard problem isn't "what should we optimize?" but "who measures it honestly?" That's an oracle design problem, addressed through multi-source verification.

5.1 Precedents That Work

Rule-based allocation: Systematic approaches consistently outperform discretionary judgment across domains. Over 15-year periods, 90%+ of active fund managers underperform benchmark indices after fees[144]. Formula-based programs like Social Security COLA remove annual political battles over adjustments.

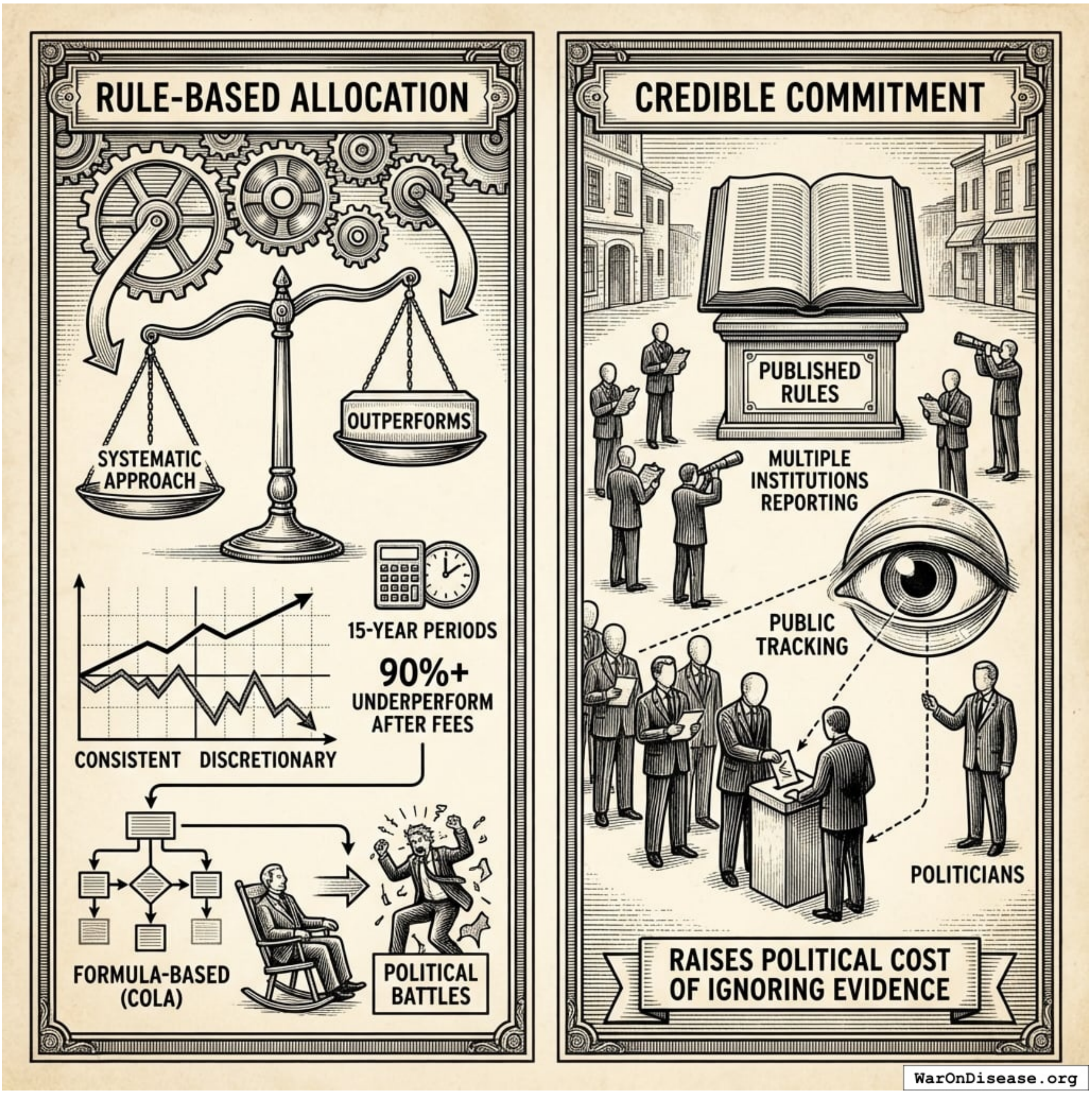

Figure 9: A comparative visualization of rule-based versus discretionary systems, illustrating the performance gap and the feedback loop where public reporting increases the political cost of ignoring evidence.

Credible commitment: Published rules make deviation visible. Multiple institutions report outcomes. Politicians vote freely, but alignment with recommendations is tracked publicly. This raises the political cost of ignoring evidence.

6 Mechanism Design

6.1 Architecture Overview

Optimocracy is purely advisory. It operates through three functions:

```
                    RECOMMEND
  - Algorithm calculates optimal policies/budgets
  - Optimizes for Health & Wealth (universal values)
  - Publishes recommendations for every major vote
```

```
                      TRACK
  - Politicians vote however they want
  - System records alignment with recommendations
  - Voting records are public
```

```
                      REWARD
  - IABs provide campaign support & career opportunities
  - 80% aligned = 2x the support of 40% aligned
  - Metric trends validate the system, attract funding
```

What about edge cases, catastrophic risks, novel situations? Politicians handle them. That's their job. The algorithm recommends; politicians decide. If a recommendation seems dangerous or wrong, they ignore it and face public accountability for that choice. The system tracks and rewards, nothing more.

6.2 The Verification Layer

Verification systems translate real-world outcomes into data that allocation algorithms can act upon. The critical design challenge is verification capture: if a single entity controls the data feed, they effectively control the allocation.

Concrete example (measuring median income growth):

Oracle	Source	Reported Value
Census Bureau	American Community Survey	+2.1%
Federal Reserve	Survey of Consumer Finances	+2.3%
Bureau of Labor Statistics	Current Population Survey	+1.9%
University of Michigan	Panel Study of Income Dynamics	+2.2%
Tax Foundation	IRS data analysis	+2.0%

Aggregation: Take the median of all five sources → **+2.1%**

If one source reported +5.0% (an outlier), it would be excluded by the median. The system doesn't require trust in any single institution, only that a majority aren't colluding. Since these institutions have different governance structures, funding sources, and methodologies, coordinated manipulation is difficult.

The correlated data problem and its solution:

An important limitation: the five sources above all ultimately rely on the same underlying data (Census surveys, tax records, employer reports). Their errors are correlated, meaning "five independent sources" may be an illusion. If the underlying methodology is captured, all five report the same biased figure.

This is a real concern. The solution is **genuinely independent data collection via decentralized citizen surveys.**

Decentralized Survey Architecture:

Component	Implementation	Why It Works
Identity verification	National digital ID, bank KYC, or similar	Prevents fake-identity attacks (one person = one response)
Data collection	Standard web/mobile interface	No technical expertise required
Storage	Secure database with audit logs	Can't be retroactively altered without detection
Aggregation	Open-source algorithm	Transparent, anyone can verify
Privacy	Anonymization after identity verification	Responses can't be linked to individuals

Why individual gaming is irrelevant:

With millions of survey responses, one person lying has approximately zero impact on the aggregate. Survey responses are aggregated statistically, so there is no strategic incentive to misreport.

6.3 The Execution Layer

Given the objective function and data feeds, the execution layer performs constrained optimization:

$$\max_{\mathbf{x}} M(\mathbf{x}) \quad \text{subject to} \quad \sum_i x_i = B, \quad x_i \geq 0$$

Where $M(\cdot)$ is the chosen metric, $\mathbf{x}$ is the allocation vector, and B is the total budget.

For simple metrics with known production functions, this optimization is straightforward. For complex metrics requiring causal inference, the execution layer may incorporate:

1. **Randomized controlled trials**: Allocate experimental budgets to estimate causal effects
2. **Prediction markets**: Aggregate distributed information about allocation effectiveness
3. **Machine learning**: Learn allocation-outcome mappings from historical data

The execution system publishes recommendations and tracks politician alignment for IAB rewards. Outcome measurement validates the system over time.

For a concrete worked example of how the optimization algorithm calculates allocations, see Appendix A: Optimization Algorithm.

7 Welfare Metrics

7.1 Default Implementation: Two-Metric Welfare Function

For practical implementation, the Optimocracy framework provides a simplified two-metric system that captures core welfare dimensions without requiring complex conversion factors or contested weighting decisions:

Metric 1: Real After-Tax Median Income Growth

- **Definition**: Year-over-year percentage change in inflation-adjusted, post-tax median household income
- **Units**: Percentage points per year (pp/year)
- **Sources**: Census Bureau, BLS, national statistical offices
- **Why median**: Captures typical citizen welfare without billionaire skew
- **Why after-tax**: Reflects actual purchasing power after government transfers
- **Why growth rate**: Enables cross-jurisdiction comparison regardless of baseline

Metric 2: Median Healthy Life Years

- **Definition**: Expected years of life in good health at the population median
- **Units**: Years
- **Sources**: WHO Global Health Observatory, national health surveys (BRFSS in US)
- **Relationship to QALYs**: Healthy life years life expectancy × average health utility

The Welfare Function

$$W_j = \alpha \cdot \text{IncomeGrowth}_j + (1 - \alpha) \cdot \text{HealthyYears}_j$$

Default weighting: $\alpha = 0.5$ (equal weight to economic and health welfare). Alternative weightings can be selected through democratic process.

Why These Two Metrics Work

Most policy effects eventually show up in one or both:

Policy Domain	How It Flows Through Real After-Tax Median Income
Employment	More jobs → higher wages → higher median income
Tax policy	Directly changes after-tax income
Transfer programs	Social Security, EITC → directly change after-tax income
Cost of living	Inflation adjustment captures housing, food, healthcare costs
Market power	Monopoly pricing → lower real income
Education quality	Better skills → better wages → higher income
Crime/instability	Lower productivity → lower wages
Trade policy	Consumer prices, job markets → flow through income

Policy Domain	How It Flows Through Real After-Tax Median Income

If GDP rises but median after-tax income doesn't, the policy correctly registers as low-welfare. Gaming this metric is hard because it requires actually improving typical household purchasing power.

Policy Domain	How It Flows Through Median Healthy Life Years
Healthcare access	More treatment → longer healthier life
Environmental regulation	Less pollution → fewer respiratory/cancer deaths
Occupational safety	Fewer workplace injuries/deaths
Mental health policy	Suicide prevention, addiction treatment → life years
Public safety	Lower homicide/accident rates → life years
Lifestyle policy	Tobacco/alcohol taxes → behavioral health
Elder care	Quality of late-life health
Food safety	Fewer foodborne illness deaths

Using *median* (not mean) avoids distortion from extremes like infant mortality or billionaire longevity clinics. It captures what a typical person can expect.

Why Two Metrics Beat Long Indicator Lists

Why not track 50 outcomes instead of 2? Three reasons:

1. **Gaming multiplies**: Each additional target creates a new way to cheat. Two broad, hard-to-fake outcomes are easier to audit than 50 narrow ones.
2. **Errors compound**: Composite indices (like HDI's 11 indicators) aggregate noise. Disagreements over weights become proxy wars for policy preferences.
3. **Veto points accumulate**: With 50 metrics, every policy "harms" something. Analysis paralysis replaces action.

Markets figured this out: firms optimize profit, not 200 sub-metrics. Profit combines customer satisfaction, efficiency, and innovation into one number. These two welfare metrics do the same for policy.

What This Covers

Most policies affect one or both metrics:

- **Economic policies** (taxes, regulations, trade) primarily affect income growth
- **Health policies** (healthcare access, public health, safety) primarily affect healthy life years
- **Many policies** (education, infrastructure) affect both

What about edge cases? Politicians handle them. Optimocracy is purely advisory. If a recommendation seems to violate rights, increase catastrophic risk, or harm minorities, politicians ignore it. That's their job. The algorithm recommends what maximizes health and wealth; politicians decide whether to follow.

This two-metric system is used by both the Optimal Policy Generator for policy recommendations and the Optimal Budget Generator for spending targets, ensuring consistency across the Optimocracy framework.

8 Advantages Over Political Governance

8.1 Capture Resistance

Under current governance, 12,000+ federal lobbyists spend $4+ billion annually with estimated 100:1 returns[145]. Every budget line item is a lobbying opportunity.

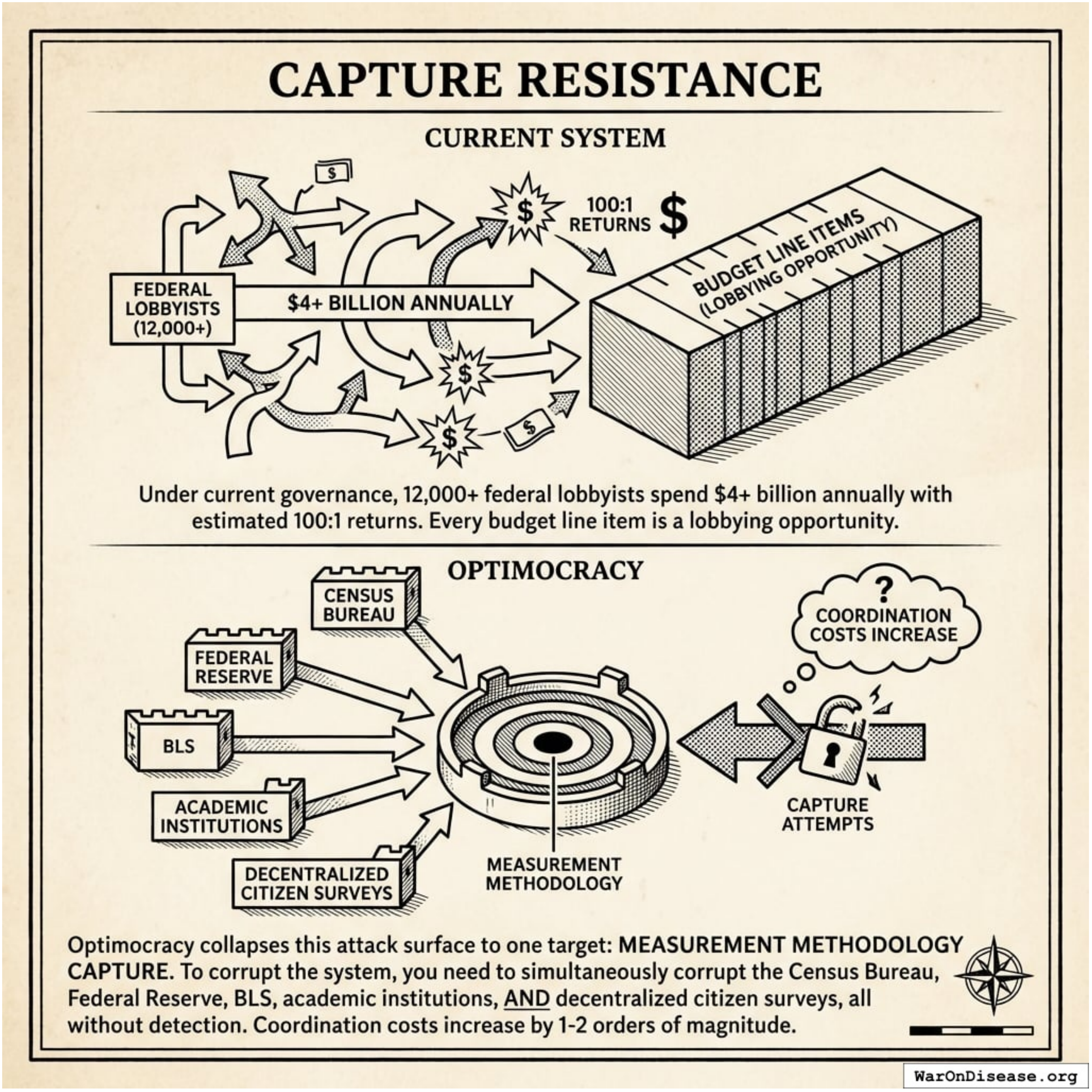

Figure 10: A comparison showing the shift from a fragmented lobbying landscape with thousands of targets to a consolidated Optimocracy model protected by multi-institutional verification layers.

Optimocracy collapses this attack surface to one target: **measurement methodology capture**. To corrupt the system, you need to simultaneously corrupt the Census Bureau, Federal Reserve, BLS, academic institutions, AND decentralized citizen surveys, all without detection. Coordination costs increase by 1-2 orders of magnitude.

8.2 Time Consistency

Politicians face 2-6 year horizons; welfare-improving investments often take 15-30 years to pay off. The algorithm weights long-term outcomes appropriately because it doesn't face reelection.

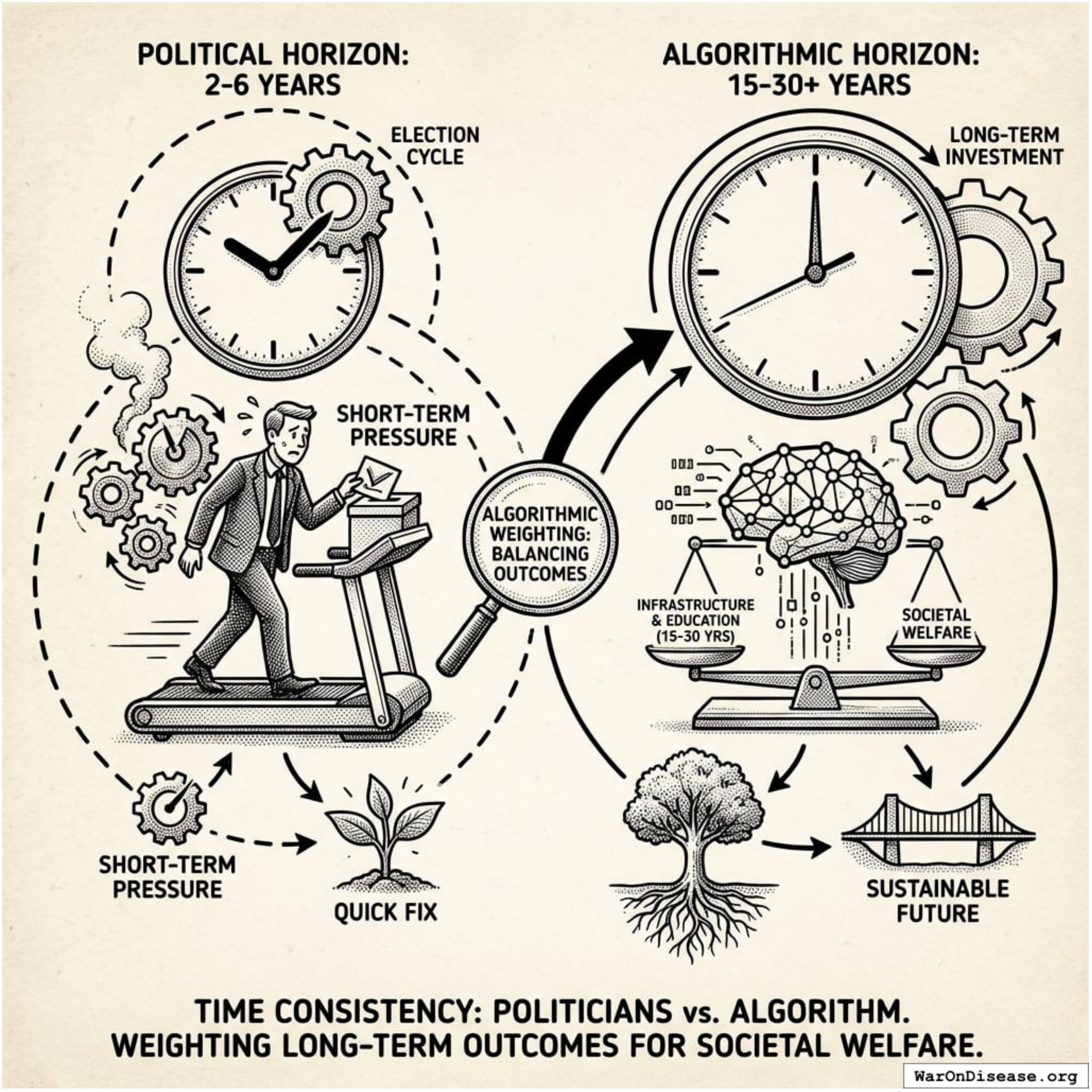

Figure 11: A comparison of decision-making timelines showing the 2-6 year political reelection cycle versus the 15-30 year payoff period for societal investments.

8.3 Transparency

Component	Political Governance	Optimocracy
Objective	Unstated	Explicit, published
Decision process	Closed-door negotiations	Open-source algorithm
Deviation detection	Requires investigative journalism	Automatic, publicly verifiable

Any citizen can verify alignment. Politicians who ignore recommendations must do so on the record.

8.4 Depolarization

Current politics rewards tribal opposition. Optimocracy shifts the debate from "whose policy?" to "what works?" The algorithm doesn't know which party proposed the budget. You can't spin the Census Bureau as a Democratic or Republican institution. The number either went up or it didn't.

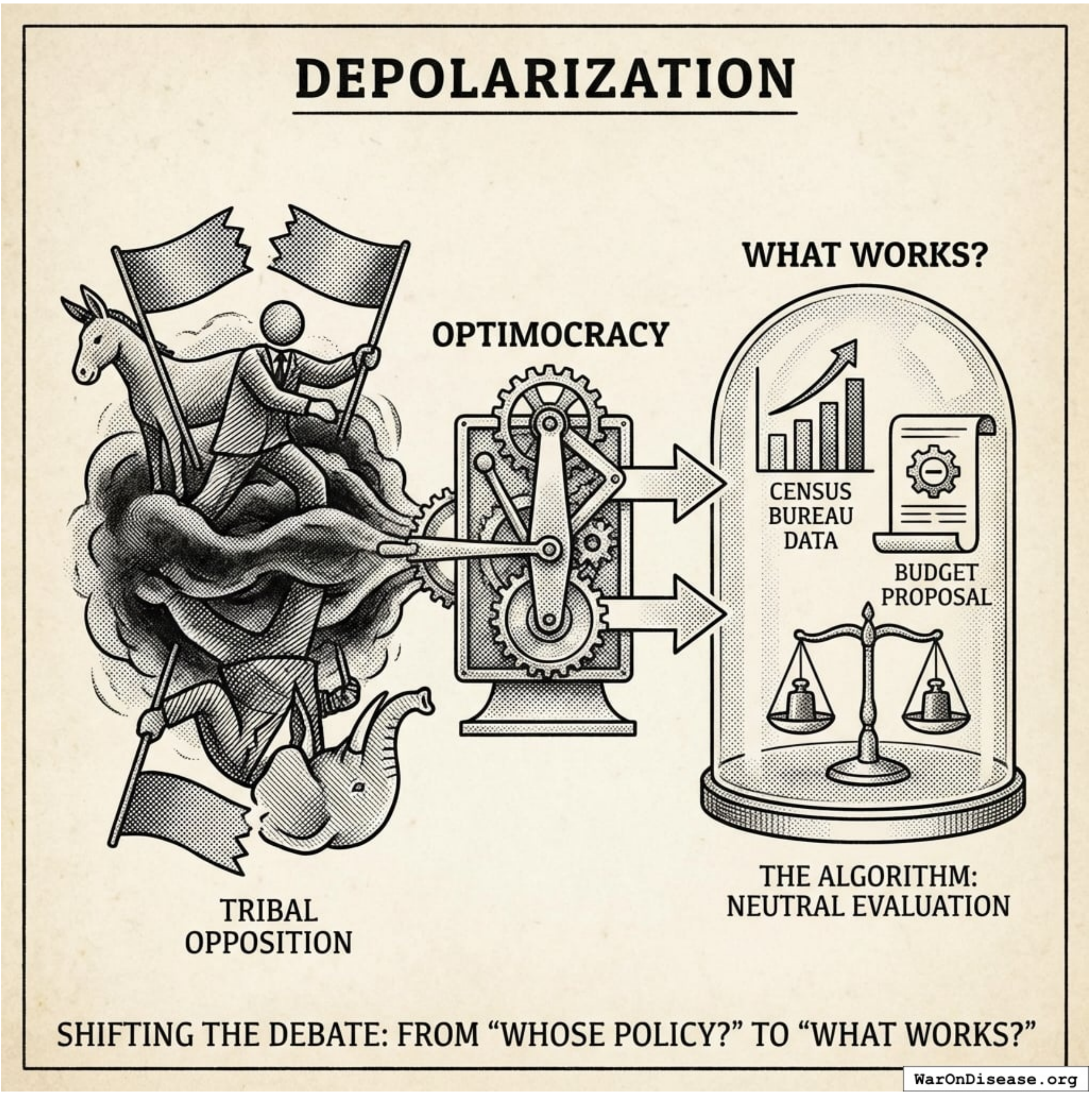

Figure 12: A comparison between tribal politics, which focuses on party identity, and Optimocracy, which focuses on objective data outcomes.

9 Challenges and Failure Modes

9.1 Why Goodhart's Law Doesn't Apply Here

"When a measure becomes a target, it ceases to be a good measure"[146]. Traditional Goodhart examples (teaching to test, hospital readmissions) involve gaming metrics *without improving underlying reality*.

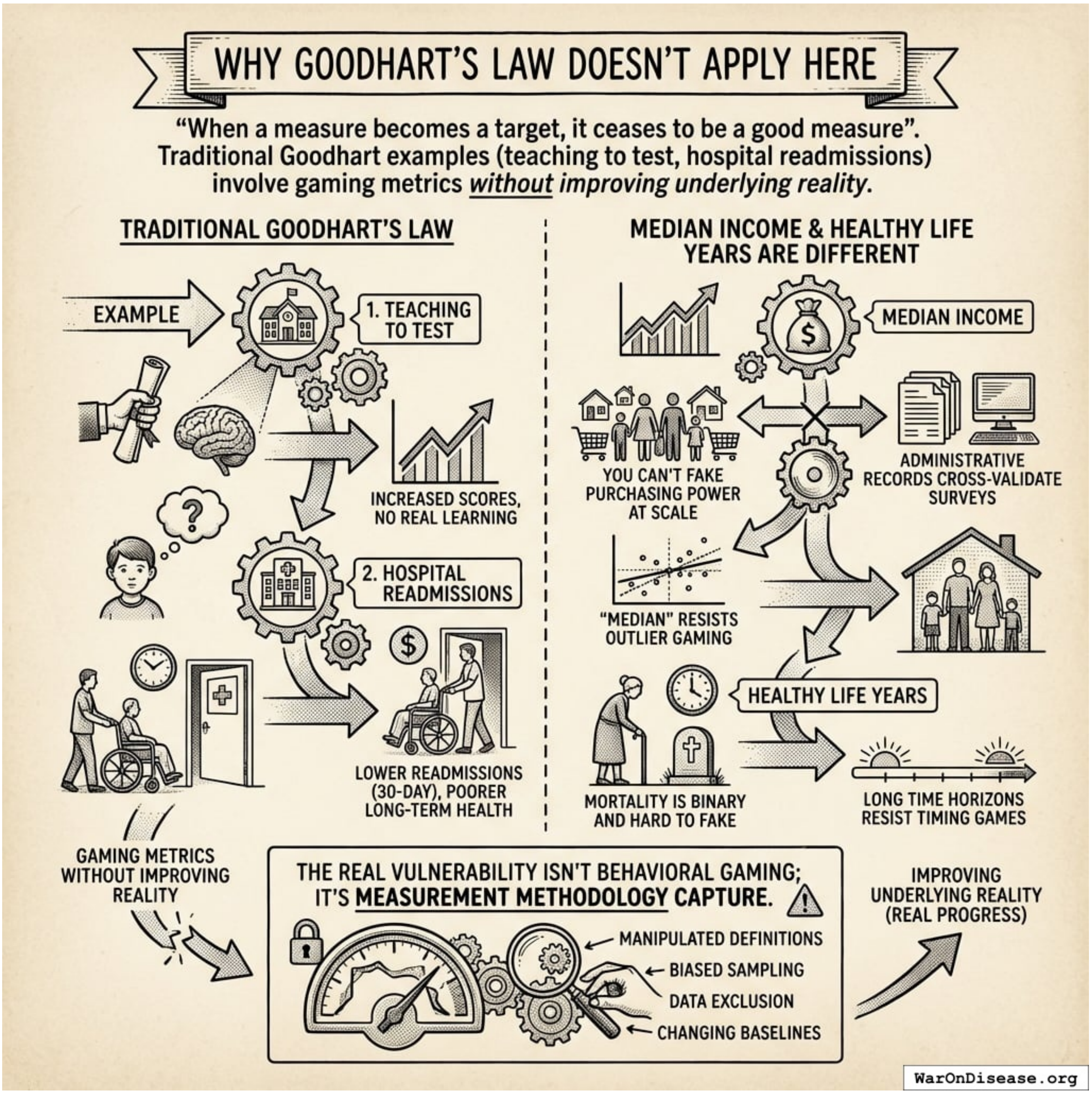

Figure 13: A comparison between traditional metrics vulnerable to behavioral gaming and resilient metrics like median income and healthy life years, highlighting why certain data structures resist Goodhart's Law.

Median income and healthy life years are different: You can't fake purchasing power at scale; administrative records cross-validate surveys. Mortality is binary and hard to fake. “Median” resists outlier gaming. Long time horizons resist timing games.

The real vulnerability isn’t behavioral gaming; it’s **measurement methodology capture**.

9.2 Oracle Capture (The Real Challenge)

If adversaries control the data feed, they control the allocation. This is the main attack surface.

Attack vectors: Direct manipulation, methodology capture (“redefine median household”), sample selection, timing manipulation, institutional collusion.

Government statistics are not neutral: Unemployment definitions (U3 vs U6) differ by 50%+. CPI methodology has changed 20+ times since 1978. With trillions at stake, the incentive to influence measurement is enormous.

Defense-in-depth:

Defense Layer	Why It Helps
Multiple independent sources (5+)	Capturing five agencies with different governance is much harder than one
Decentralized citizen surveys	Ground truth that exposes manipulation in official statistics
International benchmarking	Makes domestic manipulation visible via OECD/UN comparison
Academic replication	Adversarial verification with reputation incentives

The honest assessment: Oracle capture is not a solvable technical problem. The question is whether oracle capture is less damaging than current allocation capture. The answer may be yes: oracle capture affects measurement, while allocation capture affects outcomes directly. The wrapper architecture converts oracle capture from catastrophic failure to degraded performance, as politicians can ignore manipulated data.

9.3 Democratic Legitimacy

The concern: “The algorithm decided” lacks the felt legitimacy of “we the people decided.”

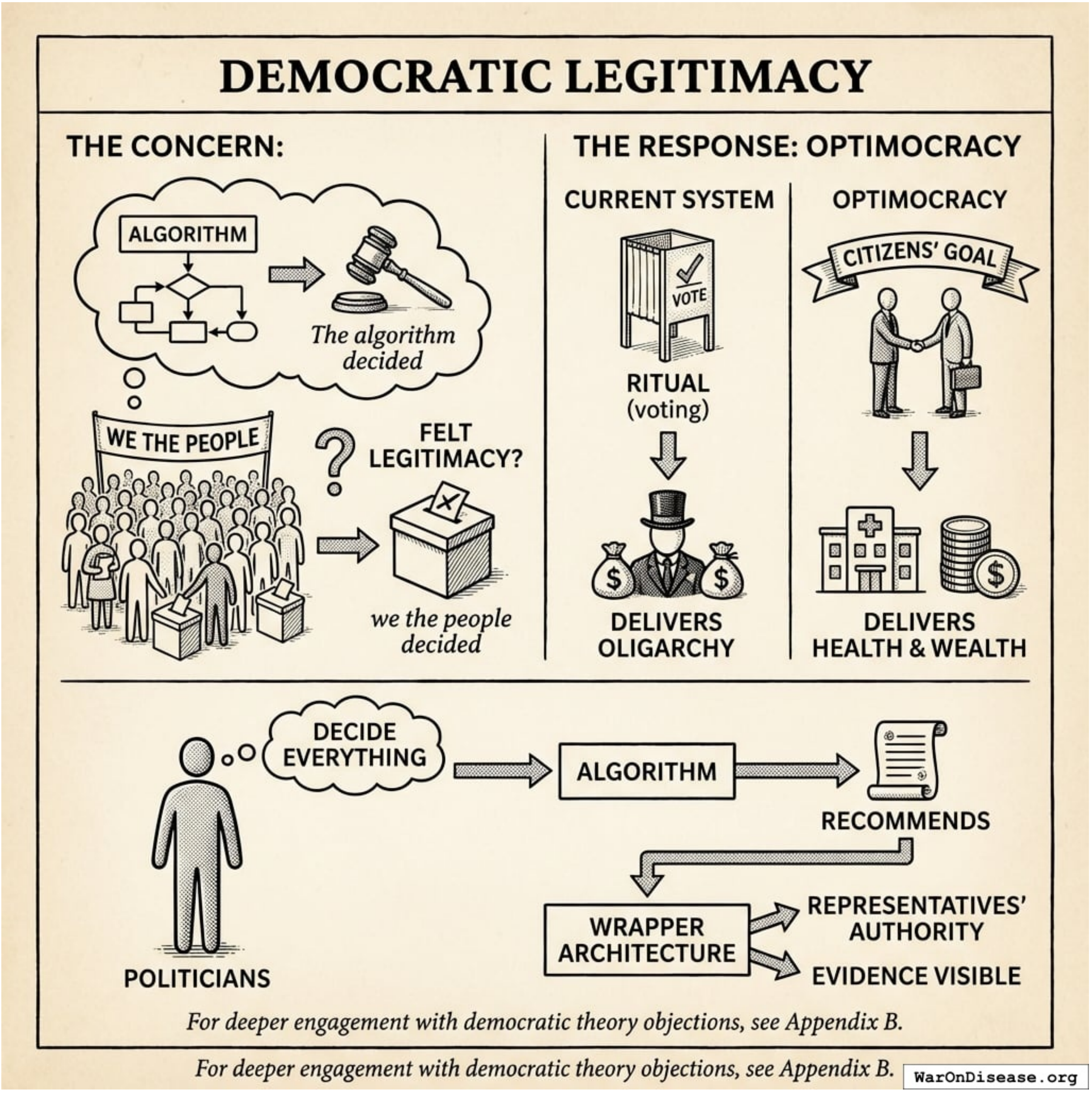

Figure 14: A conceptual diagram of the Optimocracy wrapper architecture, showing citizens at the top setting goals, politicians providing the decision-making 'wrapper' around algorithmic recommendations, and the resulting alignment with public health and wealth.

The response: Optimocracy is *more* democratic, not less. The current system offers ritual (voting) but delivers oligarchy ([138]). Optimocracy delivers what citizens actually voted for: health and wealth.

Politicians still decide everything. The algorithm only recommends. Citizens choose the goal; we already delegate implementation to the Fed and FDA. The wrapper architecture preserves representatives' authority completely while making their alignment with evidence visible.

For deeper engagement with democratic theory objections, see Appendix B.

10 Testable Predictions

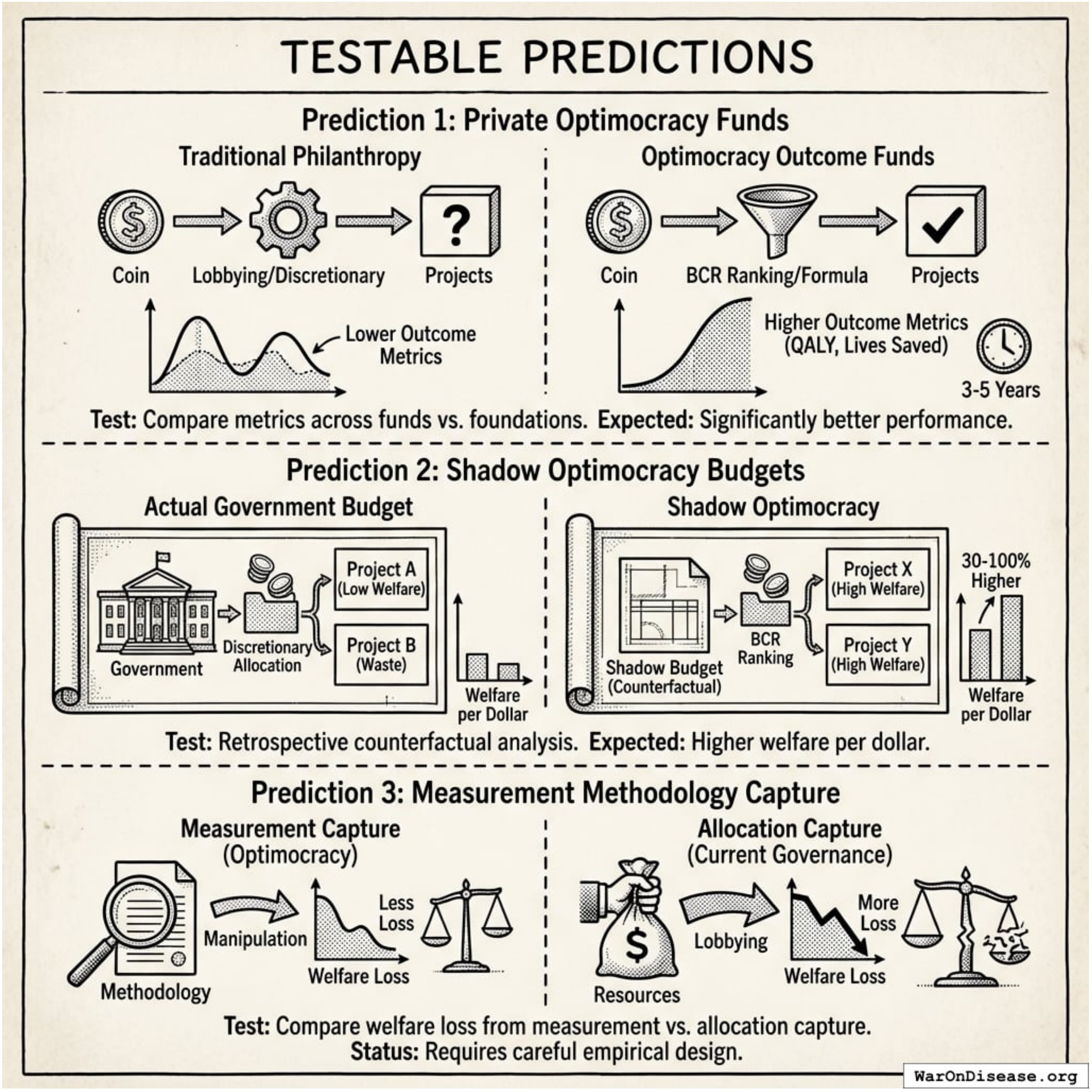

Figure 15: A comparative visualization showing the predicted performance gains of Optimocracy over traditional allocation methods across philanthropic metrics, government welfare per dollar, and resilience to institutional capture.

A publishable theory must generate falsifiable predictions. This section presents predictions that would confirm or refute the Optimocracy thesis. (Note: Existing evidence already supports the general case for algorithmic over discretionary allocation; formula programs like Social Security show lower lobbying intensity than discretionary programs[145]. The predictions below are specific to Optimocracy.)

Prediction 1: Private Optimocracy funds will outperform traditional philanthropic allocation on chosen metrics.

- *Test*: Compare QALY/, *livessaved*/, or similar metrics across Optimocracy outcome funds vs. traditional foundations
- *Expected finding*: Optimocracy allocation achieves significantly better metric performance (commensurate with eliminating the Crony Tax)
- *Timeline*: Testable within 3-5 years of deployment

Prediction 2: Shadow Optimocracy budgets will outperform actual government budgets in retrospective analysis.

- *Test*: Construct counterfactual "what if government allocated according to BCR rankings" and compare projected outcomes
- *Expected finding*: Shadow budget shows 30-100% higher welfare per dollar
- *Timeline*: Testable immediately with existing data

Prediction 3: Measurement methodology capture under Optimocracy will be less welfare-reducing than allocation capture under current governance.

- *Test*: Compare welfare loss from documented measurement manipulation vs. documented lobbying/capture
- *Expected finding*: Measurement capture is harder and less damaging than allocation capture
- *Status*: **Requires careful empirical design**

10.1 Rejection Criteria

The Optimocracy thesis would be falsified by:

1. **Consistent underperformance**: If outcome-optimizing systems consistently underperform political systems on their target metrics
2. **Measurement capture**: If oracle/methodology capture proves as easy and damaging as current allocation capture
3. **Value divergence**: If health and wealth prove NOT to be universal values (if significant populations genuinely prefer to be sicker and poorer)
4. **Legitimacy failure**: If citizens reject algorithmic governance even after demonstrated welfare improvements

We commit to updating or abandoning the proposal if evidence accumulates against these predictions.

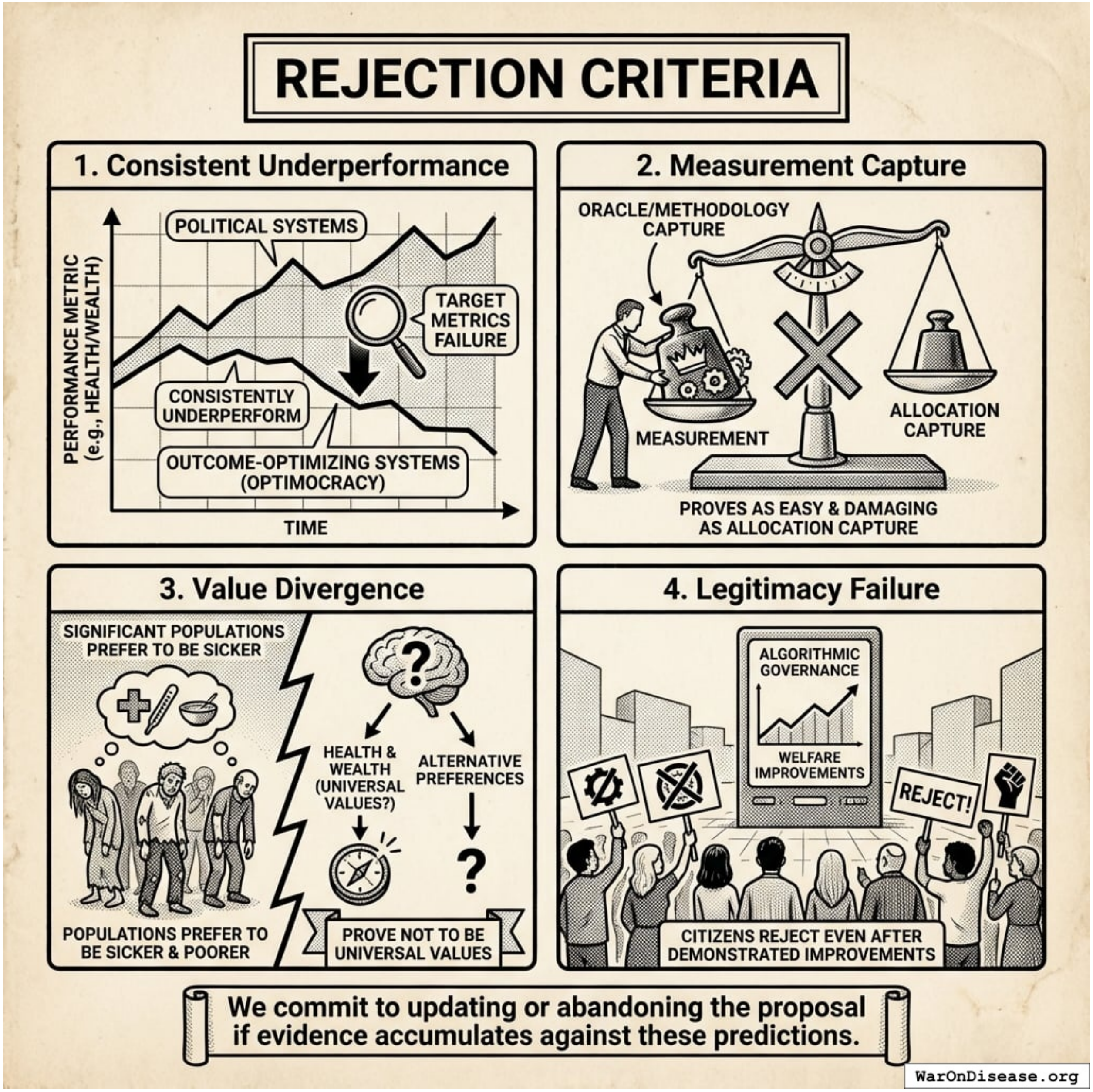

Figure 16: A conceptual map of the four falsification criteria for the Optimocracy thesis, illustrating potential failure modes across performance, systemic integrity, universal values, and social legitimacy.

11 The Wrapper Architecture: How Funding Works

Optimocracy provides the *what* (evidence-based recommendations); the SuperPAC provides the *how* (making alignment profitable). The integration works as follows:

Event	What Happens
Optimocracy publishes recommendations	"Support HR-1234 (early childhood funding increase)." "Oppose HR-5678 (regulatory capture provision)."

Event	What Happens
Votes recorded	Senator Smith votes aligned on both (2/2 = 100%). Senator Jones votes misaligned on both (0/2 = 0%).
Alignment scores calculated	End-of-quarter totals: Smith 78% aligned with recommendations. Jones 34% aligned.
SuperPAC allocates support	Algorithm weighs: alignment difference between candidates, position power, race competitiveness, marginal funding impact. Smith's competitive race with high alignment gets priority.
Public transparency	Citizens see: "Smith followed evidence-based recommendations 78% of the time."
System validation (ongoing)	Over years, metrics trend upward. This validates the recommendations, attracting more donations.

Funding sources:

1. **Donations (primary):** Donations are high-leverage. Historical lobbying ROI suggests $1 of campaign spending can shift $100+ in policy outcomes. This makes funding Optimocracy more effective than traditional charity.
2. **Incentive Alignment Bonds (for specific reallocations):** When Optimocracy recommends a specific funding reallocation (like the 1% Treaty), IABs can provide investor returns tied to policy success. Investors get a percentage of redirected funds, scaling funding beyond donations.

11.1 Campaign Funding Allocation Algorithm

The SuperPAC maximizes expected governance improvement per dollar. For each race, calculate:

$$E[\Delta G] = \text{alignment\_gap} \times \text{position\_power} \times \text{win\_probability\_shift} \times P(\text{marginal\_dollar\_matters})$$

Factors:

Factor	What it measures	Example
Alignment gap	|candidate_A_score - candidate_B_score|	78% vs 34% = 44-point gap
Position power	Committee chairs, leadership, swing votes	Health Committee chair = 3× multiplier
Race competitiveness	How much can funding shift the outcome?	48-52 polling = high; 30-70 = zero
Marginal funding effectiveness	Diminishing returns as spending increases	First $1M matters more than 10th $1M

Allocation priority examples:

Race	Alignment Gap	Competitiveness	Position Power	Priority
Senate (Health Cmte chair)	40 pts	Close (48-52)	High	**Top**
Senate (back-bencher)	40 pts	Close (48-52)	Medium	High
House (swing district)	30 pts	Close (49-51)	Low	Medium
Senate (safe seat)	50 pts	Landslide (70-30)	High	**Low** (can't shift outcome)
House (safe seat)	10 pts	Safe (60-40)	Low	Skip

A close race with a large alignment gap gets priority over a safe seat, even if the safe-seat candidate has higher absolute alignment. Funding flows where it can actually change outcomes.

Dynamic reallocation: As polling shifts during election season, the algorithm reallocates. A race that becomes uncompetitive frees funds for newly competitive races.

12 Political Economy: Making Reform Happen

The empirical case for Optimocracy is strong (Section 2 documented the Political Dysfunction Tax at 20% (95% CI: 9%-39.3%) of potential welfare). But optimal policy has been documented for decades. The question is not "what should we do?" but "how do we overcome political opposition to doing it?"

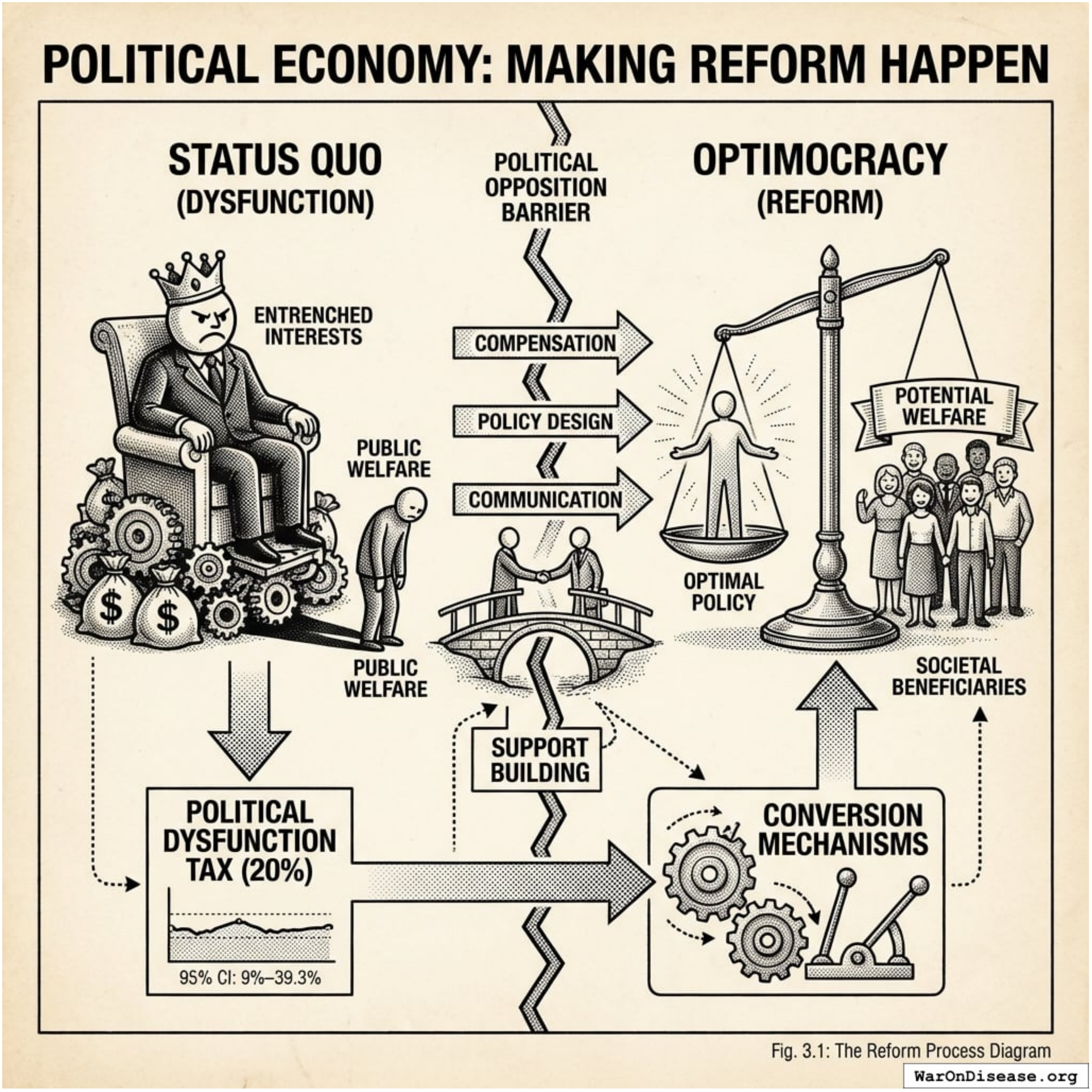

Figure 17: A conceptual framework showing the transformation of political opposition into support for reform by identifying and compensating stakeholders impacted by the shift to optimal policy.

This section develops the political economy of reform, distinguishing actors who benefit from optimal policy from those who lose, and proposing mechanisms to convert opponents into supporters.

12.1 Why Buying Off the Opposition Works

Optimal policy creates massive value ($20-50T annually). Those currently benefiting from bad policy would lose much less ($2-5T annually).

The math: Pay off all the losers and you still have $15-45T left over. This is why reform is feasible: the pie is big enough that everyone can get a bigger slice.

(This is the Coase Theorem[147] applied to governance: when gains vastly exceed losses, compensation makes everyone better off.)

12.2 How the SuperPAC Makes This Concrete

The Optimocracy SuperPAC creates a market for political support. It allocates campaign support proportional to politician alignment with evidence-based recommendations. Alignment is tracked via Optimocracy's recommendation/voting comparison. Politicians can follow donor preferences or follow evidence and receive SuperPAC support. The scoring rules are transparent and published.

A senator currently receives ~$174K salary plus ~$5-20M in career post-office value from lobbying relationships. SuperPAC campaign support for aligned politicians shifts this calculus: supporting optimal policy becomes the career-maximizing choice.

For specific policy reallocations with concrete funding sources (like the 1% Treaty redirecting military spending to medical research), Incentive Alignment Bonds can scale funding beyond donations by providing investor returns tied to policy success.

12.3 The Cost of Political Reform

Political feasibility is a cost, not a binary. In a companion analysis (see The Price of Political Change), we estimate the maximum plausible cost of achieving political reform through democratic engagement.

US Political System Reform Investment (Maximum Scenarios):

Component	Cost Estimate
Match all lobbying expenditure (1.5×)	$6.6B/year
Match all federal campaign spending	$10B/cycle
Match full Congress career incentives (535 × ~$10M NPV)[1]	$5.35B one-time
Total US maximum reform investment	**~$25B**

For reforms like the 1% Treaty ($2.5T NPV conservative estimate), this implies ~100:1 ROI. Political reform is dramatically underinvested relative to its expected value.

12.4 The Natural Reform Coalition

The political economy of reform requires identifying actors who would benefit from optimal policy and mobilizing them to overcome opposition.

Major Health Funders as Anchor Investors

The most natural funders for political reform are organizations already committed to maximizing health outcomes:

[1] Post-Congress lobbying compensation averages 10-20× congressional salary[148]; $174K × 15 × 10 years $10M NPV.

Funder	Annual Health Spending	Current Focus	Reform Alignment
Gates Foundation	~$7B	Global health, disease eradication	High: political reform unlocks orders of magnitude more impact
Wellcome Trust	~$1.5B	Biomedical research	High: regulatory reform accelerates drug development
Open Philanthropy	~$500M	GiveWell-style interventions, policy reform	Very high: already funds policy advocacy
Bloomberg Philanthropies	~$1.5B	Public health, tobacco/obesity	High: policy is their primary lever
Arnold Ventures	~$500M	Evidence-based policy	Very high: explicitly focused on policy effectiveness

These funders collectively spend $10B+ annually on health. If even 5% were redirected to political reform infrastructure (the Optimocracy SuperPAC), it would exceed all current political reform spending.

The ROI Case for Funders

The Gates Foundation achieves approximately $50-100 per DALY through direct interventions. A $25B investment in political reform, even at 5% success probability, yields expected cost-effectiveness of ~$0.025 per DALY, roughly 3,000x more cost-effective. This calculation is developed in detail in The Price of Political Change: Implications for Major Health Funders.

Why Funders Haven't Invested (Yet)

Three barriers explain underinvestment in political reform:

1. **Legibility**: Direct interventions have clear attribution ("we funded 10M bed nets"). Political reform success is diffuse and contested.
2. **Reputational risk**: Political engagement risks partisan association. Optimocracy's outcome focus (health and wealth, not ideology) mitigates this.
3. **Coordination failure**: No single funder can reform the political system. The SuperPAC provides the coordination mechanism: funders contribute to a pool that rewards outcome-aligned politicians.

Coalition Structure

The reform coalition assembles in phases:

Phase	Actors	Role	Capital
Seed	Reform advocates, EA orgs	Initial design, proof of concept	$10-50M

Phase	Actors	Role	Capital
Pilot	Tech billionaires, forward-looking foundations	First SuperPAC deployment	\$100M-500M
Scale	Major health funders	Anchor capital for SuperPAC + IABs for specific reallocations	\$1-5B
Victory	Converted politicians, broader coalition	Political implementation	\$10-25B

Reform advocates provide initial capital and legitimacy; major funders provide anchor capital for the SuperPAC; converted politicians provide political support. The arithmetic strongly favors reform: welfare gains (\$20-50T annually) vastly exceed maximum compensation costs (\$25-200B one-time).

13 Implementation Pathway

13.1 Why Government Adoption Isn't Required

Optimocracy operates as a permanent advisory layer. Politicians don't surrender power; they just face new incentives. The question changes from "Will politicians give up power?" (hard) to "Can we fund the SuperPAC enough to make ignoring recommendations unprofitable?" (tractable). This reframing converts the implementation problem from a political revolution to a fundraising challenge.

13.2 Implementation Strategy

Optimocracy does not require government permission for private fund allocation. The technology exists today. The strategy: **start private → demonstrate → scale**.

- **Private Outcome Fund**: Deploy \$10M+ for philanthropic capital, prove mechanism works
- **Shadow Tracking**: Generate counterfactual "what would Optimocracy allocate?" to demonstrate outperformance
- **Government Pilots**: Partner with reform-minded jurisdictions for limited-scope pilots
- **Broader Adoption**: Scale through demonstrated success and competitive pressure

The bottleneck is not technology. The bottleneck is coordination: assembling capital, coalition members, and governance structures. Starting small is a feature: a \$10M outcome fund with 50 committed participants can experiment, fail fast, and improve before scaling.

13.3 Concrete First Deployment: The GiveWell Outcome Fund

We propose a specific first deployment to make Optimocracy actionable rather than theoretical.

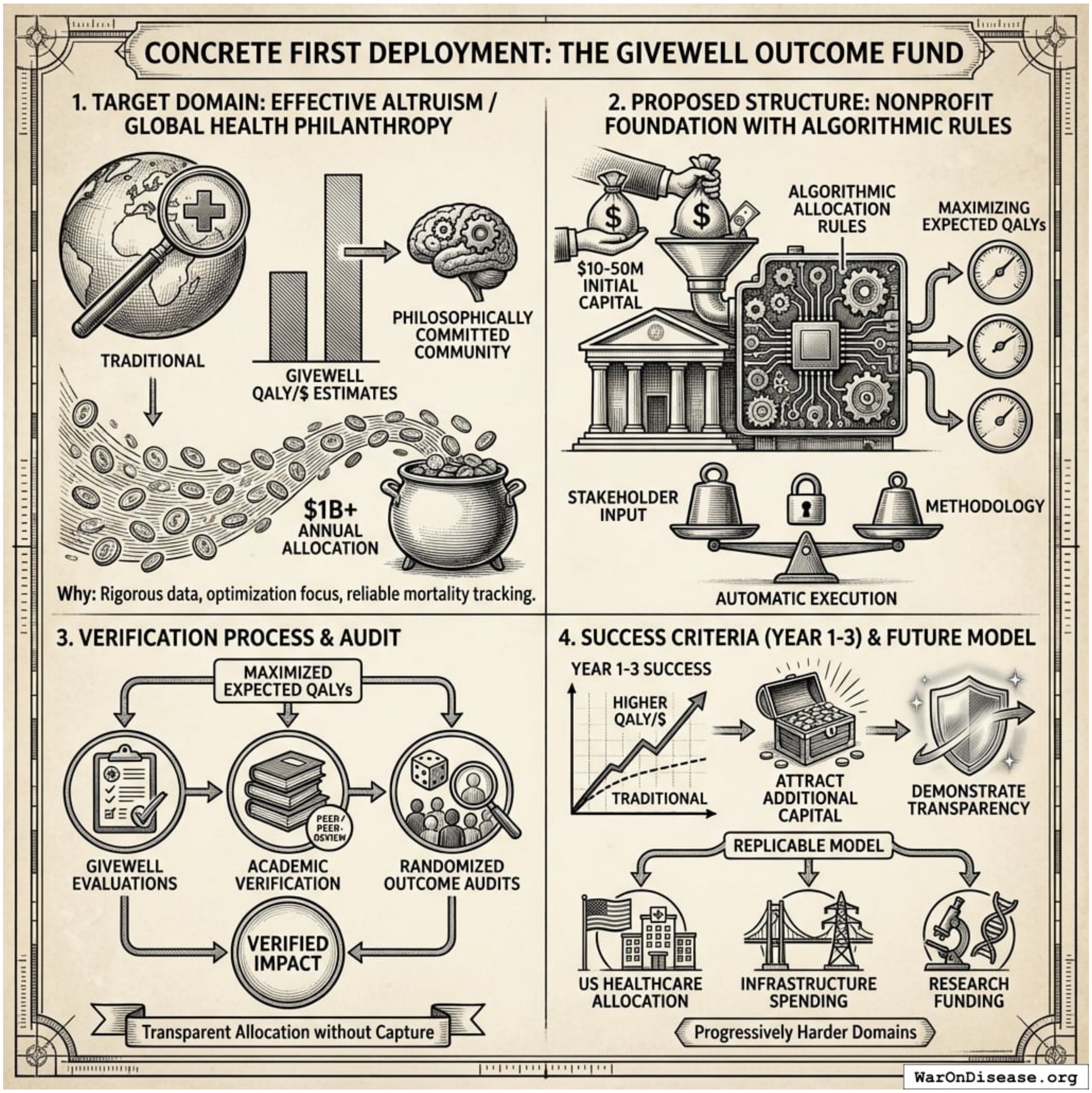

Figure 18: A structural diagram illustrating the GiveWell Outcome Fund's workflow, from initial capital and stakeholder methodology through algorithmic allocation to outcome verification and scaling.

Target domain: Effective altruism / global health philanthropy. Why: GiveWell already produces rigorous QALY/$ estimates, the EA community is philosophically committed to outcome optimization, mortality data is increasingly reliable, and the community allocates $1B+ annually.

Proposed structure: Nonprofit foundation with algorithmic allocation rules, $10-50M initial capital from aligned foundations, maximizing expected QALYs verified by GiveWell evaluations + academic verification + randomized outcome audits. Stakeholder input on methodology; algorithm execution is automatic.

Success criteria (Year 1-3): Achieve measurably higher QALY/$ than comparable traditional foundations, demonstrate transparent allocation without capture, attract additional capital, and generate replicable model for progressively harder domains (US healthcare allocation, infrastructure spending, research funding).

14 Conclusion

Political governance is not failing because politicians are evil or voters are ignorant. It is failing because the incentive structure makes capture inevitable. No amount of transparency, campaign finance reform, or civic education can eliminate the fundamental misalignment between political incentives and citizen welfare.

Optimocracy offers a different approach: rather than relying on systems that don't optimize for outcomes, introduce outcome-optimizing allocation for decisions where optimization is feasible. Define the objective democratically, measure outcomes rigorously, allocate algorithmically, and enforce via transparent rules.

The precedents are encouraging. Formula-based programs like Social Security COLA remove annual political battles. Published, transparent allocation rules enable credible commitment by making deviation visible.

Optimocracy is not utopian. It faces real challenges: measurement methodology capture, edge cases, and democratic legitimacy. But these challenges are addressable through institutional diversity, the wrapper architecture, and iterative implementation.

The question is not whether Optimocracy is perfect; no governance system is. The question is whether outcome-optimizing algorithmic allocation produces better outcomes than captured allocation. The evidence suggests it does.

14.1 What About AI?

Some may ask: won't superintelligent AI make Optimocracy obsolete? The answer is no. Optimocracy *is* the safe architecture for AI governance. "Aligned AI" requires specifying aligned *to what*. Health and wealth are the answer: universal human values, not arbitrary choices. Even a superintelligent AI should operate within an Optimocracy-like structure: optimize for what humans universally value, independent systems verify outcomes, and humans retain final approval authority. The algorithm (whether simple optimization or AGI) proposes; humans decide. Optimocracy isn't replaced by better AI; it's what makes AI governance safe. The framework scales from spreadsheet calculations to superintelligence while preserving human oversight.

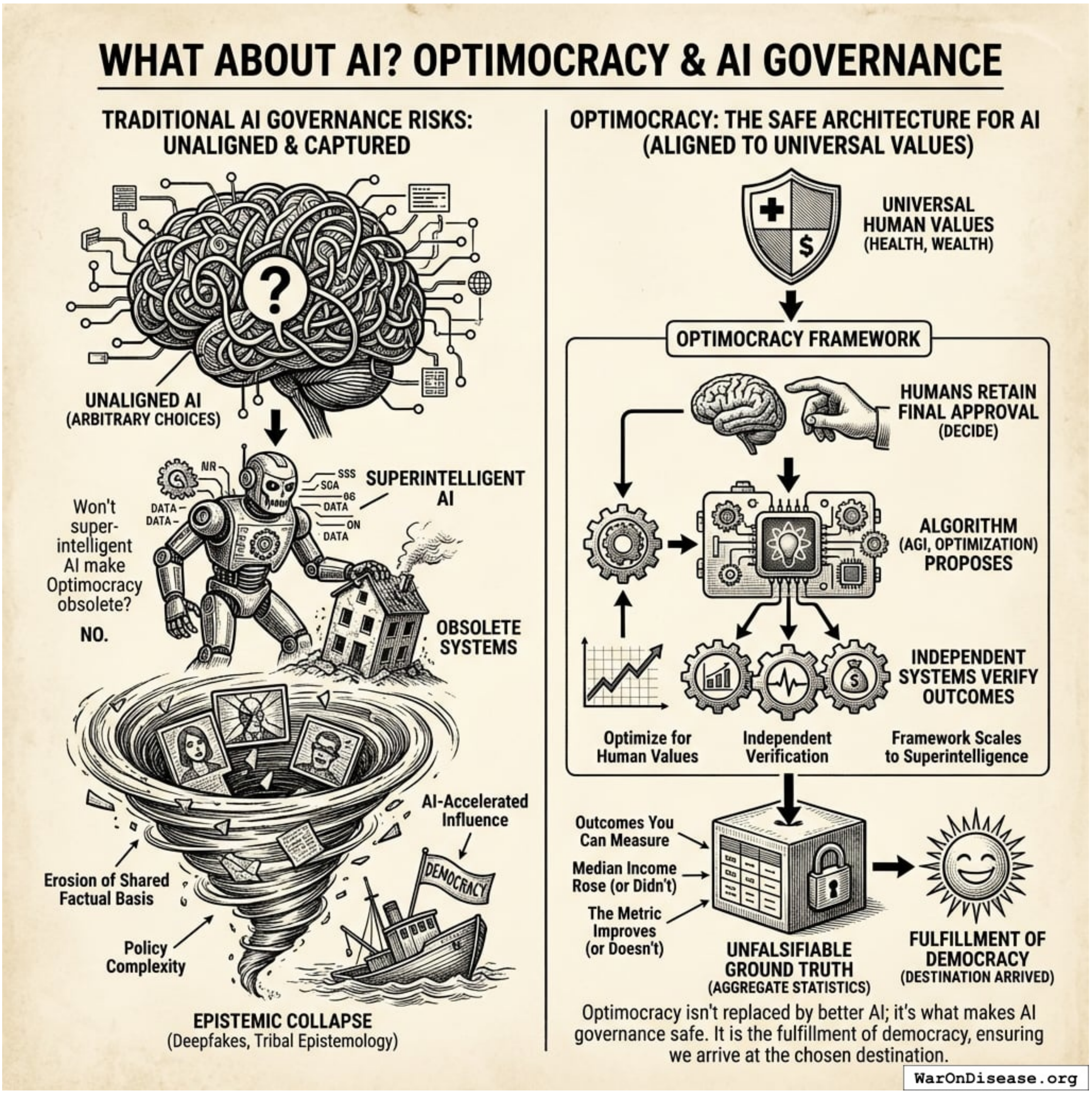

Figure 19: A diagram illustrating the Optimocracy AI governance framework where AI systems propose optimizations based on universal values, which are then verified by independent systems before final human approval.

We propose Optimocracy not as a replacement for democracy but as its fulfillment: a system where citizens genuinely choose outcomes, not just representatives who promise outcomes and deliver something else. Democracy selects the destination; Optimocracy ensures we actually arrive.

The urgency grows with each passing year. Deepfakes are eroding the shared factual basis democracy requires. Policy complexity increasingly exceeds human cognitive capacity. AI-accelerated influence operations will make capture orders of magnitude cheaper and more effective. The tribal epistemology that already interprets identical videos along partisan lines will have no ground truth whatsoever once synthetic media becomes indistinguishable from authentic footage. Optimocracy offers an exit from this epistemic collapse: outcomes you can measure, not narratives you must trust. Aggregate statistics from multiple independent sources provide unfalsifiable ground truth. Median income either rose or it didn't. The metric either improves or it doesn't. That fact provides a foundation for governance when all other foundations have eroded.

15 Appendix A: Technical Specification Sketch

15.1 Budget and Policy Optimization: Two Complementary Frameworks

Optimocracy optimizes governance through two complementary mechanisms, each addressing a different aspect of the welfare-maximization problem.

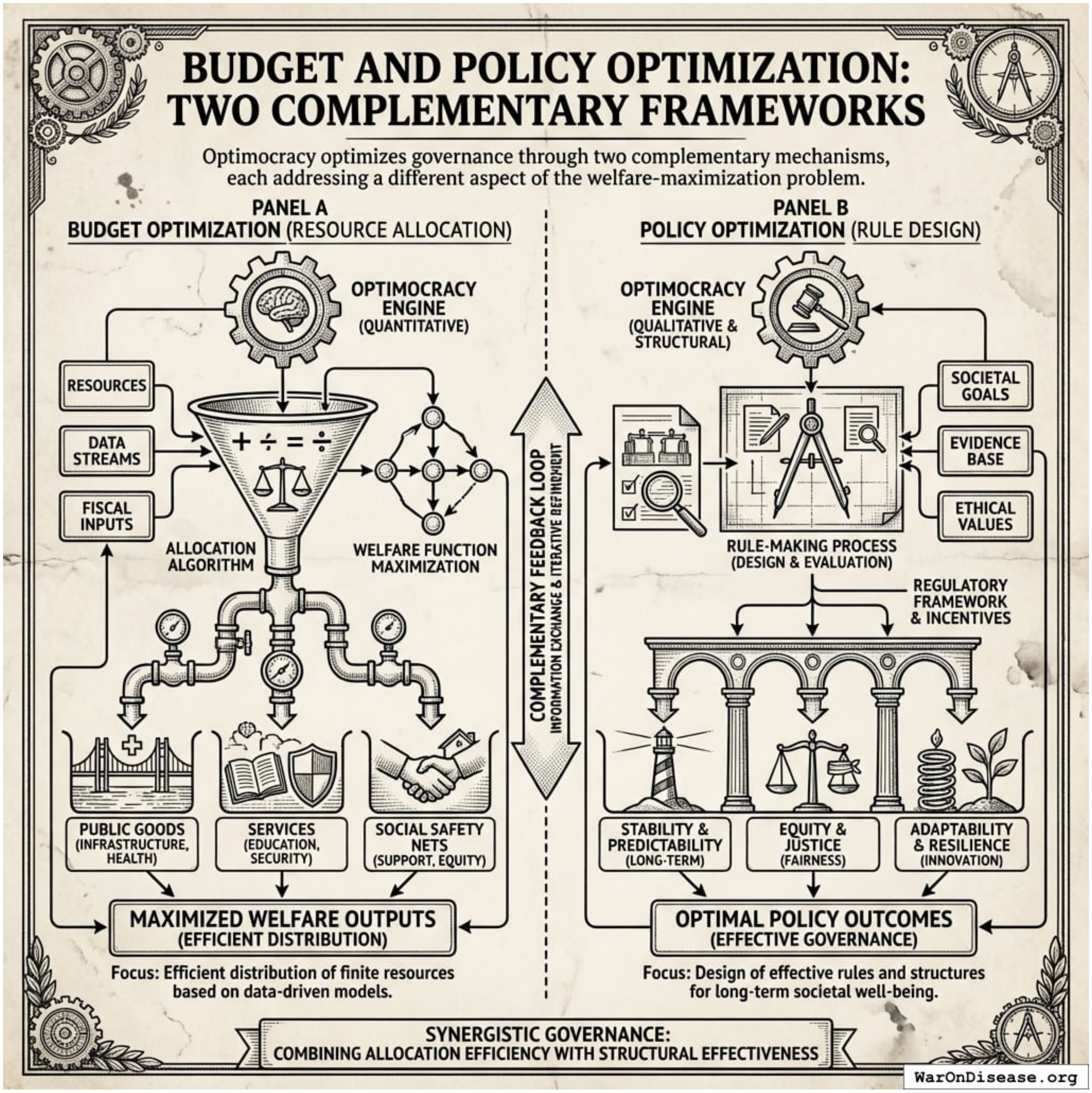

Figure 20: A conceptual diagram showing how Budget and Policy Optimization serve as the two pillars of the Optimocracy framework to achieve welfare maximization.

15.1.1 Budget Optimization: The Optimal Budget Generator (OBG) Framework

The **Optimal Budget Generator (OBG)** framework answers: "How should we allocate the budget to maximize welfare?"

Each spending category has an optimal level - not just a marginal return. Too little means underinvestment and foregone welfare gains; too much means diminishing returns. But unlike the Recommended Daily Allowance for nutrients (where you can meet all targets simultaneously), budget allocation is zero-sum: spending more on one category means less for others. OBG generates integrated recommendations that balance these tradeoffs.

Spending Level	Health Analogy	Budget Interpretation
Below optimal	Vitamin deficiency	Foregone welfare gains
At optimal	Recommended daily allowance	Maximum return per dollar
Above optimal	Diminishing returns / toxicity	Waste, opportunity cost

The OBG framework combines three evidence sources:

1. **Reference country benchmarking**: What do high-performing peer countries spend?
2. **Diminishing returns modeling**: Where is the "knee" of the spending-outcome curve?
3. **Cost-effectiveness threshold analysis**: Which interventions pass standard health economics thresholds?

The **Budget Impact Score (BIS)** measures our *confidence* in each category's target estimate based on the quality of causal evidence. Categories with strong RCT evidence have high BIS; categories with only cross-sectional correlations have low BIS.

Example output:

Category	Current	Target	Gap	Evidence
Pragmatic clinical trials	$0.5B	$50B	**+$49.5B**	A (RCTs)
Vaccinations	$8B	$35B	+$27B	A (RCTs)
Basic research	$45B	$90B	+$45B	B (spillovers)
Military	$850B	$459B	-$391B	C (benchmarks)

For the complete methodology including estimation procedures, validation framework, and worked examples, see Optimal Budget Generator Specification.

15.1.2 Policy Optimization: The Policy Impact Score (PIS) Framework

The **Policy Impact Score (PIS)** framework answers: "Which policy reforms would most improve welfare outcomes?"

This extends beyond budget allocation to evaluate *all policies*: laws, regulations, taxes, and administrative rules. Budget reallocation alone cannot fix structural inefficiencies:

Problem Type	Example	Budget Optimization Handles?
Program allocation	Too little preventive care	Yes (reallocate budget)
Administrative overhead	~$1T/year US admin costs	No (requires regulatory reform)
Drug pricing	US pays 256% of OECD average[149]	No (requires policy change)

PIS uses quasi-experimental methods (synthetic control, difference-in-differences, regression discontinuity) to estimate causal effects of policy changes across centuries of variation in hundreds of jurisdictions.

Example output:

Policy Reform	Effect on Outcome	Evidence Grade
Tobacco tax (+$1/pack)	-8.2 pp smoking rate	A (synthetic control)
Seat belt laws (primary)	-1.8 traffic deaths/100K	A (DiD, 47 states)
Occupational licensing	+2-3% consumer prices	B (cross-state variation)

For the complete methodology including database schema, Bradford Hill criteria mapping, and validation framework, see Optimal Policy Generator Specification.

15.1.3 How They Work Together

Framework	Unit of Analysis	Primary Output	Key Question
OBG/BIS	Spending category	Integrated budget recommendations	How much to spend?
PIS	Policy/regulation	Ranked reforms by impact	Which policies to adopt?

Both frameworks generate recommendations that politicians can follow or ignore:

```
+-------------------------+    +-------------------------+
|   Budget Optimization   |    |   Policy Optimization   |
|   (OBG/BIS Framework)   |    |   (PIS Framework)       |
|                         |    |                         |
|   Recommends:           |    |   Recommends:           |
|   - Optimal spending    |    |   - Which reforms       |
|     levels per category|    |     improve welfare     |
|   - Investment gaps     |    |   - Priority ranking    |
+-------------------------+    +-------------------------+
           |                               |
           +------------+  +---------------+
                        |  |
                        v  v
         +---------------------------+
         |   ORACLE VERIFICATION     |
         |   Independent measurement |
         |   of actual outcomes      |
         +---------------------------+
```

The Political Dysfunction Tax (estimated at 20% (95% CI: 9%-39.3%) of potential welfare) arises from both **misallocation** (wrong spending levels) and **bad policy** (welfare-reducing regulations). Addressing both requires both frameworks working together.

15.2 Algorithmic Governance Threat Model

Any honest assessment of algorithmic governance must acknowledge potential failure modes and attack surfaces:

15.2.1 Known Failure Modes

Attack Vector	Risk	Mitigation	Residual Risk
Verification manipulation	Corrupting data feeds to trigger favorable allocations	Multi-source aggregation, time-weighted averages, circuit breakers	High: fundamental challenge
Parameter exploitation	Gaming system rules through edge cases	Formal verification, economic audits	Medium: novel attacks possible
Administrative compromise	Capturing upgrade or amendment authority	Multi-party approval, time-locks, transparency requirements	Medium: key management remains hard
Implementation errors	Software bugs leading to unintended behavior	Code audits, gradual deployment, bug bounties	Medium: novel variants emerge

15.2.2 Realistic Security Properties

Optimocracy does not claim to eliminate all governance risk. It claims to:

1. **Raise attack costs**: Capturing multiple independent verification sources costs more than lobbying a committee
2. **Increase transparency**: All allocation rules are public and auditable
3. **Reduce political surface**: Fewer decision points where capture can occur
4. **Enable credible commitment**: Harder (not impossible) to deviate from stated rules

The honest comparison isn't "algorithmic governance vs. perfect security." It's "algorithmic governance vs. Congress." Algorithmic systems create clearer feedback loops between failure and correction: failures are visible, embarrassing, and spawn better security practices. Congressional capture is often invisible and legal. The Farm Bill has been getting worse since 1933. Neither system is perfect, and the process of updating algorithms in response to failures remains subject to political pressures, but algorithmic systems at least make failures visible.

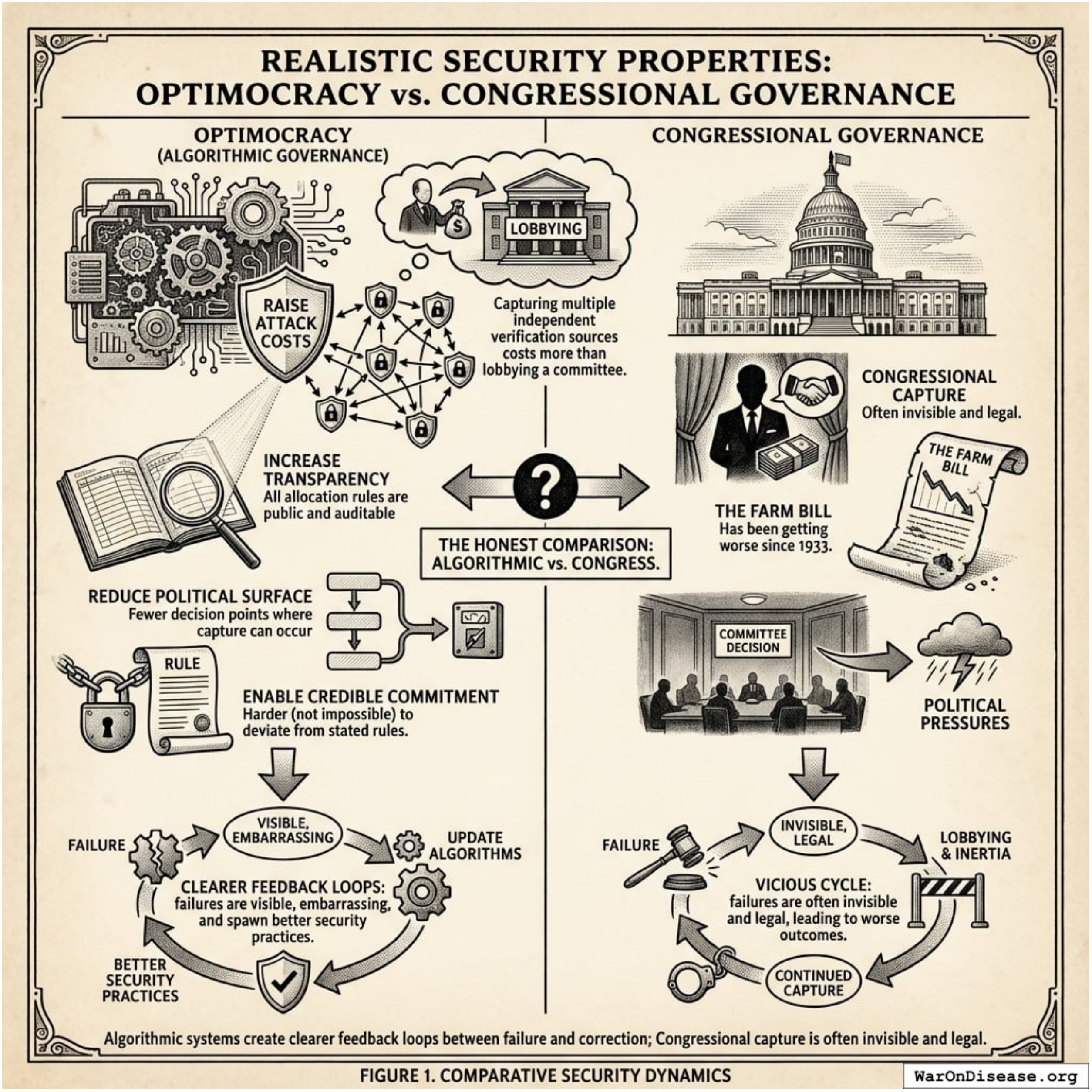

Figure 21: A comparison between algorithmic governance and traditional political systems, highlighting differences in attack costs, transparency of failure, and the feedback loops for system correction.

15.2.3 Defense-in-Depth Architecture

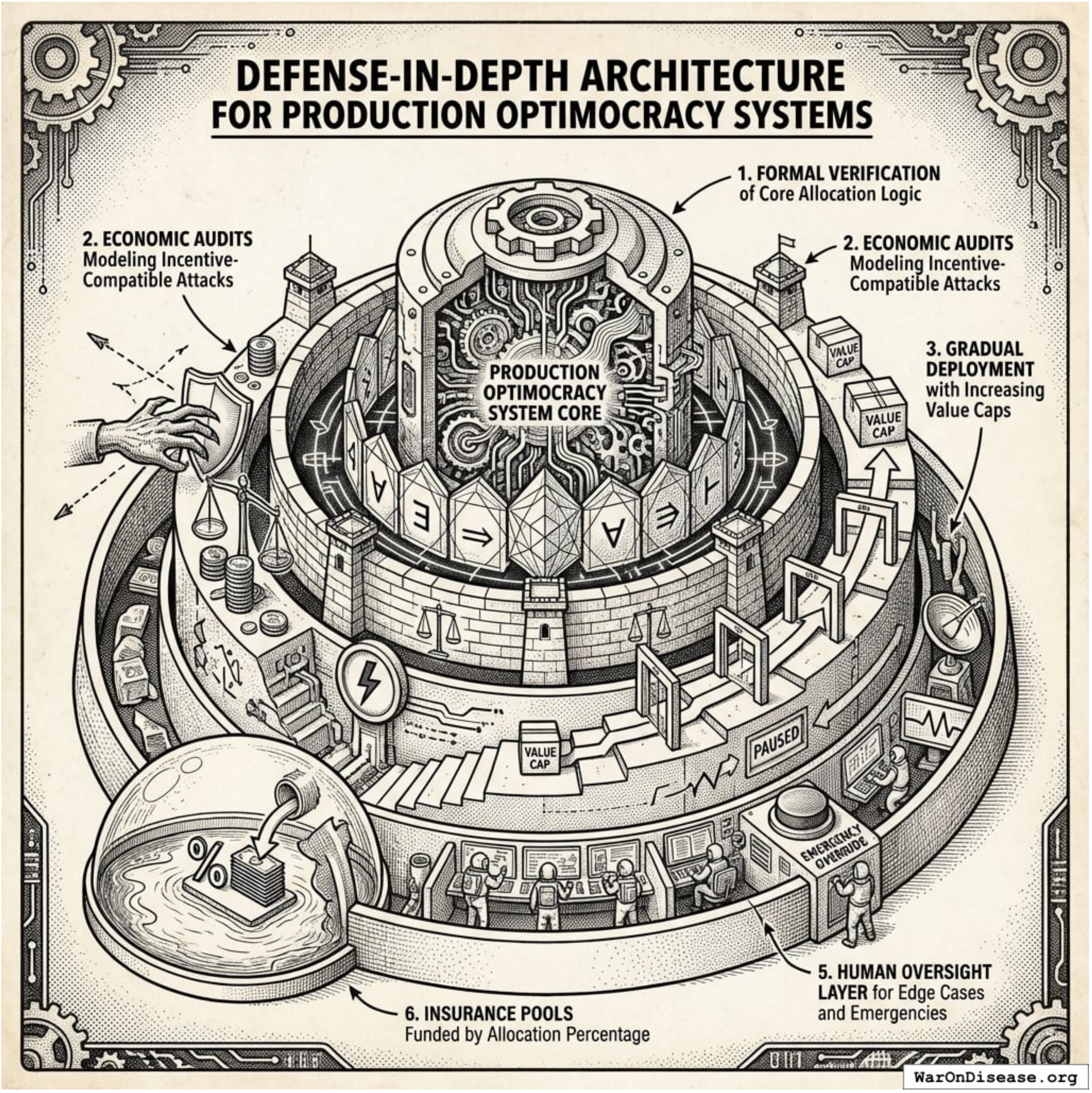

Figure 22: A layered defense-in-depth architecture diagram illustrating the six security measures protecting the core system logic.

Production Optimocracy systems should implement:

1. **Formal verification** of core allocation logic
2. **Economic audits** modeling incentive-compatible attacks
3. **Gradual deployment** with value caps that increase as the system proves reliable
4. **Circuit breakers** that pause allocation if metrics move anomalously
5. **Human oversight layer** for handling edge cases and emergencies
6. **Insurance pools** funded by a percentage of allocations

16 Appendix B: Theoretical Background

This appendix provides deeper theoretical context for readers interested in the academic foundations and historical precedents. The main text presents the mechanism; this appendix addresses why alternative approaches have failed and engages with theoretical objections.

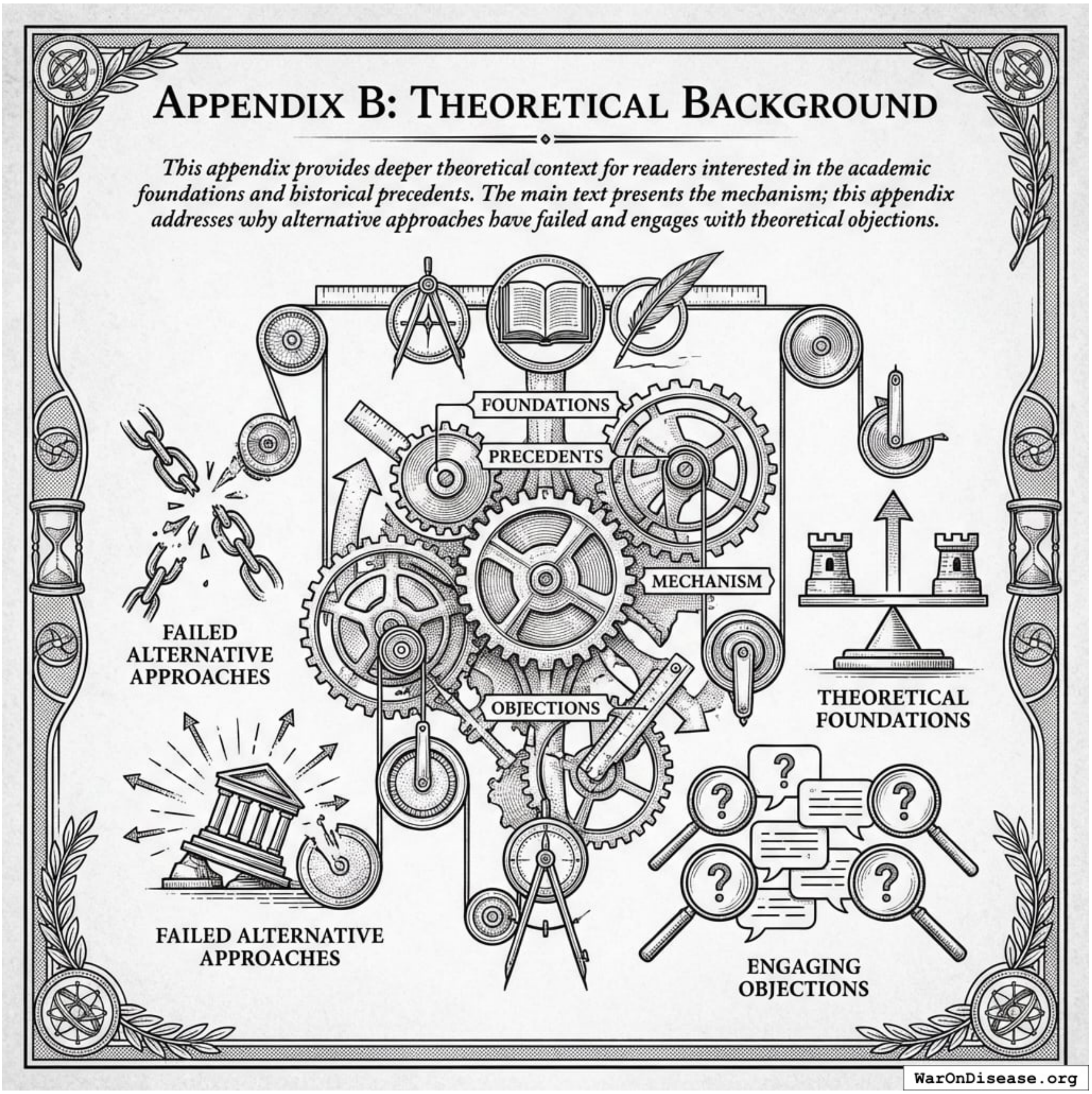

Figure 23: A conceptual framework comparing the proposed mechanism with alternative historical approaches and theoretical foundations.

16.1 Historical Precedents: Technocratic Governance Failures

16.1.1 Soviet Central Planning

The Soviet system attempted comprehensive optimization: central planners would calculate optimal production quantities and allocate resources accordingly. The failure was comprehensive:

1. **Knowledge problem**[88]: Planners lacked the distributed information that prices aggregate in markets.
2. **Incentive problem**: Planners had no personal stake in outcomes and faced perverse incentives.
3. **Scope problem**: Central planning attempted to optimize *everything*, including decisions where local knowledge is essential.

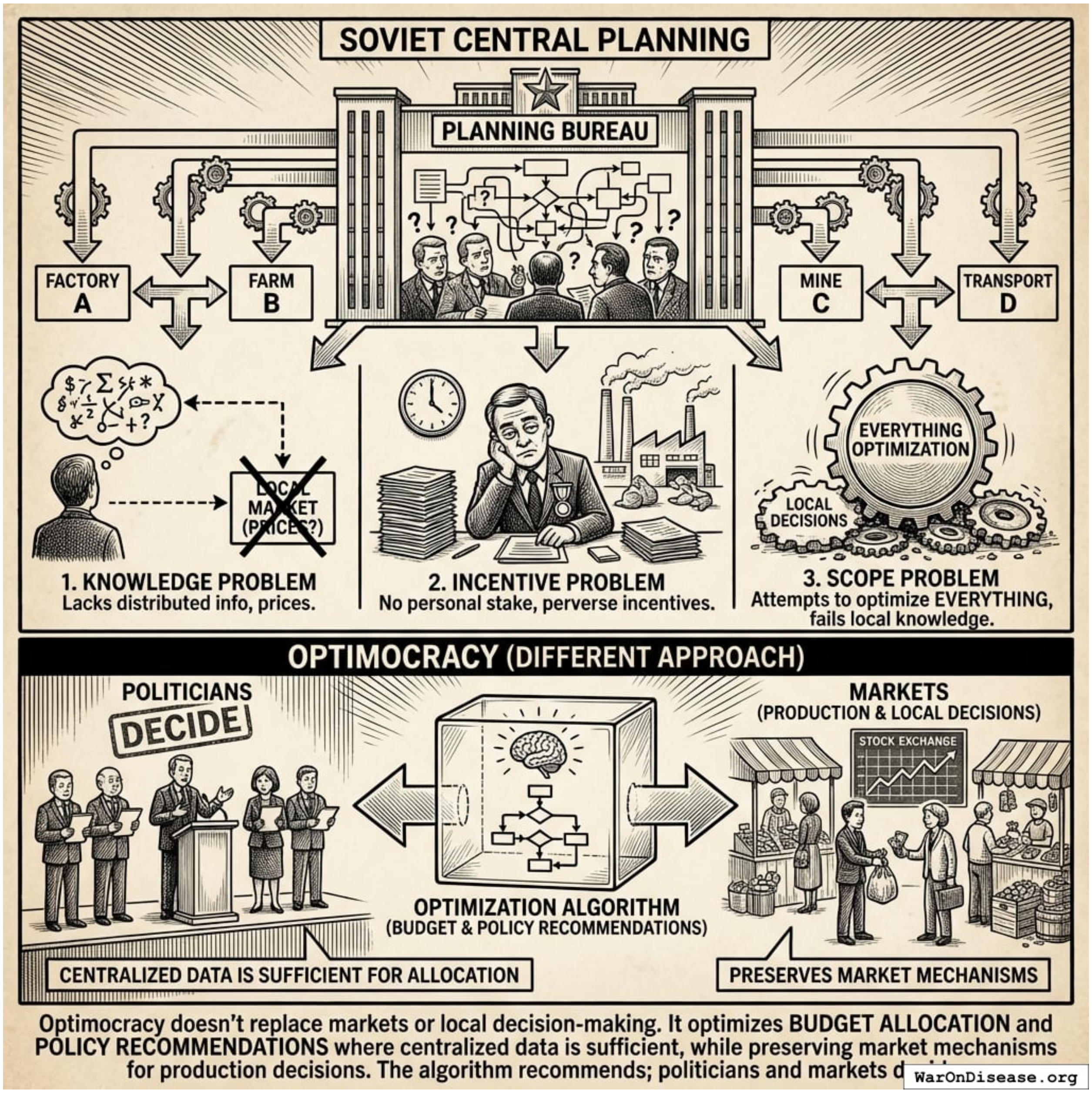

Figure 24: A comparison between Soviet Central Planning, which manages all market production, and Optimocracy, which provides algorithmic recommendations for budgets and policy while leaving production to the market.

Why Optimocracy differs: Optimocracy doesn't replace markets or local decision-making. It optimizes *budget allocation* and *policy recommendations* where centralized data is sufficient, while preserving market mechanisms for production decisions. The algorithm recommends; politicians and markets decide.

16.1.2 Credit Rating Agencies (1990s-2008)

Credit ratings were supposed to be objective, algorithmic assessments of default risk. Instead:

1. **Incentive misalignment**: Agencies were paid by issuers (those being rated)
2. **Metric gaming**: Financial engineers designed securities to maximize ratings while hiding risk
3. **Capture**: Rating models were "optimized" to give favorable ratings, not accurate risk assessment

What Optimocracy learns: Use median aggregation across multiple independent sources with different governance structures. No single entity controls measurement.

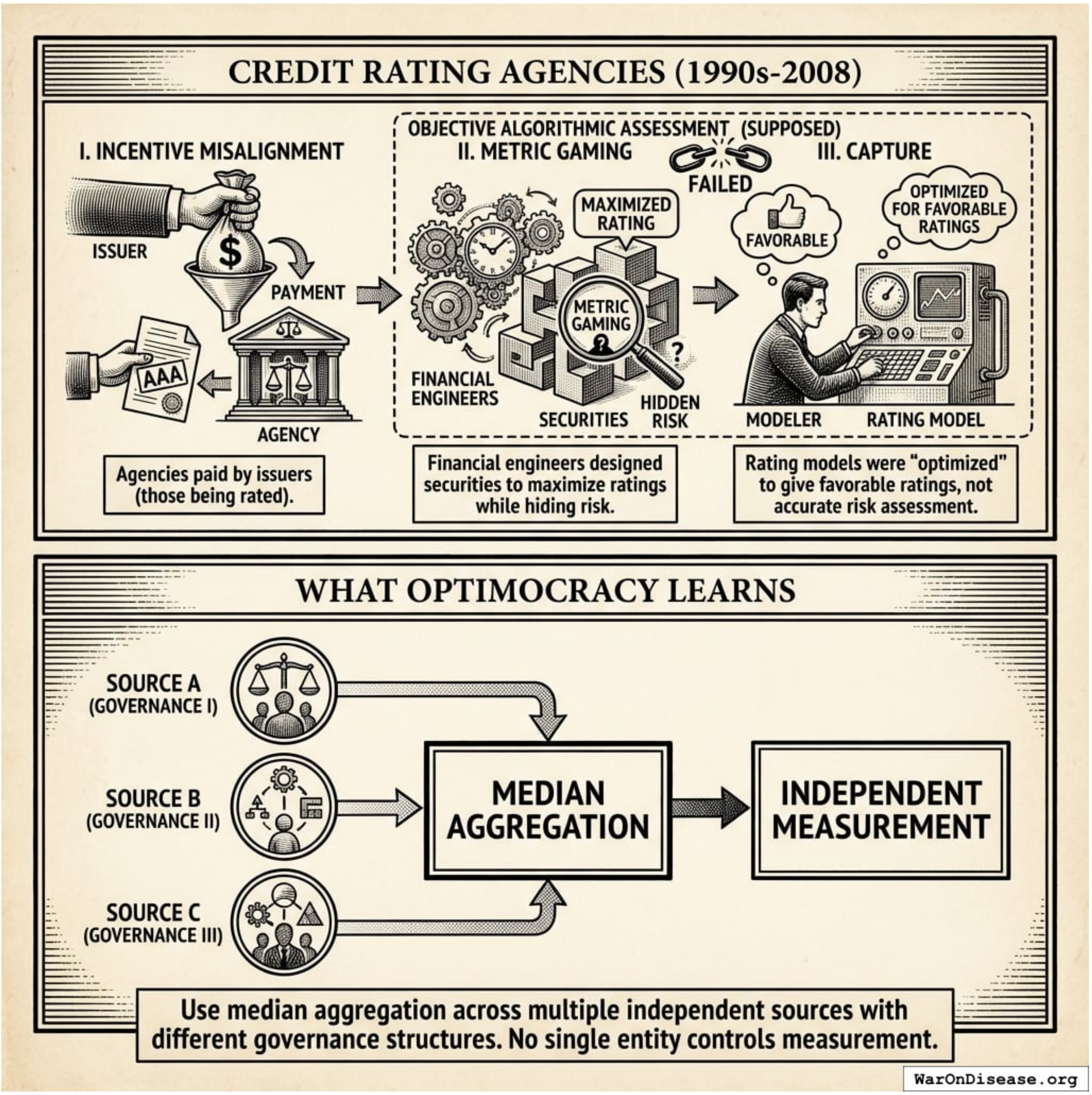

Figure 25: A conceptual diagram contrasting the centralized conflict of interest in traditional credit rating models with the decentralized median aggregation approach.

16.2 Engaging with Impossibility Theorems

16.2.1 Arrow and Gibbard-Satterthwaite

[150] proved no voting rule perfectly aggregates conflicting preferences.[151] showed any voting rule can be strategically gamed.

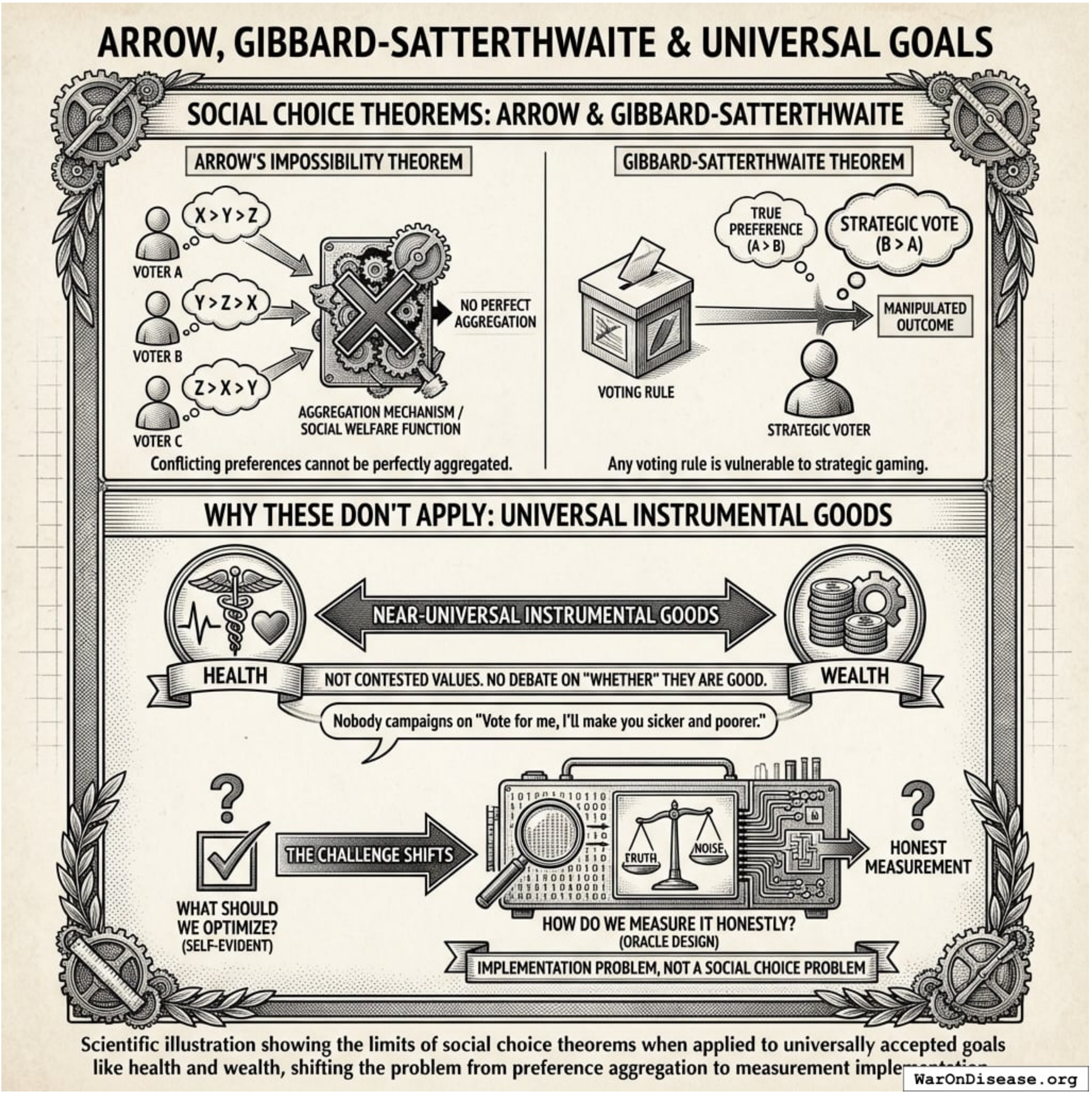

Figure 26: A comparison showing the shift from a Social Choice Problem where goals are contested to an Implementation Problem where goals are shared but measurement is the primary challenge.

Why these don't apply: Health and wealth are not contested values in the Arrow sense. They are near-universal instrumental goods. Nobody campaigns on "Vote for me, I'll make you sicker and poorer." The debate is always about *how* to achieve health and wealth, never *whether* they're good.

The challenge shifts from "what should we optimize?" (self-evident) to "how do we measure it honestly?" (oracle design). That's an implementation problem, not a social choice problem.

16.2.2 Mechanism Design: The Revelation Principle

The Revelation Principle[152,153] states that data reporters only tell the truth when lying costs more than it's worth.

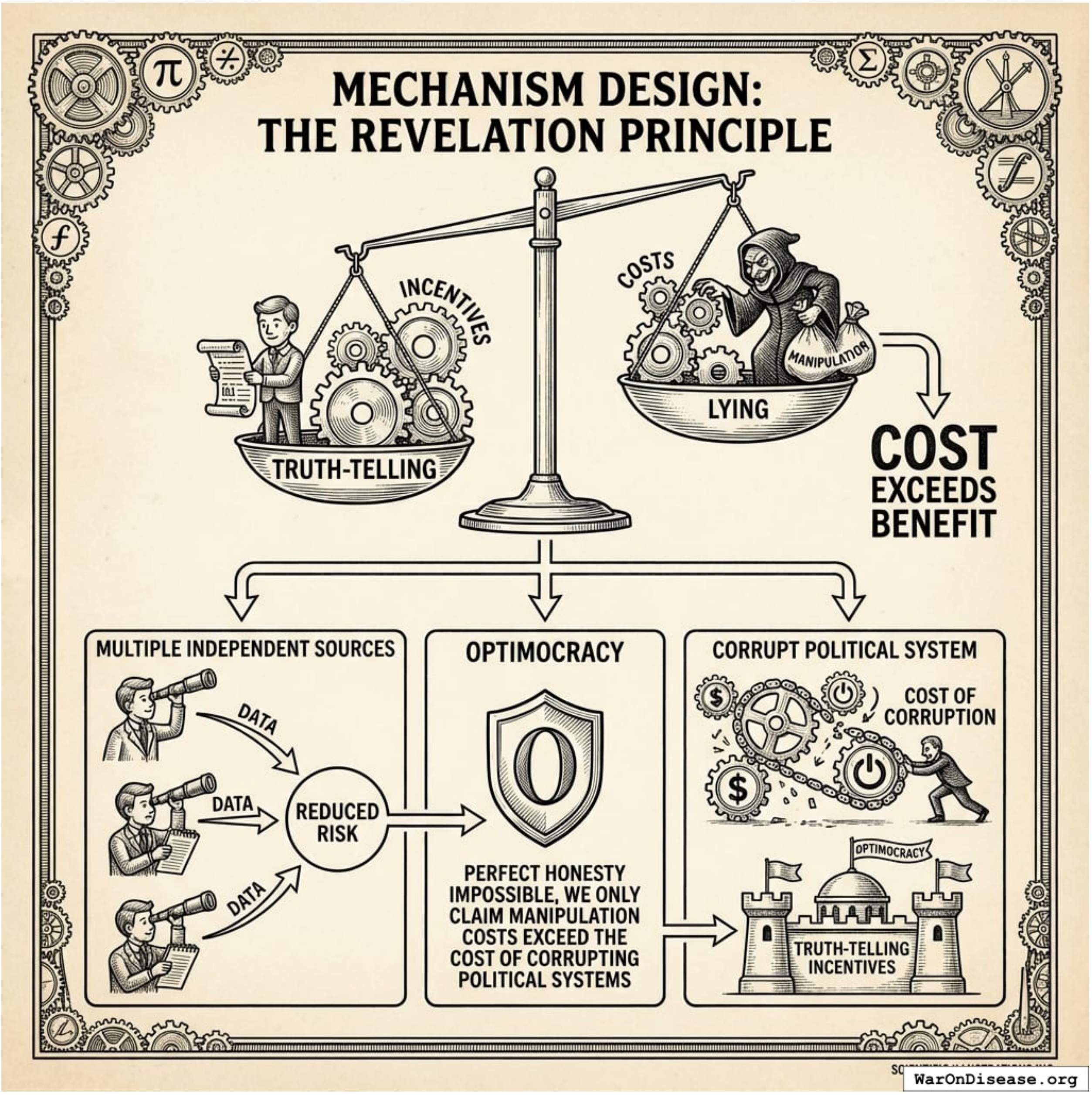

Figure 27: A conceptual diagram illustrating the trade-off between the cost of manipulation and the incentives for truth-telling within the Optimocracy framework.

What this means for Optimocracy: - Data reporters need incentives for truth-telling - Multiple independent sources reduce manipulation risk - Perfect honesty may be impossible; we only claim manipulation costs exceed the cost of corrupting political systems

16.3 Democratic Legitimacy: A Deeper Analysis

The concern is not whether Optimocracy would work technically, but whether citizens would accept it as legitimate governance. Democratic theorists from Rousseau to Habermas argue that the *process* of collective decision-making has intrinsic value.

The objection has real force. On this view, a benevolent dictator who maximized welfare would still be illegitimate because citizens didn't author their own laws.

Responses:

Response	Key Point
More democratic, not less	Current system offers ritual (voting) but delivers oligarchy ([138]). Optimocracy delivers what citizens actually voted for.
Citizens choose the goal	We already delegate implementation to the Fed and FDA.
Universal values	Health and wealth aren't contested preferences. We're measuring what humans universally value.
Legitimacy evolves	Social Security and index funds were controversial initially.
The alternative is capture	The realistic choice is captured allocation dressed in democratic ritual.

There is a genuine tension between outcome legitimacy (governance produces outcomes citizens want) and process legitimacy (citizens meaningfully participate in decisions). Optimocracy prioritizes outcome legitimacy while preserving process legitimacy through the wrapper architecture: politicians still decide everything; they just face new incentives and new information.

16.4 Boundary Conditions for Algorithmic Governance

The historical record suggests algorithmic governance works under specific conditions:

Condition	Favorable	Unfavorable
Metric clarity	Single, measurable objective	Multiple, contested objectives
Knowledge requirements	Aggregable statistics sufficient	Distributed local knowledge essential
Value consensus	Broad agreement on objective	Fundamental value conflicts
Scope	Narrow domain	Comprehensive planning
Accountability	Clear consequences for failure	Diffuse responsibility

Optimocracy operates in domains where conditions are favorable (budget allocation, policy evaluation) and defers to democratic deliberation where they are not (value conflicts, novel situations requiring judgment).

Acknowledgments

[To be added: acknowledgments for seminar participants, reviewers, and colleagues who provided feedback.]

17 References

1. Fund, N. C. NIH pragmatic trials: Minimal funding despite 30x cost advantage. *NIH Common Fund: HCS Research Collaboratory* https://commonfund.nih.gov/hcscollaboratory (2025)
*The NIH Pragmatic Trials Collaboratory funds trials at **$500K for planning phase, $1M/year. for implementation**—a tiny fraction of NIH's budget. The ADAPTABLE trial cost **$14 million** for **15,076 patients** (= **$929/patient**) versus **$420 million** for a similar traditional RCT (30x cheaper), yet pragmatic trials remain severely underfunded. PCORnet infrastructure enables real-world trials embedded in healthcare systems, but receives minimal support compared to basic research funding. Additional sources: https://commonfund.nih.gov/hcscollaboratory | https://pcornet.org/wp-content/uploads/2025/08/ADAPTABLE_Lay_Summary_21JUL2025.pdf | https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5604499/*
2. NIH. Antidepressant clinical trial exclusion rates. *Zimmerman et al.* https://pubmed.ncbi.nlm.nih.gov/26276679/ (2015)
Mean exclusion rate: 86.1% across 158 antidepressant efficacy trials (range: 44.4% to. 99.8%) More than 82% of real-world depression patients would be ineligible for antidepressant registration trials Exclusion rates increased over time: 91.4% (2010-2014) vs. 83.8% (1995-2009) Most common exclusions: comorbid psychiatric disorders, age restrictions, insufficient depression severity, medical conditions Emergency psychiatry patients: only 3.3% eligible (96.7% excluded) when applying 9 common exclusion criteria Only a minority of depressed patients seen in clinical practice are likely to be eligible for most AETs Note: Generalizability of antidepressant trials has decreased over time, with increasingly stringent exclusion criteria eliminating patients who would actually use the drugs in clinical practice Additional sources: https://pubmed.ncbi.nlm.nih.gov/26276679/ | https://pubmed.ncbi.nlm.nih.gov/26164052/ | https://www.wolterskluwer.com/en/news/antidepressant-trials-exclude-most-real-world-patients-with-depression
3. CNBC. Warren buffett's career average investment return. *CNBC* https://www.cnbc.com/2025/05/05/warren-buffetts-return-tally-after-60-years-5502284percent.html (2025)
Berkshire's compounded annual return from 1965 through 2024 was 19.9%, nearly. double the 10.4% recorded by the S&P 500. Berkshire shares skyrocketed 5,502,284% compared to the S&P 500's 39,054% rise during that period. Additional sources: https://www.cnbc.com/2025/05/05/warren-buffetts-return-tally-after-60-years-5502284percent.html | https://www.slickcharts.com/berkshire-hathaway/returns
4. Group, E. W. US farm subsidy database and analysis. *Environmental Working Group* https://farm.ewg.org/ (2024)
US agricultural subsidies total approximately $30 billion annually, but create much larger. economic distortions. Top 10% of farms receive 78% of subsidies, benefits concentrated in commodity crops (corn, soy, wheat, cotton), environmental damage from monoculture incentivized, and overall deadweight loss estimated at $50-120 billion annually. Additional sources: https://farm.ewg.org/ | https://www.ers.usda.gov/topics/farm-economy/farm-sector-income-finances/government-payments-the-safety-net/

5. Posen, B. R. *Restraint: A New Foundation for u.s. Grand Strategy*. (Posen, 2014).
The United States could maintain adequate deterrence and defense with a much smaller. military budget. Current spending levels reflect force projection capabilities far beyond what homeland security and deterrence require. A strategy of restraint could reduce defense spending by 40-50% while maintaining security through nuclear deterrence and geographic advantages. Additional sources: https://www.cornellpress.cornell.edu/book/9780801452581/restraint/

6. Alliance, D. P. The drug war by the numbers. (2021)
Since 1971, the war on drugs has cost the United States an estimated $1 trillion in enforcement. The federal drug control budget was $41 billion in 2022. Mass incarceration costs the U.S. at least $182 billion every year, with over $450 billion spent to incarcerate individuals on drug charges in federal prisons.

7. Crawford, N. C. & Lutz, C. Blood and treasure: United states budgetary costs and human costs of 20 years of war. (2023)
The total costs of the post-9/11 wars in Iraq, Afghanistan, Pakistan, and Syria are expected to exceed $8 trillion. This includes $2.89 trillion for Iraq/Syria, veterans care through 2050 projected at more than $2 trillion, and interest on war debt adding $6.5 trillion through 2050.

8. Fund, I. M. IMF fossil fuel subsidies data: 2023 update. (2023)
Globally, fossil fuel subsidies were $7 trillion in 2022 or 7.1 percent of GDP. The United States subsidies totaled $649 billion. Underpricing for local air pollution costs and climate damages are the largest contributor, accounting for about 30 percent each.

9. Papanicolas, I. et al. Health care spending in the united states and other high-income countries. *Papanicolas et al.* https://jamanetwork.com/journals/jama/article-abstract/2674671 (2018)
The US spent approximately twice as much as other high-income countries on medical. care (mean per capita: $9,892 vs $5,289), with similar utilization but much higher prices. Administrative costs accounted for 8% of US spending vs 1-3% in other countries. US spending on pharmaceuticals was $1,443 per capita vs $749 elsewhere. Despite spending more, US health outcomes are not better. Additional sources: https://jamanetwork.com/journals/jama/article-abstract/2674671

10. Hsieh, C.-T. & Moretti, E. Housing constraints and spatial misallocation. *Hsieh & Moretti* https://www.aeaweb.org/articles?id=10.1257/mac.20170388 (2019)
We quantify the amount of spatial misallocation of labor across US cities and its aggregate. costs. Tight land-use restrictions in high-productivity cities like New York, San Francisco, and Boston lowered aggregate US growth by 36% from 1964 to 2009. Local constraints on housing supply have had enormous effects on the national economy. Additional sources: https://www.aeaweb.org/articles?id=10.1257/mac.20170388

11. Justice, V. I. of. The economic burden of incarceration in the united states. *Vera Institute* https://www.vera.org/publications/the-economic-burden-of-incarceration-in-the-u-s (2024)
US incarceration costs approximately $80 billion annually in direct correctional expendi-. tures alone. Including social costs (lost earnings, family impacts, health effects, reduced child outcomes), total burden exceeds $300 billion annually. The US incarcerates at 5x the rate of other OECD countries with no corresponding reduction in crime. Evidence shows community-based alternatives cost less and reduce recidivism more effectively. Additional sources: https://www.vera.org/publications/the-economic-burden-of-incarceration-in-the-u-s | https://www.prisonpolicy.org/reports/pie2024.html | https://www.rand.org/pubs/research_reports/RRA108-3.html

12. Marron Institute, N. Transit costs project - why US infrastructure costs so much. *NYU Transit Costs Project* https://transitcosts.com/ (2024)
The United States builds transit infrastructure at dramatically higher costs than peer countries.. New York's Second Avenue Subway cost $2.5 billion per kilometer vs. $200-500 million in European cities. US highway construction similarly costs 2-5x more than comparable projects abroad. Causes include: excessive environmental review, litigation risk, lack of in-house expertise, fragmented project management, and inflated soft costs. Additional sources: https://transitcosts.com/ | https://www.brookings.edu/articles/why-does-infrastructure-cost-so-much/

13. Clemens, M. A. Economics and emigration: Trillion-dollar bills on the sidewalk? *Journal of Economic Perspectives* **25**, 83–106 (2011)
Free global labor mobility would increase gross world product by somewhere in the range of 67-147%... The gains to eliminating migration barriers amount to large fractions of world GDP—one or two orders of magnitude larger than the gains from dropping all remaining restrictions on international flows of goods and capital.

14. Kleiner, M. M. & Krueger, A. B. Analyzing the extent and influence of occupational licensing on the labor market. *Journal of Labor Economics* **31**, S173–S202 (2013)
Occupational licensing affects 29% of US workers and creates labor market distortions costing 2-3% of GDP.

15. Lab, Y. B. The fiscal, economic, and distributional effects of all u.s. tariffs. (2025)
Accounting for all the 2025 US tariffs and retaliation implemented to date, the level of real GDP is persistently -0.6% smaller in the long run, the equivalent of $160 billion 2024$ annually.

16. Foundation, T. Tax compliance costs the US economy $546 billion annually. https://taxfoundation.org/data/all/federal/irs-tax-compliance-costs/ (2024)
Americans will spend over 7.9 billion hours complying with IRS tax filing and reporting requirements in 2024. This costs the economy roughly $413 billion in lost productivity. In addition, the IRS estimates that Americans spend roughly $133 billion annually in out-of-pocket costs, bringing the total compliance costs to $546 billion, or nearly 2 percent of GDP.

17. Organization, W. H. WHO global health estimates 2024. *World Health Organization* https://www.who.int/data/gho/data/themes/mortality-and-global-health-estimates (2024)
Comprehensive mortality and morbidity data by cause, age, sex, country, and year Global. mortality: 55-60 million deaths annually Lives saved by modern medicine (vaccines, cardiovascular drugs, oncology): 12M annually (conservative aggregate) Leading causes of death: Cardiovascular disease (17.9M), Cancer (10.3M), Respiratory disease (4.0M) Note: Baseline data for regulatory mortality analysis. Conservative estimate of pharmaceutical impact based on WHO immunization data (4.5M/year from vaccines) + cardiovascular interventions (3.3M/year) + oncology (1.5M/year) + other therapies. Additional sources: https://www.who.int/data/gho/data/themes/mortality-and-global-health-estimates

18. GiveWell. GiveWell cost per life saved for top charities (2024). *GiveWell: Top Charities* https://www.givewell.org/charities/top-charities
General range: $3,000-$5,500 per life saved (GiveWell top charities) Helen Keller International. (Vitamin A): $3,500 average (2022-2024); varies $1,000-$8,500 by country Against Malaria Foundation: $5,500 per life saved New Incentives (vaccination incentives): $4,500 per life saved Malaria Consortium (seasonal malaria chemoprevention): $3,500 per life saved VAS program details: $2 to provide vitamin A supplements to child for one year Note: Figures accurate for 2024. Helen Keller VAS program has wide country variation ($1K-$8.5K) but $3,500 is accurate average. Among most cost-effective interventions globally Additional sources: https://www.givewell.org/charities/top-charities | https://www.givewell.org/charities/helen-keller-international | https://ourworldindata.org/cost-effectiveness

19. AARP. Unpaid caregiver hours and economic value. *AARP 2023* https://www.aarp.org/caregiving/financial-legal/info-2023/unpaid-caregivers-provide-billions-in-care.html (2023)
Average family caregiver: 25-26 hours per week (100-104 hours per month) 38 million. caregivers providing 36 billion hours of care annually Economic value: $16.59 per hour = $600 billion total annual value (2021) 28% of people provided eldercare on a given day, averaging 3.9 hours when providing care Caregivers living with care recipient: 37.4 hours per week Caregivers not living with recipient: 23.7 hours per week Note: Disease-related caregiving is subset of total; includes elderly care, disability care, and child care Additional sources: https://www.aarp.org/caregiving/financial-legal/info-2023/unpaid-caregivers-provide-billions-in-care.html | https://www.bls.gov/news.release/elcare.nr0.htm | https://www.caregiver.org/resource/caregiver-statistics-demographics/

20. MMWR, C. Childhood vaccination economic benefits. *CDC MMWR* https://www.cdc.gov/mmwr/volumes/73/wr/mm7331a2.htm (1994)
US programs (1994-2023): $540B direct savings, $2.7T societal savings ($18B/year direct, $90B/year societal) Global (2001-2020): $820B value for 10 diseases in 73 countries ($41B/year) ROI: $11 return per $1 invested Measles vaccination alone saved 93.7M lives (61% of 154M total) over 50 years (1974-2024) Additional sources: https://www.cdc.gov/mmwr/volumes/73/wr/mm7331a2.htm | https://www.thelancet.com/journals/lancet/article/PIIS0140-6736(24

21. CDC. Childhood vaccination (US) ROI. *CDC* https://www.cdc.gov/mmwr/preview/mmwrhtml/mm6316a4.htm (2017).

22. Labor Statistics, U. S. B. of. CPI inflation calculator. (2024)
CPI-U (1980): 82.4 CPI-U (2024): 313.5 Inflation multiplier (1980-2024): 3.80× Cumulative. inflation: 280.48% Average annual inflation rate: 3.08% Note: Official U.S. government inflation data using Consumer Price Index for All Urban Consumers (CPI-U). Additional sources: https://www.bls.gov/data/inflation_calculator.htm

23. Del Rosal, I. The empirical measurement of rent-seeking costs. *Journal of Economic Surveys* https://onlinelibrary.wiley.com/doi/abs/10.1111/j.1467-6419.2009.00621.x (2011)
A comprehensive survey of empirical estimates finds rent-seeking costs range from 0.2% to 23.7% of GDP across different methodologies and countries. Laband & Sophocleus (1988) estimated up to 45% for the US.

24. via, D. analysis. ClinicalTrials.gov cumulative enrollment data (2025). *Direct analysis via ClinicalTrials.gov API v2* https://clinicaltrials.gov/data-api/api
Analysis of 100,000 active/recruiting/completed trials on ClinicalTrials.gov (November 2025). shows cumulative enrollment of 12.2 million participants: Phase 1 (722k), Phase 2 (2.2M), Phase 3 (6.5M), Phase 4 (2.7M). Median participants per trial: Phase 1 (33), Phase 2 (60), Phase 3 (237), Phase 4 (90). Additional sources: https://clinicaltrials.gov/data-api/api

25. CAN, A. Clinical trial patient participation rate. *ACS CAN: Barriers to Clinical Trial Enrollment* https://www.fightcancer.org/policy-resources/barriers-patient-enrollment-therapeutic-clinical-trials-cancer
Only 3-5% of adult cancer patients in US receive treatment within clinical trials About 5% of. American adults have ever participated in any clinical trial Oncology: 2-3% of all oncology patients participate Contrast: 50-60% enrollment for pediatric cancer trials (<15 years old) Note: 20% of cancer trials fail due to insufficient enrollment; 11% of research sites enroll zero patients Additional sources: https://www.fightcancer.org/policy-resources/barriers-patient-enrollment-therapeutic-clinical-trials-cancer | https://hints.cancer.gov/docs/Briefs/HINTS_Brief_48.pdf

26. ScienceDaily. Global prevalence of chronic disease. *ScienceDaily: GBD 2015 Study* https://www.sciencedaily.com/releases/2015/06/150608081753.htm (2015)
2.3 billion individuals had more than five ailments (2013) Chronic conditions caused 74%. of all deaths worldwide (2019), up from 67% (2010) Approximately 1 in 3 adults suffer from multiple chronic conditions (MCCs) Risk factor exposures: 2B exposed to biomass fuel, 1B to air pollution, 1B smokers Projected economic cost: $47 trillion by 2030 Note: 2.3B with 5+ ailments is more accurate than "2B with chronic disease." One-third of all adults globally have multiple chronic conditions Additional sources: https://www.sciencedaily.com/releases/2015/06/150608081753.htm | https://pmc.ncbi.nlm.nih.gov/articles/PMC10830426/ | https://pmc.ncbi.nlm.nih.gov/articles/PMC6214883/

27. C&EN. Annual number of new drugs approved globally: 50. *C&EN* https://cen.acs.org/pharmaceuticals/50-new-drugs-received-FDA/103/i2 (2025)
50 new drugs approved annually Additional sources: https://cen.acs.org/pharmaceuticals/50-new-drugs-received-FDA/103/i2 | https://www.fda.gov/drugs/development-approval-process-drugs/novel-drug-approvals-fda

28. estimates, I. Clinical trial abandonment.
Average: 10% abandoned before completion .

29. Report, I. Global trial capacity. *IQVIA Report: Clinical Trial Subjects Number Drops Due to Decline in COVID-19 Enrollment* https://gmdpacademy.org/news/iqvia-report-clinical-trial-subjects-number-drops-due-to-decline-in-covid-19-enrollment/
1.9M participants annually (2022, post-COVID normalization from 4M peak in 2021) Additional sources: https://gmdpacademy.org/news/iqvia-report-clinical-trial-subjects-number-drops-due-to-decline-in-covid-19-enrollment/

30. Research & Markets. Global clinical trials market 2024. *Research and Markets* https://www.globenewswire.com/news-release/2024/04/19/2866012/0/en/Global-Clinical-Trials-Market-Research-Report-2024-An-83-16-Billion-Market-by-2030-AI-Machine-Learning-and-Blockchain-will-Transform-the-Clinical-Trials-Landscape.html (2024)
Global clinical trials market valued at approximately $83 billion in 2024, with projections to. reach $83-132 billion by 2030. Additional sources: https://www.globenewswire.com/news-release/2024/04/19/2866012/0/en/Global-Clinical-Trials-Market-Research-Report-2024-An-83-16-Billion-Market-by-2030-AI-Machine-Learning-and-Blockchain-will-Transform-the-Clinical-Trials-Landscape.html | https://www.precedenceresearch.com/clinical-trials-market

31. OpenSecrets. Lobbying spend (defense). *OpenSecrets* https://www.opensecrets.org/federal-lobbying/industries/summary?cycle=2024/&id=D (2024).
32. GiveWell. Cost per DALY for deworming programs. https://www.givewell.org/international/technical/programs/deworming/cost-effectiveness
Schistosomiasis treatment: $28.19-$70.48 per DALY (using arithmetic means with varying disability weights) Soil-transmitted helminths (STH) treatment: $82.54 per DALY (midpoint estimate) Note: GiveWell explicitly states this 2011 analysis is "out of date" and their current methodology focuses on long-term income effects rather than short-term health DALYs Additional sources: https://www.givewell.org/international/technical/programs/deworming/cost-effectiveness
33. Numbers, T. by. Pre-1962 drug development costs and timeline (think by numbers). *Think by Numbers: How Many Lives Does FDA Save?* https://thinkbynumbers.org/health/how-many-net-lives-does-the-fda-save/ (1962)
Historical estimates (1970-1985): USD $226M fully capitalized (2011 prices) 1980s drugs: $65M after-tax R&D (1990 dollars), $194M compounded to approval (1990 dollars) Modern comparison: $2-3B costs, 7-12 years (dramatic increase from pre-1962) Context: 1962 regulatory clampdown reduced new treatment production by 70%, dramatically increasing development timelines and costs Note: Secondary source; less reliable than Congressional testimony Additional sources: https://thinkbynumbers.org/health/how-many-net-lives-does-the-fda-save/ | https://en.wikipedia.org/wiki/Cost_of_drug_development | https://www.statnews.com/2018/10/01/changing-1962-law-slash-drug-prices/
34. (BIO), B. I. O. BIO clinical development success rates 2011-2020. *Biotechnology Innovation Organization (BIO)* https://go.bio.org/rs/490-EHZ-999/images/ClinicalDevelopmentSuccessRates2011_2020.pdf (2021)
Phase I duration: 2.3 years average Total time to market (Phase I-III + approval): 10.5 years average Phase transition success rates: Phase I→II: 63.2%, Phase II→III: 30.7%, Phase III→Approval: 58.1% Overall probability of approval from Phase I: 12% Note: Largest publicly available study of clinical trial success rates. Efficacy lag = 10.5 - 2.3 = 8.2 years post-safety verification. Additional sources: https://go.bio.org/rs/490-EHZ-999/images/ClinicalDevelopmentSuccessRates2011_2020.pdf
35. Medicine, N. Drug repurposing rate (30%). *Nature Medicine* https://www.nature.com/articles/s41591-024-03233-x (2024)
Approximately 30% of drugs gain at least one new indication after initial approval. Additional sources: https://www.nature.com/articles/s41591-024-03233-x
36. EPI. Education investment economic multiplier (2.1). *EPI: Public Investments Outside Core Infrastructure* https://www.epi.org/publication/bp348-public-investments-outside-core-infrastructure/
Early childhood education: Benefits 12X outlays by 2050; $8.70 per dollar over lifetime Educational facilities: $1 spent → $1.50 economic returns Energy efficiency comparison: 2-to-1 benefit-to-cost ratio (McKinsey) Private return to schooling: 9% per additional year (World Bank meta-analysis) Note: 2.1 multiplier aligns with benefit-to-cost ratios for educational infrastructure/energy efficiency. Early childhood education shows much higher returns (12X by 2050) Additional sources: https://www.epi.org/publication/bp348-public-investments-outside-core-infrastructure/ | https://documents1.worldbank.org/curated/en/442521523465644318/pdf/WPS8402.pdf | https://freopp.org/whitepapers/establishing-a-practical-return-on-investment-framework-for-education-and-skills-development-to-expand-economic-opportunity/

37. PMC. Healthcare investment economic multiplier (1.8). *PMC: California Universal Health Care* https://pmc.ncbi.nlm.nih.gov/articles/PMC5954824/ (2022)
Healthcare fiscal multiplier: 4.3 (95% CI: 2.5-6.1) during pre-recession period (1995-2007) Overall government spending multiplier: 1.61 (95% CI: 1.37-1.86) Why healthcare has high multipliers: No effect on trade deficits (spending stays domestic); improves productivity & competitiveness; enhances long-run potential output Gender-sensitive fiscal spending (health & care economy) produces substantial positive growth impacts Note: "1.8" appears to be conservative estimate; research shows healthcare multipliers of 4.3 Additional sources: https://pmc.ncbi.nlm.nih.gov/articles/PMC5954824/ | https://cepr.org/voxeu/columns/government-investment-and-fiscal-stimulus | https://ncbi.nlm.nih.gov/pmc/articles/PMC3849102/ | https://set.odi.org/wp-content/uploads/2022/01/Fiscal-multipliers-review.pdf

38. Bank, W. Infrastructure investment economic multiplier (1.6). *World Bank: Infrastructure Investment as Stimulus* https://blogs.worldbank.org/en/ppps/effectiveness-infrastructure-investment-fiscal-stimulus-what-weve-learned (2022)
Infrastructure fiscal multiplier: 1.6 during contractionary phase of economic cycle Average across all economic states: 1.5 (meaning $1 of public investment → $1.50 of economic activity) Time horizon: 0.8 within 1 year, 1.5 within 2-5 years Range of estimates: 1.5-2.0 (following 2008 financial crisis & American Recovery Act) Italian public construction: 1.5-1.9 multiplier US ARRA: 0.4-2.2 range (differential impacts by program type) Economic Policy Institute: Uses 1.6 for infrastructure spending (middle range of estimates) Note: Public investment less likely to crowd out private activity during recessions; particularly effective when monetary policy loose with near-zero rates Additional sources: https://blogs.worldbank.org/en/ppps/effectiveness-infrastructure-investment-fiscal-stimulus-what-weve-learned | https://www.gihub.org/infrastructure-monitor/insights/fiscal-multiplier-effect-of-infrastructure-investment/ | https://cepr.org/voxeu/columns/government-investment-and-fiscal-stimulus | https://www.richmondfed.org/publications/research/economic_brief/2022/eb_22-04

39. Mercatus. Military spending economic multiplier (0.6). *Mercatus: Defense Spending and Economy* https://www.mercatus.org/research/research-papers/defense-spending-and-economy
Ramey (2011): 0.6 short-run multiplier Barro (1981): 0.6 multiplier for WWII spending (war spending crowded out 40¢ private economic activity per federal dollar) Barro & Redlick (2011): 0.4 within current year, 0.6 over two years; increased govt spending reduces private-sector GDP portions General finding: $1 increase in deficit-financed federal military spending = less than $1 increase in GDP Variation by context: Central/Eastern European NATO: 0.6 on impact, 1.5-1.6 in years 2-3, gradual fall to zero Ramey & Zubairy (2018): Cumulative 1% GDP increase in military expenditure raises GDP by 0.7% Additional sources: https://www.mercatus.org/research/research-papers/defense-spending-and-economy | https://cepr.org/voxeu/columns/world-war-ii-america-spending-deficits-multipliers-and-sacrifice | https://www.rand.org/content/dam/rand/pubs/research_reports/RRA700/RRA739-2/RAND_RRA739-2.pdf

40. FDA. FDA-approved prescription drug products (20,000+). *FDA* https://www.fda.gov/media/143704/download
There are over 20,000 prescription drug products approved for marketing. Additional sources: https://www.fda.gov/media/143704/download

41. FDA. FDA GRAS list count (570-700). *FDA* https://www.fda.gov/food/generally-recognized-safe-gras/gras-notice-inventory
The FDA GRAS (Generally Recognized as Safe) list contains approximately 570–700 substances. Additional sources: https://www.fda.gov/food/generally-recognized-safe-gras/gras-notice-inventory
42. ACLED. Active combat deaths annually. *ACLED: Global Conflict Surged 2024* https://acleddata.com/2024/12/12/data-shows-global-conflict-surged-in-2024-the-washington-post/ (2024)
2024: 233,597 deaths (30% increase from 179,099 in 2023) Deadliest conflicts: Ukraine (67,000), Palestine (35,000) Nearly 200,000 acts of violence (25% higher than 2023, double from 5 years ago) One in six people globally live in conflict-affected areas Additional sources: https://acleddata.com/2024/12/12/data-shows-global-conflict-surged-in-2024-the-washington-post/ | https://acleddata.com/media-citation/data-shows-global-conflict-surged-2024-washington-post | https://acleddata.com/conflict-index/index-january-2024/
43. UCDP. State violence deaths annually. *UCDP: Uppsala Conflict Data Program* https://ucdp.uu.se/
Uppsala Conflict Data Program (UCDP): Tracks one-sided violence (organized actors attacking unarmed civilians) UCDP definition: Conflicts causing at least 25 battle-related deaths in calendar year 2023 total organized violence: 154,000 deaths; Non-state conflicts: 20,900 deaths UCDP collects data on state-based conflicts, non-state conflicts, and one-sided violence Specific "2,700 annually" figure for state violence not found in recent UCDP data; actual figures vary annually Additional sources: https://ucdp.uu.se/ | https://en.wikipedia.org/wiki/Uppsala_Conflict_Data_Program | https://ourworldindata.org/grapher/deaths-in-armed-conflicts-by-region
44. Data, O. W. in. Terror attack deaths (8,300 annually). *Our World in Data: Terrorism* https://ourworldindata.org/terrorism (2024)
2023: 8,352 deaths (22% increase from 2022, highest since 2017) 2023: 3,350 terrorist incidents (22% decrease), but 56% increase in avg deaths per attack Global Terrorism Database (GTD): 200,000+ terrorist attacks recorded (2021 version) Maintained by: National Consortium for Study of Terrorism & Responses to Terrorism (START), U. of Maryland Geographic shift: Epicenter moved from Middle East to Central Sahel (sub-Saharan Africa) - now >50% of all deaths Additional sources: https://ourworldindata.org/terrorism | https://reliefweb.int/report/world/global-terrorism-index-2024 | https://www.start.umd.edu/gtd/ | https://ourworldindata.org/grapher/fatalities-from-terrorism
45. Health Metrics, I. for & (IHME), E. IHME global burden of disease 2021 (2.88B DALYs, 1.13B YLD). *Institute for Health Metrics and Evaluation (IHME)* https://vizhub.healthdata.org/gbd-results/ (2024)
In 2021, global DALYs totaled approximately 2.88 billion, comprising 1.75 billion Years of Life Lost (YLL) and 1.13 billion Years Lived with Disability (YLD). This represents a 13% increase from 2019 (2.55B DALYs), largely attributable to COVID-19 deaths and aging populations. YLD accounts for approximately 39% of total DALYs, reflecting the substantial burden of non-fatal chronic conditions. Additional sources: https://vizhub.healthdata.org/gbd-results/ | https://www.thelancet.com/journals/lancet/article/PIIS0140-6736(24 | https://www.healthdata.org/research-analysis/about-gbd

46. War, B. W. C. of. Environmental cost of war ($100B annually). *Brown Watson Costs of War: Environmental Cost* https://watson.brown.edu/costsofwar/costs/social/environment
War on Terror emissions: 1.2B metric tons GHG (equivalent to 257M cars/year) Military: 5.5% of global GHG emissions (2X aviation + shipping combined) US DoD: World's single largest institutional oil consumer, 47th largest emitter if nation Cleanup costs: $500B+ for military contaminated sites Gaza war environmental damage: $56.4B; landmine clearance: $34.6B expected Climate finance gap: Rich nations spend 30X more on military than climate finance Note: Military activities cause massive environmental damage through GHG emissions, toxic contamination, and long-term cleanup costs far exceeding current climate finance commitments Additional sources: https://watson.brown.edu/costsofwar/costs/social/environment | https://earth.org/environmental-costs-of-wars/ | https://transformdefence.org/transformdefence/stats/

47. ScienceDaily. Medical research lives saved annually (4.2 million). *ScienceDaily: Physical Activity Prevents 4M Deaths* https://www.sciencedaily.com/releases/2020/06/200617194510.htm (2020)
Physical activity: 3.9M early deaths averted annually worldwide (15% lower premature deaths than without) COVID vaccines (2020-2024): 2.533M deaths averted, 14.8M life-years preserved; first year alone: 14.4M deaths prevented Cardiovascular prevention: 3 interventions could delay 94.3M deaths over 25 years (antihypertensives alone: 39.4M) Pandemic research response: Millions of deaths averted through rapid vaccine/drug development Additional sources: https://www.sciencedaily.com/releases/2020/06/200617194510.htm | https://pmc.ncbi.nlm.nih.gov/articles/PMC9537923/ | https://www.ahajournals.org/doi/10.1161/CIRCULATIONAHA.118.038160 | https://pmc.ncbi.nlm.nih.gov/articles/PMC9464102/

48. SIPRI. 36:1 disparity ratio of spending on weapons over cures. *SIPRI: Military Spending* https://www.sipri.org/commentary/blog/2016/opportunity-cost-world-military-spending (2016)
Global military spending: $2.7 trillion (2024, SIPRI) Global government medical research: $68 billion (2024) Actual ratio: 39.7:1 in favor of weapons over medical research Military R&D alone: $85B (2004 data, 10% of global R&D) Military spending increases crowd out health: 1% ↑ military = 0.62% ↓ health spending Note: Ratio actually worse than 36:1. Each 1% increase in military spending reduces health spending by 0.62%, with effect more intense in poorer countries (0.962% reduction) Additional sources: https://www.sipri.org/commentary/blog/2016/opportunity-cost-world-military-spending | https://pmc.ncbi.nlm.nih.gov/articles/PMC9174441/ | https://www.congress.gov/crs-product/R45403

49. Numbers, T. by. Lost human capital due to war ($270B annually). *Think by Numbers: War Costs $74* %3Chttps://thinkbynumbers.org/military/war/the-economic-case-for-peace-a-comprehensive-financial-analysis/%3E (2021)
Lost human capital from war: $300B annually (economic impact of losing skilled/productive individuals to conflict) Broader conflict/violence cost: $14T/year globally 1.4M violent deaths/year; conflict holds back economic development, causes instability, widens inequality, erodes human capital 2002: 48.4M DALYs lost from 1.6M violence deaths = $151B economic value (2000 USD) Economic toll includes: commodity prices, inflation, supply chain disruption, declining output, lost human capital Additional sources: <https://thinkbynumbers.org/military/war/the-economic-case-for-peace-a-comprehensive-financial-analysis/> | https://www.weforum.org/stories/2021/02/war-violence-costs-each-human-5-a-day/ | https://pubmed.ncbi.nlm.nih.gov/19115548/

50. PubMed. Psychological impact of war cost ($100B annually). *PubMed: Economic Burden of PTSD* https://pubmed.ncbi.nlm.nih.gov/35485933/
PTSD economic burden (2018 U.S.): $232.2B total ($189.5B civilian, $42.7B military) Civilian costs driven by: Direct healthcare ($66B), unemployment ($42.7B) Military costs driven by: Disability ($17.8B), direct healthcare ($10.1B) Exceeds costs of other mental health conditions (anxiety, depression) War-exposed populations: 2-3X higher rates of anxiety, depression, PTSD; women and children most vulnerable Note: Actual burden $232B, significantly higher than "$100B" claimed Additional sources: https://pubmed.ncbi.nlm.nih.gov/35485933/ | https://news.va.gov/103611/study-national-economic-burden-of-ptsd-staggering/ | https://pmc.ncbi.nlm.nih.gov/articles/PMC9957523/

51. CGDev. UNHCR average refugee support cost. *CGDev* https://www.cgdev.org/blog/costs-hosting-refugees-oecd-countries-and-why-uk-outlier (2024)
The average cost of supporting a refugee is $1,384 per year. This represents total host country costs (housing, healthcare, education, security). OECD countries average $6,100 per refugee (mean 2022-2023), with developing countries spending $700-1,000. Global weighted average of $1,384 is reasonable given that 75-85% of refugees are in low/middle-income countries. Additional sources: https://www.cgdev.org/blog/costs-hosting-refugees-oecd-countries-and-why-uk-outlier | https://www.unhcr.org/sites/default/files/2024-11/UNHCR-WB-global-cost-of-refugee-inclusion-in-host-country-health-systems.pdf

52. Bank, W. World bank trade disruption cost from conflict. *World Bank* https://www.worldbank.org/en/topic/trade/publication/trading-away-from-conflict
Estimated $616B annual cost from conflict-related trade disruption. World Bank research shows civil war costs an average developing country 30 years of GDP growth, with 20 years needed for trade to return to pre-war levels. Trade disputes analysis shows tariff escalation could reduce global exports by up to $674 billion. Additional sources: https://www.worldbank.org/en/topic/trade/publication/trading-away-from-conflict | https://www.nber.org/papers/w11565 | http://blogs.worldbank.org/en/trade/impacts-global-trade-and-income-current-trade-disputes

53. VA. Veteran healthcare cost projections. *VA* https://department.va.gov/wp-content/uploads/2025/06/2026-Budget-in-Brief.pdf (2026)
VA budget: $441.3B requested for FY 2026 (10% increase). Disability compensation: $165.6B in FY 2024 for 6.7M veterans. PACT Act projected to increase spending by $300B between 2022-2031. Costs under Toxic Exposures Fund: $20B (2024), $30.4B (2025), $52.6B (2026). Additional sources: https://department.va.gov/wp-content/uploads/2025/06/2026-Budget-in-Brief.pdf | https://www.cbo.gov/publication/45615 | https://www.legion.org/information-center/news/veterans-healthcare/2025/june/va-budget-tops-400-billion-for-2025-from-higher-spending-on-mandated-benefits-medical-care

54. IQVIA Institute for Human Data Science. The global use of medicines 2024: Outlook to 2028. *IQVIA Institute Report* https://www.iqvia.com/insights/the-iqvia-institute/reports-and-publications/reports/the-global-use-of-medicines-2024-outlook-to-2028 (2024)
Global days of therapy reached 1.8 trillion in 2019 (234 defined daily doses per person). Diabetes, respiratory, CVD, and cancer account for 71 percent of medicine use. Projected to reach 3.8 trillion DDDs by 2028.

55. size, D. from global market & ratios, public/private funding. Private industry clinical trial spending.
Private pharmaceutical and biotech industry spends approximately $75-90 billion annually on clinical trials, representing roughly 90% of global clinical trial spending.

56. IHME Global Burden of Disease (2.55B DALYs), C. from & GDP per capita valuation, global. $109 trillion annual global disease burden.
*The global economic burden of disease, including direct healthcare costs (*8.2*trillion)andlostproductivity(*100.9 *trillion from 2.55 billion DALYs* × 39, 570*perDALY*), *totalsapproximately*109.1 *trillion annually.*
57. Trials, A. C. Global government spending on interventional clinical trials: $3-6 billion/year. *Applied Clinical Trials* https://www.appliedclinicaltrialsonline.com/view/sizing-clinical-research-market
Estimated range based on NIH ($0.8-5.6B), NIHR ($1.6B total budget), and EU funding. ($1.3B/year). Roughly 5-10% of global market. Additional sources: https://www.appliedclinicaltrialsonline.com/view/sizing-clinical-research-market | https://www.thelancet.com/journals/langlo/article/PIIS2214-109X(20
58. Suisse/UBS, C. Credit suisse global wealth report 2023. *Credit Suisse/UBS* https://www.ubs.com/global/en/family-office-uhnw/reports/global-wealth-report-2023.html (2023)
Total global household wealth: USD 454.4 trillion (2022) Wealth declined by USD 11.3 trillion. (-2.4%) in 2022, first decline since 2008 Wealth per adult: USD 84,718 Additional sources: https://www.ubs.com/global/en/family-office-uhnw/reports/global-wealth-report-2023.html
59. budgets:, S. component country. Global government medical research spending ($67.5B, 2023–2024). *See component country budgets: NIH Budget* #nih-budget-fy2025.
60. SIPRI. Global military spending ($2.72T, 2024). *SIPRI* https://www.sipri.org/publications/2025/sipri-fact-sheets/trends-world-military-expenditure-2024 (2025).
61. budgets, E. from major foundation & activities. Nonprofit clinical trial funding estimate.
Nonprofit foundations spend an estimated $2-5 billion annually on clinical trials globally, representing approximately 2-5% of total clinical trial spending.
62. IQVIA, I. reports: Global pharmaceutical r&d spending.
Total global pharmaceutical R&D spending is approximately $300 billion annually. Clinical trials represent 15-20% of this total ($45-60B), with the remainder going to drug discovery, preclinical research, regulatory affairs, and manufacturing development.
63. UN. Global population reaches 8 billion. *UN: World Population 8 Billion Nov 15 2022* https://www.un.org/en/desa/world-population-reach-8-billion-15-november-2022 (2022)
Milestone: November 15, 2022 (UN World Population Prospects 2022) Day of. Eight Billion" designated by UN Added 1 billion people in just 11 years (2011-2022) Growth rate: Slowest since 1950; fell under 1% in 2020 Future: 15 years to reach 9B (2037); projected peak 10.4B in 2080s Projections: 8.5B (2030), 9.7B (2050), 10.4B (2080-2100 plateau) Note: Milestone reached Nov 2022. Population growth slowing; will take longer to add next billion (15 years vs 11 years) Additional sources: https://www.un.org/en/desa/world-population-reach-8-billion-15-november-2022 | https://www.un.org/en/dayof8billion | https://en.wikipedia.org/wiki/Day_of_Eight_Billion

64. School, H. K. 3.5% participation tipping point. *Harvard Kennedy School* https://www.hks.harvard.edu/centers/carr/publications/35-rule-how-small-minority-can-change-world (2020)
The research found that nonviolent campaigns were twice as likely to succeed as violent ones, and once 3.5% of the population were involved, they were always successful. Chenoweth and Maria Stephan studied the success rates of civil resistance efforts from 1900 to 2006, finding that nonviolent movements attracted, on average, four times as many participants as violent movements and were more likely to succeed. Key finding: Every campaign that mobilized at least 3.5% of the population in sustained protest was successful (in their 1900-2006 dataset) Note: The 3.5% figure is a descriptive statistic from historical analysis, not a guaranteed threshold. One exception (Bahrain 2011-2014 with 6%+ participation) has been identified. The rule applies to regime change, not policy change in democracies. Additional sources: https://www.hks.harvard.edu/centers/carr/publications/35-rule-how-small-minority-can-change-world | https://www.hks.harvard.edu/sites/default/files/2024-05/Erica%20Chenoweth_2020-005.pdf | https://www.bbc.com/future/article/20190513-it-only-takes-35-of-people-to-change-the-world | https://en.wikipedia.org/wiki/3.5%25_rule

65. NHGRI. Human genome project and CRISPR discovery. *NHGRI* https://www.genome.gov/11006929/2003-release-international-consortium-completes-hgp (2003)
Your DNA is 3 billion base pairs Read the entire code (Human Genome Project, completed 2003) Learned to edit it (CRISPR, discovered 2012) Additional sources: https://www.genome.gov/11006929/2003-release-international-consortium-completes-hgp | https://www.nobelprize.org/prizes/chemistry/2020/press-release/

66. PMC. Only 12% of human interactome targeted. *PMC* https://pmc.ncbi.nlm.nih.gov/articles/PMC10749231/ (2023)
Mapping 350,000+ clinical trials showed that only 12% of the human interactome has ever been targeted by drugs. Additional sources: https://pmc.ncbi.nlm.nih.gov/articles/PMC10749231/

67. WHO. ICD-10 code count (14,000). *WHO* https://icd.who.int/browse10/2019/en (2019)
The ICD-10 classification contains approximately 14,000 codes for diseases, signs and symptoms. Additional sources: https://icd.who.int/browse10/2019/en

68. Wikipedia. Longevity escape velocity (LEV) - maximum human life extension potential. *Wikipedia: Longevity Escape Velocity* https://en.wikipedia.org/wiki/Longevity_escape_velocity
Longevity escape velocity: Hypothetical point where medical advances extend life expectancy. faster than time passes Term coined by Aubrey de Grey (biogerontologist) in 2004 paper; concept from David Gobel (Methuselah Foundation) Current progress: Science adds 3 months to lifespan per year; LEV requires adding >1 year per year Sinclair (Harvard): "There is no biological upper limit to age" - first person to live to 150 may already be born De Grey: 50% chance of reaching LEV by mid-to-late 2030s; SENS approach = damage repair rather than slowing damage Kurzweil (2024): LEV by 2029-2035, AI will simulate biological processes to accelerate solutions George Church: LEV "in a decade or two" via age-reversal clinical trials Natural lifespan cap: 120-150 years (Jeanne Calment record: 122); engineering approach could bypass via damage repair Key mechanisms: Epigenetic reprogramming, senolytic drugs, stem cell therapy, gene therapy, AI-driven drug discovery Current record: Jeanne Calment (122 years, 164 days) - record unbroken since 1997 Note: LEV is theoretical but increasingly plausible given demonstrated age reversal in mice (109% lifespan extension) and human cells (30-year epigenetic age reversal) Additional sources: https://en.wikipedia.org/wiki/Longevity_escape_velocity | https://pmc.ncbi.nlm.nih.gov/articles/PMC423155/ | https://www.popularmechanics.com/science/a36712084/can-science-cure-death-longevity/ | https://www.diamandis.com/blog/longevity-escape-velocity

69. OpenSecrets. Lobbyist statistics for washington d.c. *OpenSecrets: Lobbying in US* https://en.wikipedia.org/wiki/Lobbying_in_the_United_States
Registered lobbyists: Over 12,000 (some estimates); 12,281 registered (2013) Former government employees as lobbyists: 2,200+ former federal employees (1998-2004), including 273 former White House staffers, 250 former Congress members & agency heads Congressional revolving door: 43% (86 of 198) lawmakers who left 1998-2004 became lobbyists; currently 59% leaving to private sector work for lobbying/consulting firms/trade groups Executive branch: 8% were registered lobbyists at some point before/after government service Additional sources: https://en.wikipedia.org/wiki/Lobbying_in_the_United_States | https://www.opensecrets.org/revolving-door | https://www.citizen.org/article/revolving-congress/ | https://www.propublica.org/article/we-found-a-staggering-281-lobbyists-whove-worked-in-the-trump-administration

70. Vaccines, M. Measles vaccination ROI. *MDPI Vaccines* https://www.mdpi.com/2076-393X/12/11/1210 (2024)
Single measles vaccination: 167:1 benefit-cost ratio. MMR (measles-mumps-rubella) vaccination: 14:1 ROI. Historical US elimination efforts (1966-1974): benefit-cost ratio of 10.3:1 with net benefits exceeding USD 1.1 billion (1972 dollars, or USD 8.0 billion in 2023 dollars). 2-dose MMR programs show direct benefit/cost ratio of 14.2 with net savings of $5.3 billion, and 26.0 from societal perspectives with net savings of $11.6 billion. Additional sources: https://www.mdpi.com/2076-393X/12/11/1210 | https://www.tandfonline.com/doi/full/10.1080/14760584.2024.2367451

71. Organization, W. H. Mental health global burden. *World Health Organization* https://www.who.int/news/item/28-09-2001-the-world-health-report-2001-mental-disorders-affect-one-in-four-people (2022)
One in four people in the world will be affected by mental or neurological disorders at some point in their lives, representing [approximately] 30% of the global burden of disease. Additional sources: https://www.who.int/news/item/28-09-2001-the-world-health-report-2001-mental-disorders-affect-one-in-four-people

72. Institute, S. I. P. R. Trends in world military expenditure, 2023. (2024).

73. Orphanet Journal of Rare Diseases (2024), C. from. Diseases getting first effective treatment each year. *Calculated from Orphanet Journal of Rare Diseases (2024)* https://ojrd.biomedcentral.com/articles/10.1186/s13023-024-03398-1 (2024)
Under the current system, approximately 10-15 diseases per year receive their FIRST effective treatment. Calculation: 5% of 7,000 rare diseases (350) have FDA-approved treatment, accumulated over 40 years of the Orphan Drug Act = 9 rare diseases/year. Adding 5-10 non-rare diseases that get first treatments yields 10-20 total. FDA approves 50 drugs/year, but many are for diseases that already have treatments (me-too drugs, second-line therapies). Only 15 represent truly FIRST treatments for previously untreatable conditions.

74. NIH. NIH budget (FY 2025). *NIH* https://www.nih.gov/about-nih/organization/budget (2024)
The budget total of 47.7*billionalsoincludes*1.412 *billion derived from PHS Evaluation financing... Additional sources: https://www.nih.gov/about-nih/organization/budget | https://officeofbudget.od.nih.gov/*

75. al., B. et. NIH spending on clinical trials: 3.3%. *Bentley et al.* https://www.fiercebiotech.com/biotech/nih-spending-clinical-trials-reached-81b-over-decade (2023)
NIH spent $8.1 billion on clinical trials for approved drugs (2010-2019), representing 3.3% of relevant NIH spending. Additional sources: https://www.fiercebiotech.com/biotech/nih-spending-clinical-trials-reached-81b-over-decade | https://www.fiercebiotech.com/biotech/nih-spending-clinical-trials-reached-81b-over-decade

76. PMC. Standard medical research ROI ($20k-$100k/QALY). *PMC: Cost-effectiveness Thresholds Used by Study Authors* https://pmc.ncbi.nlm.nih.gov/articles/PMC10114019/ (1990)
Typical cost-effectiveness thresholds for medical interventions in rich countries range from $50,000 to $150,000 per QALY. The Institute for Clinical and Economic Review (ICER) uses a $100,000-$150,000/QALY threshold for value-based pricing. Between 1990-2021, authors increasingly cited $100,000 (47% by 2020-21) or $150,000 (24% by 2020-21) per QALY as benchmarks for cost-effectiveness. Additional sources: https://pmc.ncbi.nlm.nih.gov/articles/PMC10114019/ | https://icer.org/our-approach/methods-process/cost-effectiveness-the-qaly-and-the-evlyg/

77. Institute, M. RECOVERY trial 82× cost reduction. *Manhattan Institute: Slow Costly Trials* https://manhattan.institute/article/slow-costly-clinical-trials-drag-down-biomedical-breakthroughs
RECOVERY trial: \$500 per patient (\$20M for 48,000 patients = \$417/patient) Typical clinical trial: \$41,000 median per-patient cost Cost reduction: 80-82× cheaper (\$41,000 ÷ \$500 82×) Efficiency: \$50 per patient per answer (10 therapeutics tested, 4 effective) Dexamethasone estimated to save >630,000 lives Additional sources: https://manhattan.institute/article/slow-costly-clinical-trials-drag-down-biomedical-breakthroughs | https://pmc.ncbi.nlm.nih.gov/articles/PMC9293394/

78. Trials. Patient willingness to participate in clinical trials. *Trials: Patients' Willingness Survey* https://trialsjournal.biomedcentral.com/articles/10.1186/s13063-015-1105-3
Recent surveys: 49-51% willingness (2020-2022) - dramatic drop from 85% (2019). during COVID-19 pandemic Cancer patients when approached: 88% consented to trials (Royal Marsden Hospital) Study type variation: 44.8% willing for drug trial, 76.2% for diagnostic study Top motivation: "Learning more about my health/medical condition" (67.4%) Top barrier: "Worry about experiencing side effects" (52.6%) Additional sources: https://trialsjournal.biomedcentral.com/articles/10.1186/s13063-015-1105-3 | https://www.appliedclinicaltrialsonline.com/view/industry-forced-to-rethink-patient-participation-in-trials | https://pmc.ncbi.nlm.nih.gov/articles/PMC7183682/

79. CSDD, T. Cost of drug development.
Various estimates suggest \$1.0 - \$2.5 billion to bring a new drug from discovery through FDA approval, spread across 10 years. Tufts Center for the Study of Drug Development often cited for \$1.0 - \$2.6 billion/drug. Industry reports (IQVIA, Deloitte) also highlight \$2+ billion figures.

80. Health, V. in. Average lifetime revenue per successful drug. *Value in Health: Sales Revenues for New Therapeutic Agents02754-2/fulltext)* https://www.valueinhealthjournal.com/article/S1098-3015(24
Study of 361 FDA-approved drugs from 1995-2014 (median follow-up 13.2 years): Mean lifetime revenue: \$15.2 billion per drug Median lifetime revenue: \$6.7 billion per drug Revenue after 5 years: \$3.2 billion (mean) Revenue after 10 years: \$9.5 billion (mean) Revenue after 15 years: \$19.2 billion (mean) Distribution highly skewed: top 25 drugs (7%) accounted for 38% of total revenue (\$2.1T of \$5.5T) Additional sources: https://www.valueinhealthjournal.com/article/S1098-3015(24 | https://www.sciencedirect.com/science/article/pii/S1098301524027542

81. Lichtenberg, F. R. How many life-years have new drugs saved? A three-way fixed-effects analysis of 66 diseases in 27 countries, 2000-2013. *International Health* **11**, 403–416 (2019)
Using 3-way fixed-effects methodology (disease-country-year) across 66 diseases in 22 countries, this study estimates that drugs launched after 1981 saved 148.7 million life-years in 2013 alone. The regression coefficients for drug launches 0-11 years prior (beta=-0.031, SE=0.008) and 12+ years prior (beta=-0.057, SE=0.013) on years of life lost are highly significant (p<0.0001). Confidence interval for life-years saved: 79.4M-239.8M (95 percent CI) based on propagated standard errors from Table 2.

82. Deloitte. Pharmaceutical r&d return on investment (ROI). *Deloitte: Measuring Pharmaceutical Innovation 2025* https://www.deloitte.com/ch/en/Industries/life-sciences-health-care/research/measuring-return-from-pharmaceutical-innovation.html (2025)
Deloitte's annual study of top 20 pharma companies by R&D spend (2010-2024): 2024 ROI: 5.9% (second year of growth after decade of decline) 2023 ROI: 4.3% (estimated from trend) 2022 ROI: 1.2% (historic low since study began, 13-year low) 2021 ROI: 6.8% (record high, inflated by COVID-19 vaccines/treatments) Long-term trend: Declining for over a decade before 2023 recovery Average R&D cost per asset: $2.3B (2022), $2.23B (2024) These returns (1.2-5.9% range) fall far below typical corporate ROI targets (15-20%) Additional sources: https://www.deloitte.com/ch/en/Industries/life-sciences-health-care/research/measuring-return-from-pharmaceutical-innovation.html | https://www.prnewswire.com/news-releases/deloittes-13th-annual-pharmaceutical-innovation-report-pharma-rd-return-on-investment-falls-in-post-pandemic-market-301738807.html | https://hitconsultant.net/2023/02/16/pharma-rd-roi-falls-to-lowest-level-in-13-years/
83. Discovery, N. R. D. Drug trial success rate from phase i to approval. *Nature Reviews Drug Discovery: Clinical Success Rates* https://www.nature.com/articles/nrd.2016.136 (2016)
Overall Phase I to approval: 10-12.8% (conventional wisdom 10%, studies show 12.8%). Recent decline: Average LOA now 6.7% for Phase I (2014-2023 data) Leading pharma companies: 14.3% average LOA (range 8-23%) Varies by therapeutic area: Oncology 3.4%, CNS/cardiovascular lowest at Phase III Phase-specific success: Phase I 47-54%, Phase II 28-34%, Phase III 55-70% Note: 12% figure accurate for historical average. Recent data shows decline to 6.7%, with Phase II as primary attrition point (28% success) Additional sources: https://www.nature.com/articles/nrd.2016.136 | https://pmc.ncbi.nlm.nih.gov/articles/PMC6409418/ | https://academic.oup.com/biostatistics/article/20/2/273/4817524
84. SofproMed. Phase 3 cost per trial range. *SofproMed* https://www.sofpromed.com/how-much-does-a-clinical-trial-cost
Phase 3 clinical trials cost between $20 million and $282 million per trial, with significant. variation by therapeutic area and trial complexity. Additional sources: https://www.sofpromed.com/how-much-does-a-clinical-trial-cost | https://www.cbo.gov/publication/57126
85. PMC. Pragmatic trial cost per patient (median $97). *PMC: Costs of Pragmatic Clinical Trials* https://pmc.ncbi.nlm.nih.gov/articles/PMC6508852/
The median cost per participant was $97 (IQR $19–$478), based on 2015 dollars. Systematic. review of 64 embedded pragmatic clinical trials. 25% of trials cost <$19/patient; 10 trials exceeded $1,000/patient. U.S. studies median $187 vs non-U.S. median $27. Additional sources: https://pmc.ncbi.nlm.nih.gov/articles/PMC6508852/
86. WHO. Polio vaccination ROI. *WHO* https://www.who.int/news-room/feature-stories/detail/sustaining-polio-investments-offers-a-high-return (2019)
For every dollar spent, the return on investment is nearly US$ 39." Total investment cost. of US$ 7.5 billion generates projected economic and social benefits of US$ 289.2 billion from sustaining polio assets and integrating them into expanded immunization, surveillance and emergency response programmes across 8 priority countries (Afghanistan, Iraq, Libya, Pakistan, Somalia, Sudan, Syria, Yemen). Additional sources: https://www.who.int/news-room/feature-stories/detail/sustaining-polio-investments-offers-a-high-return
87. Olson, M. Big bills left on the sidewalk: Why some nations are rich, and others poor. *Journal of Economic Perspectives* **10**, 3–24 (1996)
Differences between rich and poor countries are primarily due to institutions and policies, not factors of production.

88. Hayek, F. A. The use of knowledge in society. *American Economic Review* **35**, 519–530 (1945)
The knowledge of the circumstances which we must make use of never exists in concentrated or integrated form but solely as dispersed bits of incomplete and frequently contradictory knowledge which all the separate individuals possess.

89. Kydland, F. E. & Prescott, E. C. Rules rather than discretion: The inconsistency of optimal plans. *Journal of Political Economy* **85**, 473–492 (1977)
Time-inconsistency describes situations where, with the passing of time, policies that were determined to be optimal yesterday are no longer perceived to be optimal today and are not implemented... This insight shifted the focus of policy analysis from the study of individual policy decisions to the design of institutions that mitigate the time consistency problem.

90. ICRC. International campaign to ban landmines (ICBL) - ottawa treaty (1997). *ICRC* https://www.icrc.org/en/doc/resources/documents/article/other/57jpjn.htm (1997)
ICBL: Founded 1992 by 6 NGOs (Handicap International, Human Rights Watch, Medico. International, Mines Advisory Group, Physicians for Human Rights, Vietnam Veterans of America Foundation) Started with ONE staff member: Jody Williams as founding coordinator Grew to 1,000+ organizations in 60 countries by 1997 Ottawa Process: 14 months (October 1996 - December 1997) Convention signed by 122 states on December 3, 1997; entered into force March 1, 1999 Achievement: Nobel Peace Prize 1997 (shared by ICBL and Jody Williams) Government funding context: Canada established $100M CAD Canadian Landmine Fund over 10 years (1997); International donors provided $169M in 1997 for mine action (up from $100M in 1996) Additional sources: https://www.icrc.org/en/doc/resources/documents/article/other/57jpjn.htm | https://en.wikipedia.org/wiki/International_Campaign_to_Ban_Landmines | https://www.nobelprize.org/prizes/peace/1997/summary/ | https://un.org/press/en/1999/19990520.MINES.BRF.html | https://www.the-monitor.org/en-gb/reports/2003/landmine-monitor-2003/mine-action-funding.aspx

91. OpenSecrets. Revolving door: Former members of congress. (2024)
388 former members of Congress are registered as lobbyists. Nearly 5,400 former congressional. staffers have left Capitol Hill to become federal lobbyists in the past 10 years. Additional sources: https://www.opensecrets.org/revolving-door

92. Kinch, M. S. & Griesenauer, R. H. Lost medicines: A longer view of the pharmaceutical industry with the potential to reinvigorate discovery. *Drug Discovery Today* **24**, 875–880 (2019)
Research identified 1,600+ medicines available in 1962. The 1950s represented industry high-water mark with >30 new products in five of ten years; this rate would not be replicated until late 1990s. More than half (880) of these medicines were lost following implementation of Kefauver-Harris Amendment. The peak of 1962 would not be seen again until early 21st century. By 2016 number of organizations actively involved in R&D at level not seen since 1914.

93. Wikipedia. US military spending reduction after WWII. *Wikipedia* https://en.wikipedia.org/wiki/Demobilization_of_United_States_Armed_Forces_after_World_War_II (2020)
Peaking at over $81 billion in 1945, the U.S. military budget plummeted to approximately $13 billion by 1948, representing an 84% decrease. The number of personnel was reduced almost 90%, from more than 12 million to about 1.5 million between mid-1945 and mid-1947. Defense spending exceeded 41 percent of GDP in 1945. After World War II, the US reduced military spending to 7.2 percent of GDP by 1948. Defense spending doubled from the 1948 low to 15 percent at the height of the Korean War in 1953. Additional sources: https://en.wikipedia.org/wiki/Demobilization_of_United_States_Armed_Forces_after_World_War_II | https://www.americanprogress.org/article/a-historical-perspective-on-military-budgets/ | https://www.stlouisfed.org/on-the-economy/2020/february/war-highest-military-spending-measured | https://www.usgovernmentspending.com/defense_spending_history

94. Baily, M. N. Pre-1962 drug development costs (baily 1972). *Baily (1972)* https://samizdathealth.org/wp-content/uploads/2020/12/hlthaff.1.2.6.pdf (1972)
Pre-1962: Average cost per new chemical entity (NCE) was $6.5 million (1980 dollars). Inflation-adjusted to 2024 dollars: $6.5M (1980) $22.5M (2024), using CPI multiplier of 3.46× Real cost increase (inflation-adjusted): $22.5M (pre-1962) → $2,600M (2024) = 116× increase Note: This represents the most comprehensive academic estimate of pre-1962 drug development costs based on empirical industry data Additional sources: https://samizdathealth.org/wp-content/uploads/2020/12/hlthaff.1.2.6.pdf

95. Numbers, T. by. Pre-1962 physician-led clinical trials. *Think by Numbers: How Many Lives Does FDA Save?* https://thinkbynumbers.org/health/how-many-net-lives-does-the-fda-save/ (1966)
Pre-1962: Physicians could report real-world evidence directly 1962 Drug Amendments replaced "premarket notification" with "premarket approval", requiring extensive efficacy testing Impact: New regulatory clampdown reduced new treatment production by 70%; lifespan growth declined from 4 years/decade to 2 years/decade Drug Efficacy Study Implementation (DESI): NAS/NRC evaluated 3,400+ drugs approved 1938-1962 for safety only; reviewed >3,000 products, >16,000 therapeutic claims FDA has had authority to accept real-world evidence since 1962, clarified by 21st Century Cures Act (2016) Note: Specific "144,000 physicians" figure not verified in sources Additional sources: https://thinkbynumbers.org/health/how-many-net-lives-does-the-fda-save/ | https://www.fda.gov/drugs/enforcement-activities-fda/drug-efficacy-study-implementation-desi | http://www.nasonline.org/about-nas/history/archives/collections/des-1966-1969-1.html

96. GAO. 95% of diseases have 0 FDA-approved treatments. *GAO* https://www.gao.gov/products/gao-25-106774 (2025)
95% of diseases have no treatment Additional sources: https://www.gao.gov/products/gao-25-106774 | https://globalgenes.org/rare-disease-facts/

97. Oren Cass, M. I. RECOVERY trial cost per patient. *Oren Cass* https://manhattan.institute/article/slow-costly-clinical-trials-drag-down-biomedical-breakthroughs (2023)
The RECOVERY trial, for example, cost only about. 500*perpatient...Bycontrast,themedianper−patientcostofapivotaltrialforanewtherapeuticisaround*41,00
Additional sources: https://manhattan.institute/article/slow-costly-clinical-trials-drag-down-biomedical-breakthroughs

98. al., N. E. Á. et. RECOVERY trial global lives saved (1 million). *NHS England: 1 Million Lives Saved* https://www.england.nhs.uk/2021/03/covid-treatment-developed-in-the-nhs-saves-a-million-lives/ (2021)
Dexamethasone saved 1 million lives worldwide (NHS England estimate, March 2021, 9 months after discovery). UK alone: 22,000 lives saved. Methodology: Águas et al. Nature Communications 2021 estimated 650,000 lives (range: 240,000-1,400,000) for July-December 2020 alone, based on RECOVERY trial mortality reductions (36% for ventilated, 18% for oxygen-only patients) applied to global COVID hospitalizations. June 2020 announcement: Dexamethasone reduced deaths by up to 1/3 (ventilated patients), 1/5 (oxygen patients). Impact immediate: Adopted into standard care globally within hours of announcement. Additional sources: https://www.england.nhs.uk/2021/03/covid-treatment-developed-in-the-nhs-saves-a-million-lives/ | https://www.nature.com/articles/s41467-021-21134-2 | https://pharmaceutical-journal.com/article/news/steroid-has-saved-the-lives-of-one-million-covid-19-patients-worldwide-figures-show | https://www.recoverytrial.net/news/recovery-trial-celebrates-two-year-anniversary-of-life-saving-dexamethasone-result
99. Museum, N. S. 11. M. &. September 11 attack facts. (2024)
2,977 people were killed in the September 11, 2001 attacks: 2,753 at the World Trade Center, 184 at the Pentagon, and 40 passengers and crew on United Flight 93 in Shanksville, Pennsylvania.
100. Bank, W. World bank singapore economic data. *World Bank* https://data.worldbank.org/country/singapore (2024)
Singapore GDP per capita (2023): $82,000 - among highest in the world Government spending: 15% of GDP (vs US 38%) Life expectancy: 84.1 years (vs US 77.5 years) Singapore demonstrates that low government spending can coexist with excellent outcomes Additional sources: https://data.worldbank.org/country/singapore
101. Fund, I. M. IMF singapore government spending data. (2024)
Singapore government spending is approximately 15% of GDP This is 23 percentage points lower than the United States (38%) Despite lower spending, Singapore achieves excellent outcomes: - Life expectancy: 84.1 years (vs US 77.5) - Low crime, world-class infrastructure, AAA credit rating Additional sources: https://www.imf.org/en/Countries/SGP
102. Organization, W. H. WHO life expectancy data by country. (2024)
Life expectancy at birth varies significantly among developed nations: Switzerland: 84.0 years (2023) Singapore: 84.1 years (2023) Japan: 84.3 years (2023) United States: 77.5 years (2023) - 6.5 years below Switzerland, Singapore Global average: 73 years Note: US spends more per capita on healthcare than any other nation, yet achieves lower life expectancy Additional sources: https://www.who.int/data/gho/data/themes/mortality-and-global-health-estimates/ghe-life-expectancy-and-healthy-life-expectancy
103. CSIS. Smallpox eradication ROI. *CSIS* https://www.csis.org/analysis/smallpox-eradication-model-global-cooperation.

104. PMC. Contribution of smoking reduction to life expectancy gains. *PMC: Benefits Smoking Cessation Longevity* https://www.ncbi.nlm.nih.gov/pmc/articles/PMC1447499/ (2012)
Population-level: Up to 14% (9% men, 14% women) of total life expectancy gain since. 1960 due to tobacco control efforts Individual cessation benefits: Quitting at age 35 adds 6.9-8.5 years (men), 6.1-7.7 years (women) vs continuing smokers By cessation age: Age 25-34 = 10 years gained; age 35-44 = 9 years; age 45-54 = 6 years; age 65 = 2.0 years (men), 3.7 years (women) Cessation before age 40: Reduces death risk by 90% Long-term cessation: 10+ years yields survival comparable to never smokers, averts 10 years of life lost Recent cessation: <3 years averts 5 years of life lost Additional sources: https://www.ncbi.nlm.nih.gov/pmc/articles/PMC1447499/ | https://www.cdc.gov/pcd/issues/2012/11_0295.htm | https://www.ajpmonline.org/article/S0749-3797(24 | https://www.nejm.org/doi/full/10.1056/NEJMsa1211128

105. ICER. Value per QALY (standard economic value). *ICER* https://icer.org/wp-content/uploads/2024/02/Reference-Case-4.3.25.pdf (2024)
Standard economic value per QALY: $100,000–$150,000. This is the US and global standard. willingness-to-pay threshold for interventions that add costs. Dominant interventions (those that save money while improving health) are favorable regardless of this threshold. Additional sources: https://icer.org/wp-content/uploads/2024/02/Reference-Case-4.3.25.pdf

106. GAO. Annual cost of u.s. Sugar subsidies. *GAO: Sugar Program* https://www.gao.gov/products/gao-24-106144
Consumer costs: $2.5-3.5 billion per year (GAO estimate) Net economic cost: $1. billion per year 2022: US consumers paid 2X world price for sugar Program costs $3-4 billion/year but no federal budget impact (costs passed directly to consumers via higher prices) Employment impact: 10,000-20,000 manufacturing jobs lost annually in sugar-reliant industries (confectionery, etc.) Multiple studies confirm: Sweetener Users Association ($2.9-3.5B), AEI ($2.4B consumer cost), Beghin & Elobeid ($2.9-3.5B consumer surplus) Additional sources: https://www.gao.gov/products/gao-24-106144 | https://www.heritage.org/agriculture/report/the-us-sugar-program-bad-consumers-bad-agriculture-and-bad-america | https://www.aei.org/articles/the-u-s-spends-4-billion-a-year-subsidizing-stalinist-style-domestic-sugar-production/

107. Bank, W. Swiss military budget as percentage of GDP. *World Bank: Military Expenditure* https://data.worldbank.org/indicator/MS.MIL.XPND.GD.ZS?locations=CH
2023: 0.70272% of GDP (World Bank) 2024: CHF 5.95 billion official military spending. When including militia system costs: 1% GDP (CHF 8.75B) Comparison: Near bottom in Europe; only Ireland, Malta, Moldova spend less (excluding microstates with no armies) Additional sources: https://data.worldbank.org/indicator/MS.MIL.XPND.GD.ZS?locations=CH | https://www.avenir-suisse.ch/en/blog-defence-spending-switzerland-is-in-better-shape-than-it-seems/ | https://tradingeconomics.com/switzerland/military-expenditure-percent-of-gdp-wb-data.html

108. Bank, W. Switzerland vs. US GDP per capita comparison. *World Bank: Switzerland GDP Per Capita* https://data.worldbank.org/indicator/NY.GDP.PCAP.CD?locations=CH
2024 GDP per capita (PPP-adjusted): Switzerland $93,819 vs United States $75,492 Switzer-. land's GDP per capita 24% higher than US when adjusted for purchasing power parity Nominal 2024: Switzerland $103,670 vs US $85,810 Additional sources: https://data.worldbank.org/indicator/NY.GDP.PCAP.CD?locations=CH | https://tradingeconomics.com/switzerland/gdp-per-capita-ppp | https://www.theglobaleconomy.com/USA/gdp_per_capita_ppp/

109. Economic Co-operation, O. for & Development. OECD government spending as percentage of GDP. (2024)
OECD government spending data shows significant variation among developed nations: United. States: 38.0% of GDP (2023) Switzerland: 35.0% of GDP - 3 percentage points lower than US Singapore: 15.0% of GDP - 23 percentage points lower than US (per IMF data) OECD average: approximately 40% of GDP Additional sources: https://data.oecd.org/gga/general-government-spending.htm

110. Economic Co-operation, O. for & Development. OECD median household income comparison. (2024)
Median household disposable income varies significantly across OECD nations: United. States: $77,500 (2023) Switzerland: $55,000 PPP-adjusted (lower nominal but comparable purchasing power) Singapore: $75,000 PPP-adjusted Additional sources: https://data.oecd.org/hha/household-disposable-income.htm

111. Institute, C. Chance of dying from terrorism statistic. *Cato Institute: Terrorism and Immigration Risk Analysis* https://www.cato.org/policy-analysis/terrorism-immigration-risk-analysis
Chance of American dying in foreign-born terrorist attack: 1 in 3.6 million per year (1975-. 2015) Including 9/11 deaths; annual murder rate is 253x higher than terrorism death rate More likely to die from lightning strike than foreign terrorism Note: Comprehensive 41-year study shows terrorism risk is extremely low compared to everyday dangers Additional sources: https://www.cato.org/policy-analysis/terrorism-immigration-risk-analysis | https://www.nbcnews.com/news/us-news/you-re-more-likely-die-choking-be-killed-foreign-terrorists-n715141

112. Wikipedia. Thalidomide scandal: Worldwide cases and mortality. *Wikipedia* https://en.wikipedia.org/wiki/Thalidomide_scandal
The total number of embryos affected by the use of thalidomide during pregnancy is esti-. mated at 10,000, of whom about 40% died around the time of birth. More than 10,000 children in 46 countries were born with deformities such as phocomelia. Additional sources: https://en.wikipedia.org/wiki/Thalidomide_scandal

113. One, P. Health and quality of life of thalidomide survivors as they age. *PLOS One* https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0210222 (2019)
Study of thalidomide survivors documenting ongoing disability impacts, quality of life, and. long-term health outcomes. Survivors (now in their 60s) continue to experience significant disability from limb deformities, organ damage, and other effects. Additional sources: https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0210222

114. Bureau, U. C. Historical world population estimates. *US Census Bureau* https://www.census.gov/data/tables/time-series/demo/international-programs/historical-est-worldpop.html
US Census Bureau historical estimates of world population by country and region. (1950-2050). US population in 1960: 180 million of 3 billion worldwide (6%). Additional sources: https://www.census.gov/data/tables/time-series/demo/international-programs/historical-est-worldpop.html

115. NCBI, F. S. via. Trial costs, FDA study. *FDA Study via NCBI* https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6248200/
Overall, the 138 clinical trials had an estimated median (IQR) cost of 19.0*million*(*12.2. million*-33.1*million*)...*Theclinicaltrialscostamedian*(*IQR*)*of41,117* (31,802−*82,362) per patient. Additional sources: https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6248200/*

116. GBD 2019 Diseases and Injuries Collaborators. Global burden of disease study 2019: Disability weights. *The Lancet* **396**, 1204–1222 (2020)
Disability weights for 235 health states used in Global Burden of Disease calculations. Weights range from 0 (perfect health) to 1 (death equivalent). Chronic conditions like diabetes (0.05-0.35), COPD (0.04-0.41), depression (0.15-0.66), and cardiovascular disease (0.04-0.57) show substantial variation by severity. Treatment typically reduces disability weights by 50-80 percent for manageable chronic conditions.

117. WHO. Annual global economic burden of alzheimer's and other dementias. *WHO: Dementia Fact Sheet* https://www.who.int/news-room/fact-sheets/detail/dementia (2019)
Global cost: $1.3 trillion (2019 WHO-commissioned study) 50% from informal caregivers. (family/friends, 5 hrs/day) 74% of costs in high-income countries despite 61% of patients in LMICs $818B (2010) → $1T (2018) → $1.3T (2019) - rapid growth Note: Costs increased 35% from 2010-2015 alone. Informal care represents massive hidden economic burden Additional sources: https://www.who.int/news-room/fact-sheets/detail/dementia | https://alz-journals.onlinelibrary.wiley.com/doi/10.1002/alz.12901

118. Oncology, J. Annual global economic burden of cancer. *JAMA Oncology: Global Cost 2020-2050* https://jamanetwork.com/journals/jamaoncology/fullarticle/2801798 (2020)
2020-2050 projection: $25.2 trillion total ($840B/year average) 2010 annual cost: $1.16. trillion (direct costs only) Recent estimate: $3 trillion/year (all costs included) Top 5 cancers: lung (15.4%), colon/rectum (10.9%), breast (7.7%), liver (6.5%), leukemia (6.3%) Note: China/US account for 45% of global burden; 75% of deaths in LMICs but only 50.0% of economic cost Additional sources: https://jamanetwork.com/journals/jamaoncology/fullarticle/2801798 | https://www.nature.com/articles/d41586-023-00634-9

119. CDC. U.s. Chronic disease healthcare spending. *CDC* https://www.cdc.gov/chronic-disease/data-research/facts-stats/index.html
Chronic diseases account for 90% of U.S. healthcare spending ($3.7T/year). Additional. sources: https://www.cdc.gov/chronic-disease/data-research/facts-stats/index.html

120. Care, D. Annual global economic burden of diabetes. *Diabetes Care: Global Economic Burden* https://diabetesjournals.org/care/article/41/5/963/36522/Global-Economic-Burden-of-Diabetes-in-Adults
2015: $1.3 trillion (1.8% of global GDP) 2030 projections: $2.1T-2.5T depending on. scenario IDF health expenditure: $760B (2019) → $845B (2045 projected) 2/3 direct medical costs ($857B), 1/3 indirect costs (lost productivity) Note: Costs growing rapidly; expected to exceed $2T by 2030 Additional sources: https://diabetesjournals.org/care/article/41/5/963/36522/Global-Economic-Burden-of-Diabetes-in-Adults | https://www.thelancet.com/journals/landia/article/PIIS2213-8587(17

121. World Bank, B. of E. A. US GDP 2024 ($28.78 trillion). *World Bank* https://data.worldbank.org/indicator/NY.GDP.MKTP.CD?locations=US (2024)
US GDP reached $28.78 trillion in 2024, representing approximately 26% of global. GDP. Additional sources: https://data.worldbank.org/indicator/NY.GDP.MKTP.CD?locations=US | https://www.bea.gov/news/2024/gross-domestic-product-fourth-quarter-and-year-2024-advance-estimate

122. Cardiology, I. J. of. Annual global economic burden of heart disease. *Int'l Journal of Cardiology: Global Heart Failure Burden02238-9/abstract)* https://www.internationaljournalofcardiology.com/article/S0167-5273(13 (2050)
Heart failure alone: $108 billion/year (2012 global analysis, 197 countries) US CVD: $555B. (2016) → projected $1.8T by 2050 LMICs total CVD loss: $3.7T cumulative (2011-2015, 5-year period) CVD is costliest disease category in most developed nations Note: No single $2.1T global figure found; estimates vary widely by scope and year Additional sources: https://www.internationaljournalofcardiology.com/article/S0167-5273(13 | https://www.ahajournals.org/doi/10.1161/CIR.0000000000001258

123. CSV, S. U. L. E. F. B. 1543-2019. US life expectancy growth 1880-1960: 3.82 years per decade. (2019)
Pre-1962: 3.82 years/decade Post-1962: 1.54 years/decade Reduction: 60% decline in. life expectancy growth rate Additional sources: https://ourworldindata.org/life-expectancy | https://www.mortality.org/ | https://www.cdc.gov/nchs/nvss/mortality_tables.htm

124. CSV, S. U. L. E. F. B. 1543-2019. Post-1962 slowdown in life expectancy gains. (2019)
Pre-1962 (1880-1960): 3.82 years/decade Post-1962 (1962-2019): 1.54 years/decade Reduc-. tion: 60% decline Temporal correlation: Slowdown occurred immediately after 1962 Kefauver-Harris Amendment See detailed calculation: [life-expectancy-increase-pre-1962](#life-expectancy-increase-pre-1962) Additional sources: https://ourworldindata.org/life-expectancy | https://www.mortality.org/ | https://www.cdc.gov/nchs/nvss/mortality_tables.htm

125. Disease Control, C. for & Prevention. US life expectancy 2023. (2024)
US life expectancy at birth was 77.5 years in 2023 Male life expectancy: 74.8 years Female. life expectancy: 80.2 years This is 6-7 years lower than peer developed nations despite higher healthcare spending Additional sources: https://www.cdc.gov/nchs/fastats/life-expectancy.htm

126. Bureau, U. C. US median household income 2023. (2024)
US median household income was $77,500 in 2023 Real median household income de-. clined 0.8% from 2022 Gini index: 0.467 (income inequality measure) Additional sources: https://www.census.gov/library/publications/2024/demo/p60-282.html

127. Statista. US military budget as percentage of GDP. *Statista* https://www.statista.com/statistics/262742/countries-with-the-highest-military-spending/ (2024)
U.S. military spending amounted to 3.5% of GDP in 2024. In 2024, the U.S. spent. nearly $1 trillion on its military budget, equal to 3.4% of GDP. Additional sources: https://www.statista.com/statistics/262742/countries-with-the-highest-military-spending/ | https://www.sipri.org/sites/default/files/2025-04/2504_fs_milex_2024.pdf

128. Bureau, U. C. Number of registered or eligible voters in the u.s. *US Census Bureau* https://www.census.gov/newsroom/press-releases/2025/2024-presidential-election-voting-registration-tables.html (2024)
73.6% (or 174 million people) of the citizen voting-age population was registered to vote. in 2024 (Census Bureau). More than 211 million citizens were active registered voters (86.6% of citizen voting age population) according to the Election Assistance Commission. Additional sources: https://www.census.gov/newsroom/press-releases/2025/2024-presidential-election-voting-registration-tables.html | https://www.eac.gov/news/2025/06/30/us-election-assistance-commission-releases-2024-election-administration-and-voting

129. Senate, U. S. Treaties. *U.S. Senate* https://www.senate.gov/about/powers-procedures/treaties.htm
The Constitution provides that the president 'shall have Power, by and with the Advice and Consent of the Senate, to make Treaties, provided two-thirds of the Senators present concur' (Article II, section 2). Treaties are formal agreements with foreign nations that require two-thirds Senate approval. 67 senators (two-thirds of 100) must vote to ratify a treaty for it to take effect. Additional sources: https://www.senate.gov/about/powers-procedures/treaties.htm

130. Commission, F. E. Statistical summary of 24-month campaign activity of the 2023-2024 election cycle. (2023)
Presidential candidates raised $2 billion; House and Senate candidates raised $3.8 billion and spent $3.7 billion; PACs raised $15.7 billion and spent $15.5 billion. Total federal campaign spending approximately $20 billion. Additional sources: https://www.fec.gov/updates/statistical-summary-of-24-month-campaign-activity-of-the-2023-2024-election-cycle/

131. OpenSecrets. Federal lobbying hit record $4.4 billion in 2024. (2024)
Total federal lobbying reached record $4.4 billion in 2024. The $150 million increase in lobbying continues an upward trend that began in 2016. Additional sources: https://www.opensecrets.org/news/2025/02/federal-lobbying-set-new-record-in-2024/

132. Kirk (2011), H. &. Valley of death in drug development. (2011)
The overall failure rate of drugs that passed into Phase 1 trials to final approval is 90%. This lack of translation from promising preclinical findings to success in human trials is known as the "valley of death." Estimated 30-50% of promising compounds never proceed to Phase 2/3 trials primarily due to funding barriers rather than scientific failure. The late-stage attrition rate for oncology drugs is as high as 70% in Phase II and 59% in Phase III trials.

133. DOT. DOT value of statistical life ($13.6M). *DOT: VSL Guidance 2024* https://www.transportation.gov/office-policy/transportation-policy/revised-departmental-guidance-on-valuation-of-a-statistical-life-in-economic-analysis (2024)
Current VSL (2024): $13.7 million (updated from $13.6M) Used in cost-benefit analyses for transportation regulations and infrastructure Methodology updated in 2013 guidance, adjusted annually for inflation and real income VSL represents aggregate willingness to pay for safety improvements that reduce fatalities by one Note: DOT has published VSL guidance periodically since 1993. Current $13.7M reflects 2024 inflation/income adjustments Additional sources: https://www.transportation.gov/office-policy/transportation-policy/revised-departmental-guidance-on-valuation-of-a-statistical-life-in-economic-analysis | https://www.transportation.gov/regulations/economic-values-used-in-analysis

134. ONE, P. Cost per DALY for vitamin a supplementation. *PLOS ONE: Cost-effectiveness of "Golden Mustard" for Treating Vitamin A Deficiency in India (2010)* https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0012046 (2010)
India: $23-$50 per DALY averted (least costly intervention, $1,000-$6,100 per death averted) Sub-Saharan Africa (2022): $220-$860 per DALY (Burkina Faso: $220, Kenya: $550, Nigeria: $860) WHO estimates for Africa: $40 per DALY for fortification, $255 for supplementation Uganda fortification: $18-$82 per DALY (oil: $18, sugar: $82) Note: Wide variation reflects differences in baseline VAD prevalence, coverage levels, and whether intervention is supplementation or fortification Additional sources: https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0012046 | https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0266495

135. News, U. Clean water & sanitation (LMICs) ROI. *UN News* https://news.un.org/en/story/2014/11/484032 (2014).

136. PMC. Cost-effectiveness threshold ($50,000/QALY). *PMC* https://pmc.ncbi.nlm.nih.gov/articles/PMC5193154/
The $50,000/QALY threshold is widely used in US health economics literature, originating. from dialysis cost benchmarks in the 1980s. In US cost-utility analyses, 77.5% of authors use either $50,000 or $100,000 per QALY as reference points. Most successful health programs cost $3,000-10,000 per QALY. WHO-CHOICE uses GDP per capita multiples (1× GDP/capita = "very cost-effective", 3× GDP/capita = "cost-effective"), which for the US ($70,000 GDP/capita) translates to $70,000-$210,000/QALY thresholds. Additional sources: https://pmc.ncbi.nlm.nih.gov/articles/PMC5193154/ | https://pmc.ncbi.nlm.nih.gov/articles/PMC9278384/

137. Institute, I. B. Chronic illness workforce productivity loss. *Integrated Benefits Institute 2024* https://www.ibiweb.org/resources/chronic-conditions-in-the-us-workforce-prevalence-trends-and-productivity-impacts (2024)
78.4% of U.S. employees have at least one chronic condition (7% increase since 2021). 58% of employees report physical chronic health conditions 28% of all employees experience productivity loss due to chronic conditions Average productivity loss: $4,798 per employee per year Employees with 3+ chronic conditions miss 7.8 days annually vs 2.2 days for those without Note: 28% productivity loss translates to roughly 11 hours per week (28% of 40-hour workweek) Additional sources: https://www.ibiweb.org/resources/chronic-conditions-in-the-us-workforce-prevalence-trends-and-productivity-impacts | https://www.onemedical.com/mediacenter/study-finds-more-than-half-of-employees-are-living-with-chronic-conditions-including-1-in-3-gen-z-and-millennial-employees/ | https://debeaumont.org/news/2025/poll-the-toll-of-chronic-health-conditions-on-employees-and-workplaces/

138. Gilens, M. & Page, B. I. Testing theories of american politics: Elites, interest groups, and average citizens. *Perspectives on Politics* **12**, 564–581 (2014).

139. Coffey, B., McLaughlin, P. A. & Peretto, P. F. The cumulative cost of regulations. *Review of Economic Dynamics* **38**, 1–21 (2020)
Regulatory restrictions have had a net effect of dampening economic growth by approximately 0.8 percent per annum since 1980... Had regulation been held constant at levels observed in 1980, the economy would have been nearly 25 percent larger by 2012 (i.e., regulatory growth since 1980 cost GDP $4 trillion in 2012, or about $13,000 per capita).

140. Dawson, J. W. & Seater, J. J. Federal regulation and aggregate economic growth. *Journal of Economic Growth* **18**, 137–177 (2013)
Federal regulations added over the past fifty years have reduced real output growth by about two percentage points on average annually over the period 1949-2005... GDP at the end of 2011 would have been $53.9 trillion instead of $15.1 trillion if regulation had remained at its 1949 level.

141. Nations, U. Corruption costs global economy $2.6 trillion annually. https://www.unodc.org/unodc/en/frontpage/2018/December/corruption-costs-the-global-economy-us-2_6-trillion-annually-says-un-anti-corruption-day.html (2018)
Corruption costs 5% of global GDP annually ($2.6 trillion) .

142. Philipson, T., Berndt, E. R., Gottschalk, A. H. & Sun, E. Cost-benefit analysis of the FDA: The case of the prescription drug user fee acts. *Journal of Public Economics* **92**, 1306–1325 (2008)
Faster drug approvals under PDUFA saved 140,000 to 310,000 life years from patients who had more rapid access to drugs, though this benefit was partially offset by an estimated 56,000 life years lost due to adverse events... Consumer welfare increased between $7 billion and $20 billion.
143. Costinot, A. & Rodriguez-Clare, A. Trade theory with numbers: Quantifying the consequences of globalization. *Handbook of International Economics* **4**, 197–261 (2015)
Gravity models estimate trade barriers cost 5-10% of GDP.
144. Indices, S. D. J. SPIVA u.s. Scorecard year-end 2023. (2023)
Over the 15-year period ending Dec. 2023, 87.98% of large-cap U.S. equity funds underperformed the S&P 500. The longer the time horizon, the more difficult it becomes for active managers to outperform. SPIVA (S&P Indices Versus Active) provides the de facto scorecard for the active vs. passive debate. Additional sources: https://www.spglobal.com/spdji/en/research-insights/spiva/
145. Drutman, L. *The Business of America Is Lobbying.* (Oxford University Press, 2015). doi:10.1093/acprof:oso/9780190215514.001.0001.
146. Goodhart, C. A. E. Problems of monetary management: The UK experience. *Monetary Theory and Practice* 91–121 (1984).
147. Coase, R. H. The problem of social cost. *Ronald H. Coase* **3**, 1–44 (1960)
The Problem of Social Cost" established that in the absence of transaction costs, parties will bargain to efficient outcomes regardless of initial property rights. This foundational insight—known as the Coase Theorem—demonstrates that inefficient allocation persists due to transaction costs (information, negotiation, enforcement), not due to the allocation problem itself. The theorem implies that welfare-improving reforms are theoretically achievable through compensation schemes that make all parties better off. Additional sources: https://www.jstor.org/stable/724810
148. Nation, T. The pentagon's revolving door keeps spinning. *The Nation* https://www.thenation.com/article/politics/pentagon-revolving-door-congress/ (2021)
From 2009 to 2019, approximately 380 high-ranking Department of Defense officials and military officers transitioned to become lobbyists, board members, executives, or consultants for defense contractors. Former members of Congress routinely transition to lucrative lobbying and consulting careers in the defense sector, with compensation often 10-20 times their congressional salary. Additional sources: https://www.thenation.com/article/politics/pentagon-revolving-door-congress/
149. Mulcahy, D. E., Andrew W.; Schwam. International prescription drug price comparisons. (2021)
US drug prices were 256% of prices in 32 OECD comparison countries. For brand-name drugs, US prices were 344% of comparison country prices. For unbranded generics, US prices were 84% of comparison prices. Potential annual savings from reference pricing policies estimated at $100 billion or more. Additional sources: https://www.rand.org/pubs/research_reports/RR2956.html

150. Arrow, K. J. *Social Choice and Individual Values*. (Kenneth J. Arrow, 1951).
Arrow's Impossibility Theorem proves that no rank-order voting system can satisfy all of four. "fairness" criteria: unrestricted domain, non-dictatorship, Pareto efficiency, and independence of irrelevant alternatives. This foundational result in social choice theory demonstrates that there is no perfect method for aggregating individual preferences into collective decisions—all voting systems involve tradeoffs. The theorem has profound implications for democratic theory, welfare economics, and mechanism design, showing that preference aggregation is irreducibly political. Additional sources: https://www.amazon.com/Social-Choice-Individual-Values-Monograph/dp/0300013647

151. Gibbard, A. Manipulation of voting schemes: A general result. *Econometrica* **41**, 587–601 (1973)
Gibbard proved that any deterministic voting scheme with at least three possible outcomes is either dictatorial or manipulable (i.e., voters can sometimes benefit by misrepresenting their preferences). This result, proven independently by Satterthwaite (1975), establishes fundamental limits on voting mechanisms: strategic voting cannot be eliminated from any non-trivial democratic system. The theorem has profound implications for mechanism design and social choice theory.

152. Myerson, R. B. Optimal auction design. *Mathematics of Operations Research* **6**, 58–73 (1981)
This seminal paper establishes the Revenue Equivalence Theorem and introduces the "virtual valuation" concept for mechanism design. Myerson shows how to design auctions that maximize expected revenue given incentive-compatible reporting constraints. The paper, along with the Revelation Principle, provides foundational tools for designing mechanisms where agents truthfully report private information. Essential for understanding incentive-compatible oracle design in algorithmic governance systems.

153. Maskin, E. Nash equilibrium and welfare optimality. *Review of Economic Studies* **66**, 23–38 (1999).